



B. N. M. Institute of Technology

An Autonomous Institution under VTU.



Bhageerathi Bai Narayana Rao Maanay
Educational Institutions
1972-2022

DEPARTMENT OF MECHANICAL ENGINEERING

ENGINEERING GRAPHICS & DESIGN

25EGD114/124



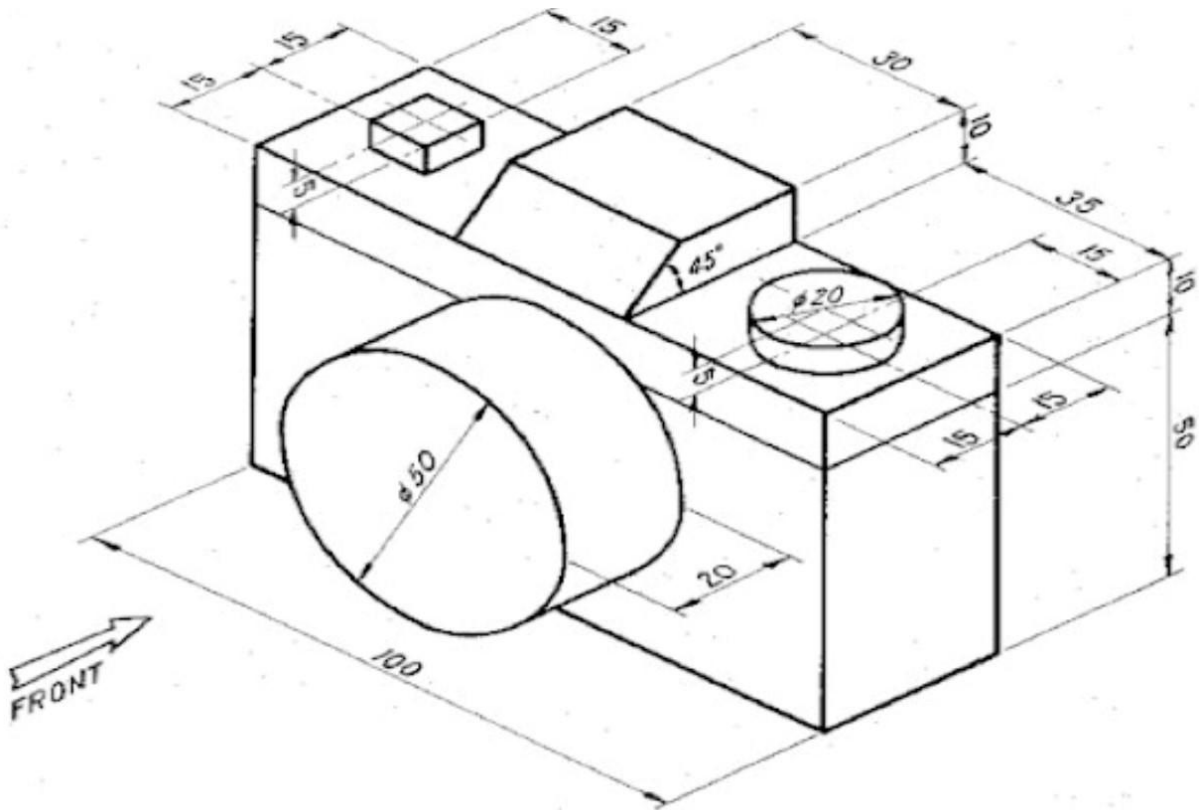
Department of Mechanical Engineering

Engineering Graphics and Design [25EDG114/124]

LIST OF EXERCISES (2025-26)

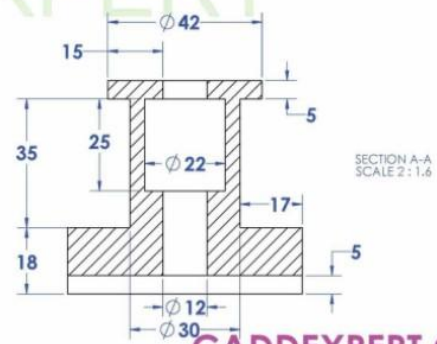
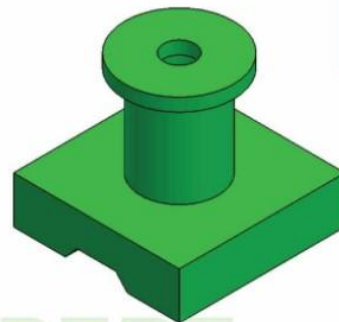
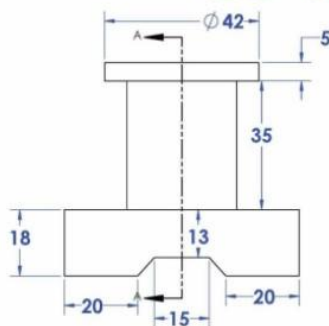
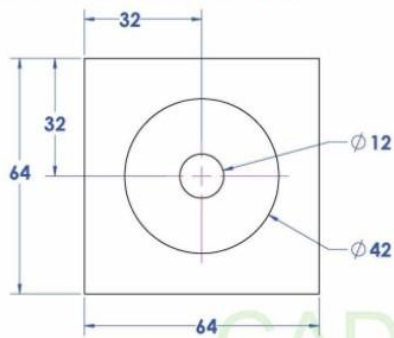
Textbook: Engineering Graphics and Design BNMIT-WILEY

Sl. No	Exercise	CW	HW	CPO
1	Use of Instruments, Dimensioning, Polygons	Types of Lines, Dimensioning	Regular Polygons	Regular Polygons
2	Projection of Points - 1	1, 2	3, 4	5
3	Projection of Straight Lines-I	1, 4	2, 9	3, 12
4	Projection of Straight Lines-II	8, 11	6, 13	5, 10
5	Projection of Planes-I	1, 4	2, 3	5
6	Projection of Planes-II	7, 8, 11	6	8, 12
7	Projection of Planes-III	13, 10	9, 17	15, 17
8	Projection of Solids I	1, 4	7, 14	15
9	Projection of Solids II	6, 8	13, 12	12
10	Projection of Solids III	3, 5, 10	9	2, 10
11	Development of lateral Surfaces-I	2, 6	1, 7	8
12	Development of lateral Surfaces-II	4, 3	5, 10	11, 12
13	Isometric-I	1, 3, 6	5, 11	4, 8
14	Isometric-II	7, 13, 12	9, 15	14, 16
15	Solid Modeling	Below Figures		
16	PCB Drawing and Floor Planning	Below Figures		



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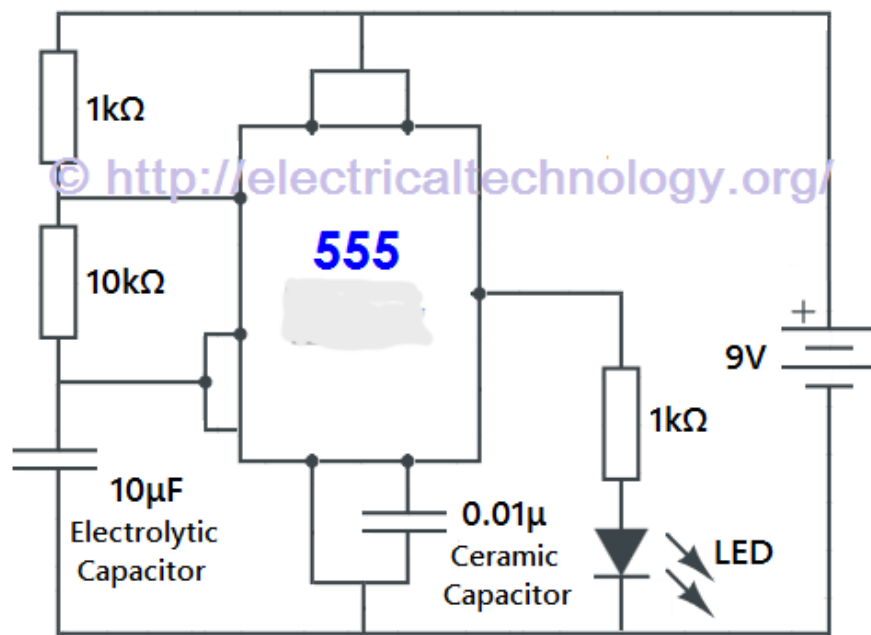
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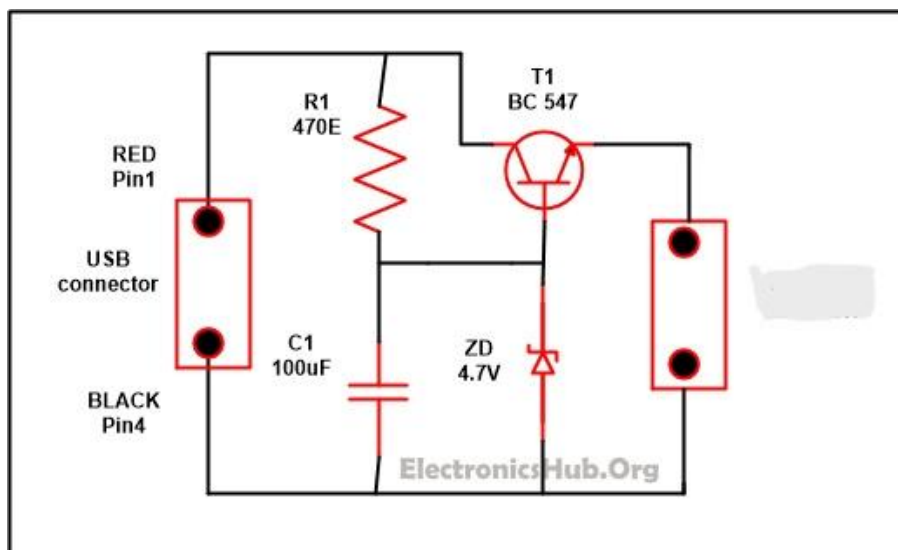
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PCB Drawing

Timer Circuit



USB Mobile Charger Circuit



Floor Plan

Lab Layout

Window and Door Symbols

WINDOW SYMBOLS



FIXED WINDOW



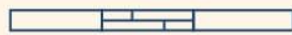
SINGLE CASEMENT WINDOW



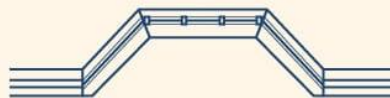
VERTICAL SLIDING WINDOW



DOUBLE CASEMENT WINDOW



HORIZONTAL SLIDING WINDOW

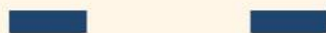


BAY WINDOW



DOUBLE-HUNG WINDOW

DOOR SYMBOLS



OPEN DOORWAY



SINGLE DOOR



SLIDING DOOR



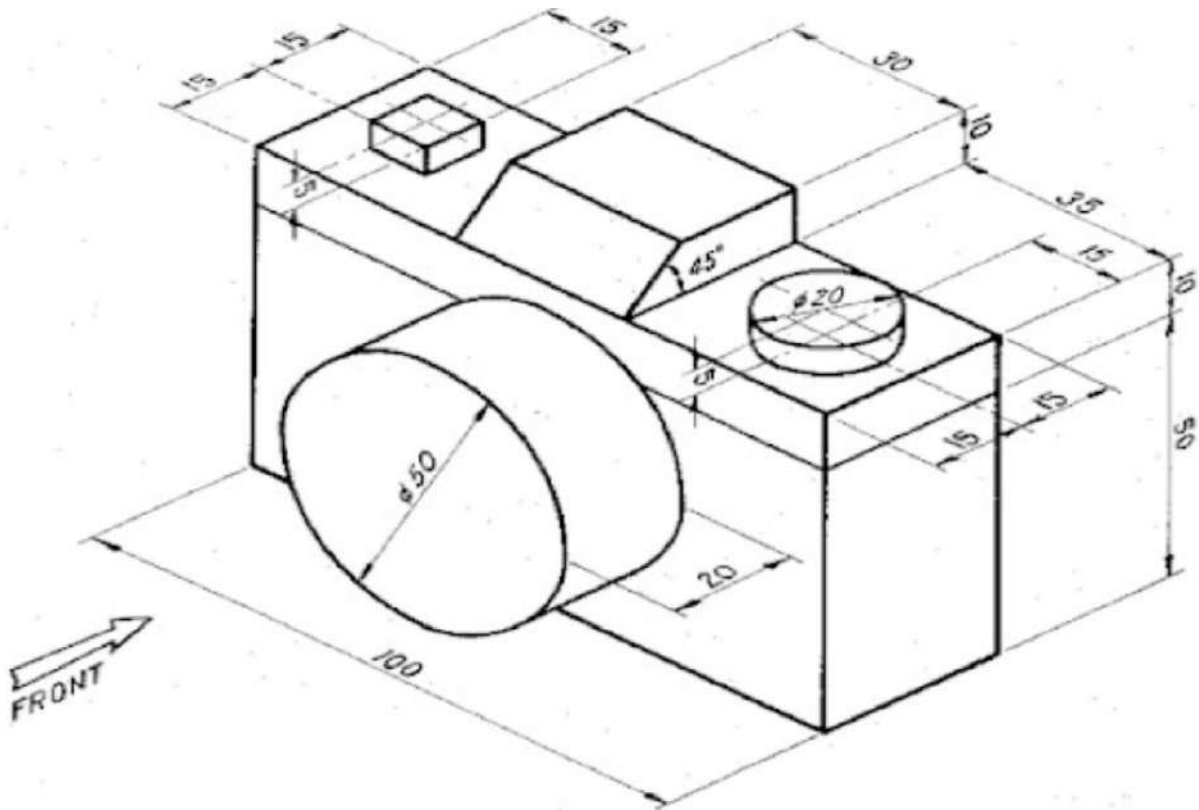
DOUBLE DOOR



POCKET DOOR

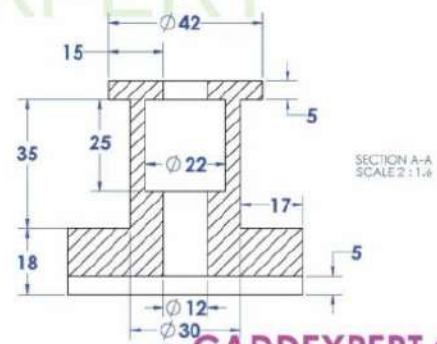
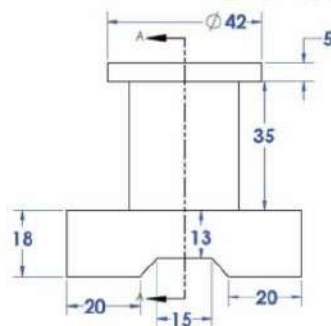
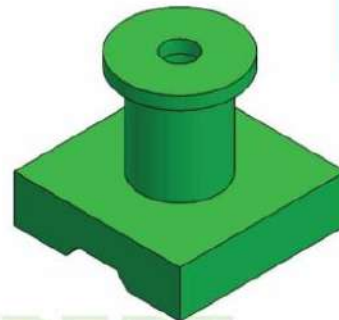
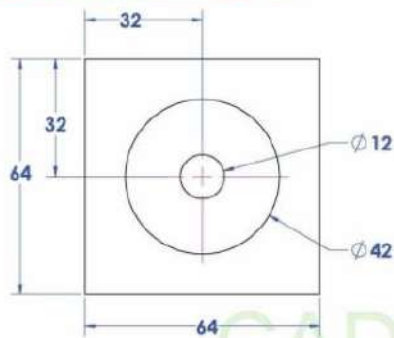


BIFOLD DOOR



CADDEXPERT.COM

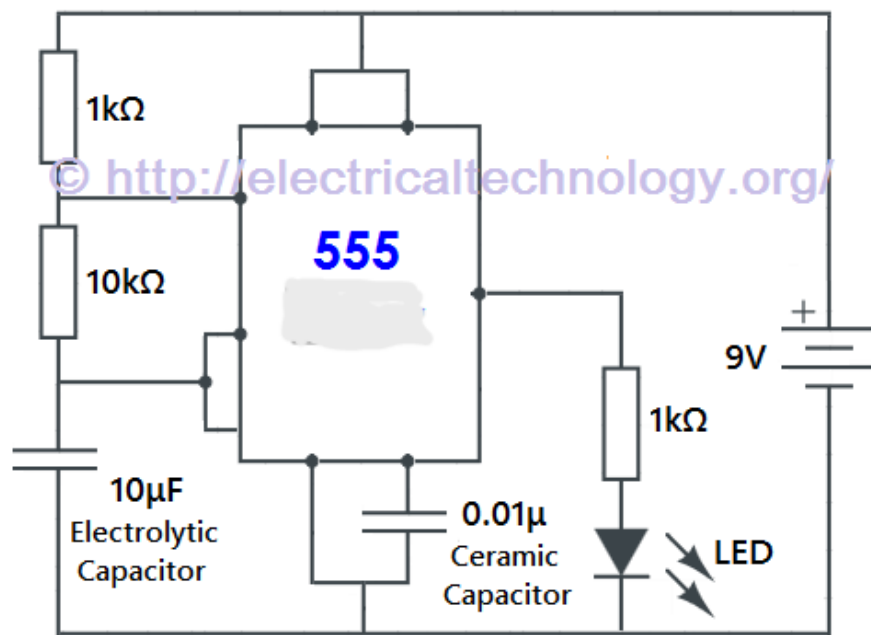
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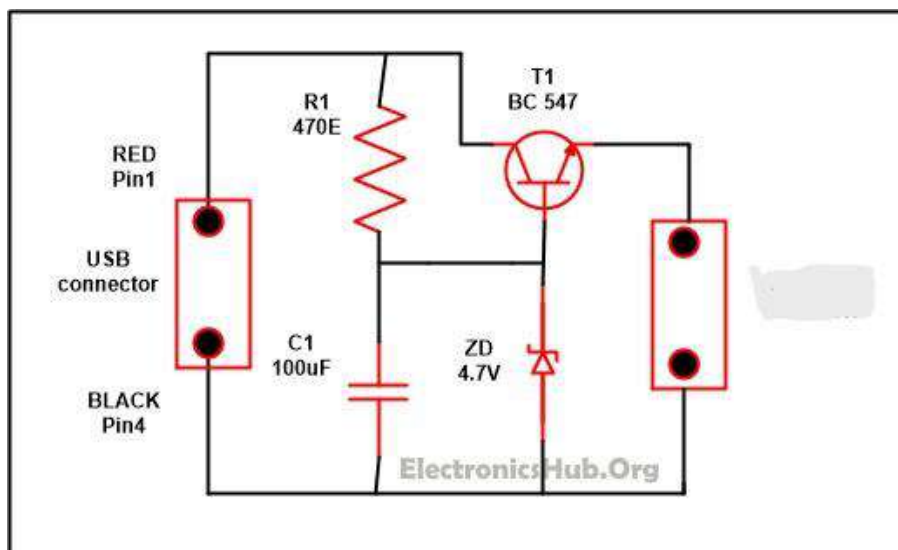
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PCB Drawing

Timer Circuit



USB Mobile Charger Circuit



Floor Plan

Lab Layout

Window and Door Symbols

WINDOW SYMBOLS



FIXED WINDOW



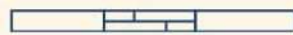
SINGLE CASEMENT WINDOW



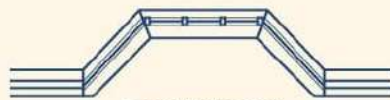
VERTICAL SLIDING WINDOW



DOUBLE CASEMENT WINDOW



HORIZONTAL SLIDING WINDOW

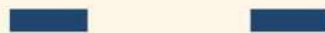


BAY WINDOW



DOUBLE-HUNG WINDOW

DOOR SYMBOLS



OPEN DOORWAY



SINGLE DOOR



SLIDING DOOR



DOUBLE DOOR



POCKET DOOR

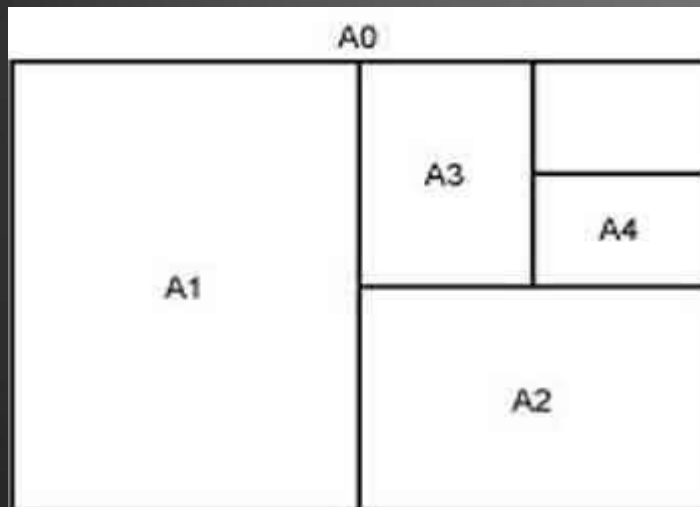


BIFOLD DOOR

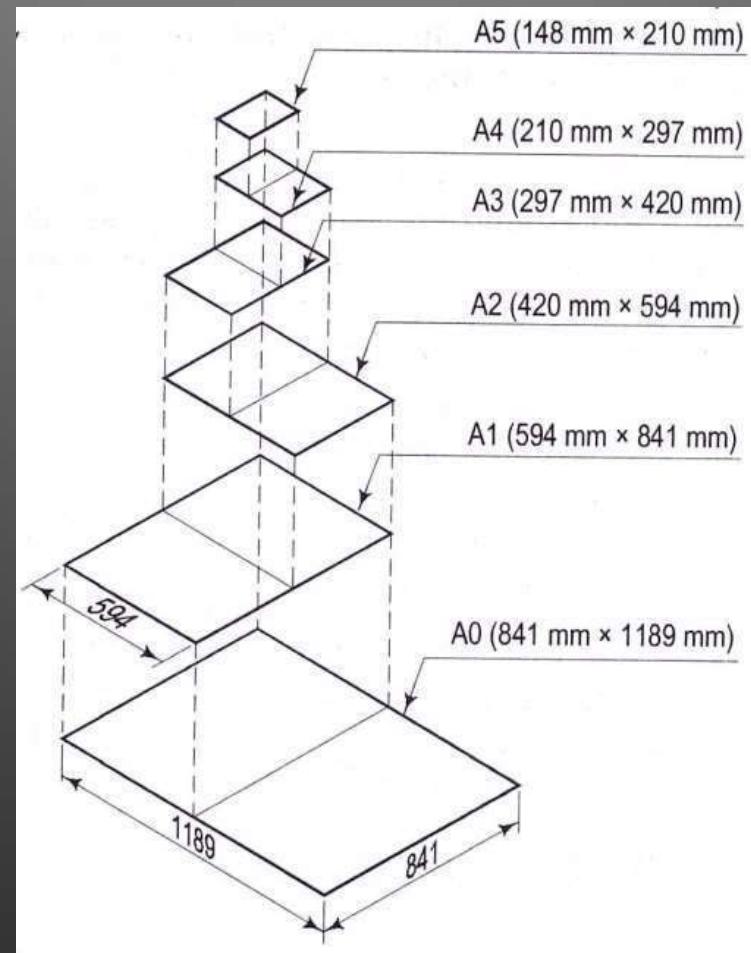
Drawing Instruments

1. Drawing Board
2. Drawing Sheet
3. Drawing Sheet Holder
4. Set-squares – 45° and 30°–60°
5. Large size Compass
6. Small bow Compass
7. Large size Divider
8. Small bow Divider
9. Scales – 6" and 12"
10. Protractor
11. French Curve
12. Drawing Pencils – H, 2H, HB
13. Sand Paper
14. Eraser (Rubber)
15. Drawing Pins and Clips
16. Cello Tape
17. Duster or Handkerchief
18. Drafting Machine / Mini Drafter
19. Sketch Book (Medium size)
20. Roller Scale
21. Pencil Sharpener
22. Sheet Folder

Type of Drawing Sheets

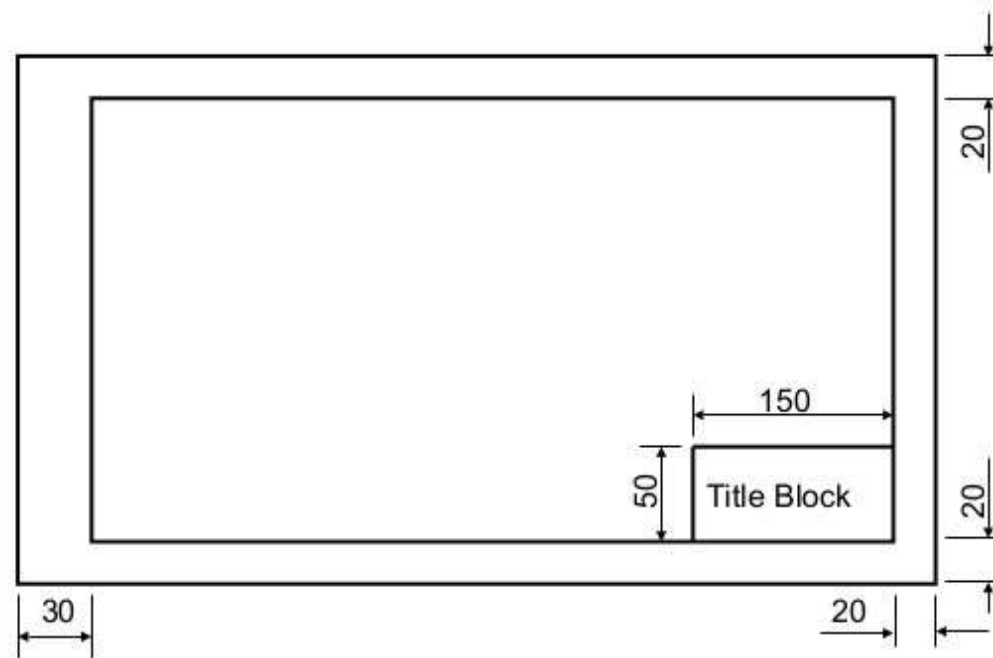


Designation	Trimmed Size (mm)	Untrimmed size (mm)
A0	841 x 1189	880 x 1230
A1	594 x 841	625 x 880
A2	420 x 594	450 x 625
A3	297 x 420	330 x 450
A4	210 x 297	240 x 330



Layout of a Drawing Sheet

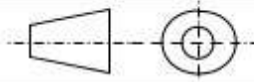
Layout of Drawing Sheet



All the dimensions are in millimeters.

Title Block

Title Block (Sample)

10	NAME:	UPES DEHRADUN		20
10	ROLL NO.:			
10	SEM. & SEC.:	LINES & LETTERING		10
10	SUB. DATE:		SHEET NO.: 01	10
10	GRADE:		CHECKED BY:	10
	50	50	50	

NOTES:

All the dimensions are in millimeters.

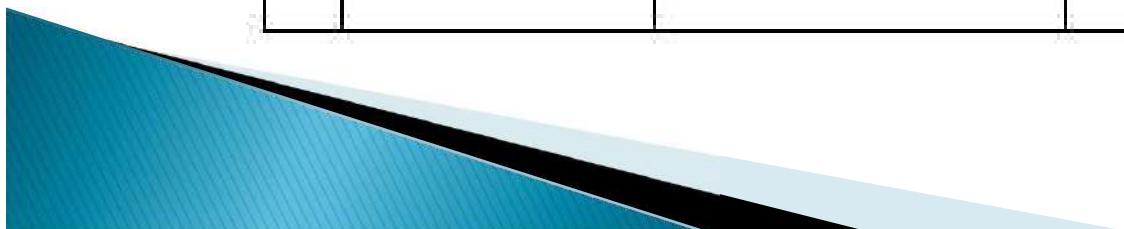


Types of Lines

Line		Description	General Application	
A		Continuous thick	A1 A2	Visible outlines. Visible edges.
B		Continuous thin (straight or curved)	B1 B2 B3 B4 B5 B6 B7	Imaginary lines of intersection. Dimension lines. Projection lines. Leader lines. Hatching lines. Outlines of revolved sections in place. Short centre lines
C		Continuous thin free hand	C1	Limits of partial or interrupted views and sections, If the limit is not a chain thin.
D		Continuous thin (straight) with zigzags	D1	Long break line
E		Dashed thick	E1 E2	Hidden outlines. Hidden edges.

Types of Lines

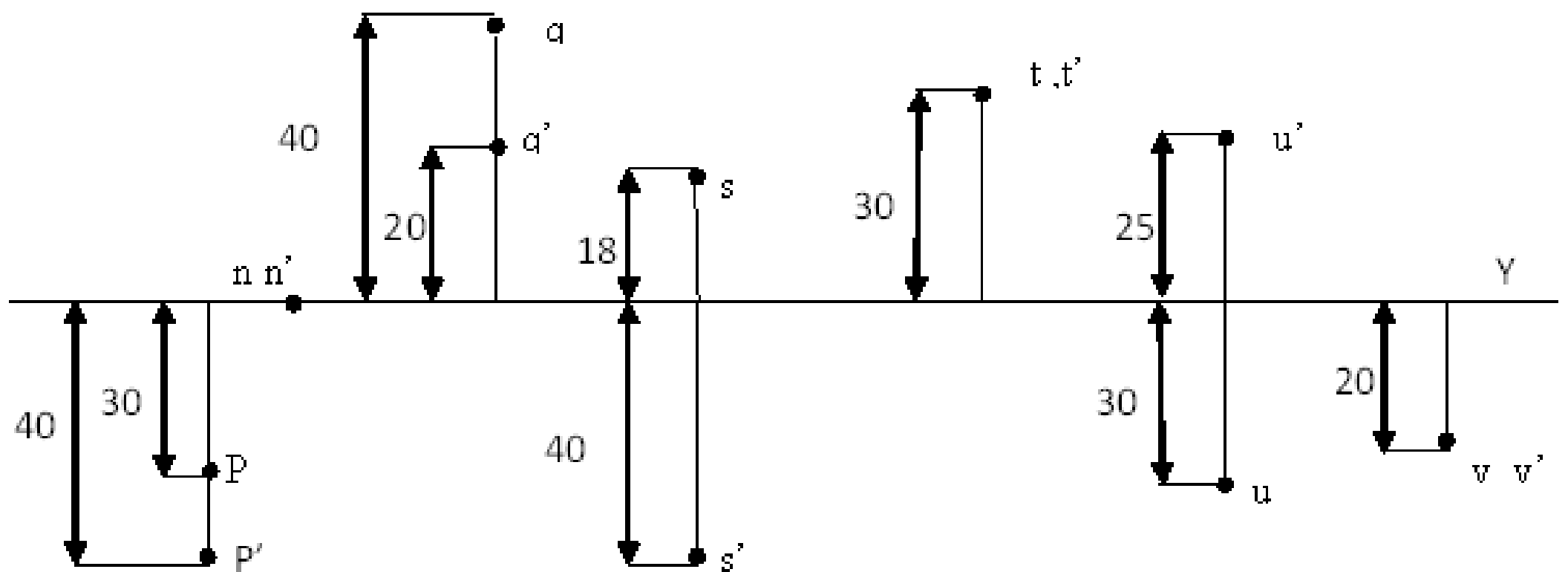
F		Dashed thin	F1 F2	Hidden outlines. Hidden edges.
G		Chain thin	G1 G2 G3	Center lines. Lines of symmetry. Trajectories
H		Chain thin, thick at ends and changes of direction	H1	Cutting planes.
J		Chain thick	J1	Indication of lines or surfaces to which a special requirement applies
K		Chain thin double dashed	K1 K1 K3 K4 K5	Outlines of adjacent parts. Alternative or extreme position of movable parts. Centroidal lines. Initial outlines prior to forming Parts situated in front of the cutting plane



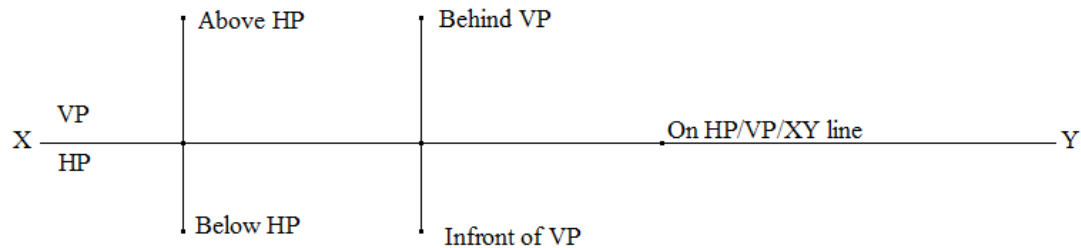
DEPARTMENT OF MECHANICAL ENGINEERING

PROJECTION OF POINTS

25EGD114/124

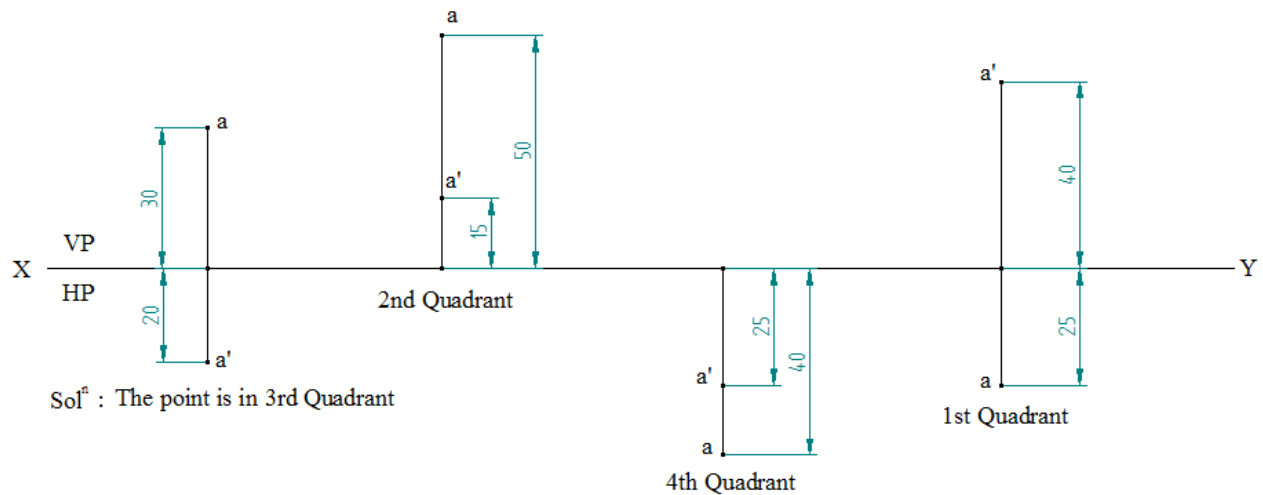


Note :

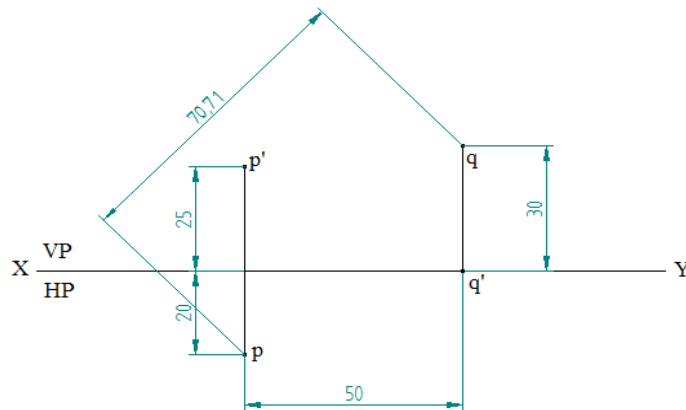


1. Draw the projections of point A in the following positions.

- 20mm below HP and 30mm behind VP.
- 15mm above HP and 50mm behind VP.
- 40mm below HP and 25mm in front of VP.
- 40mm above HP and 25mm in front of VP.

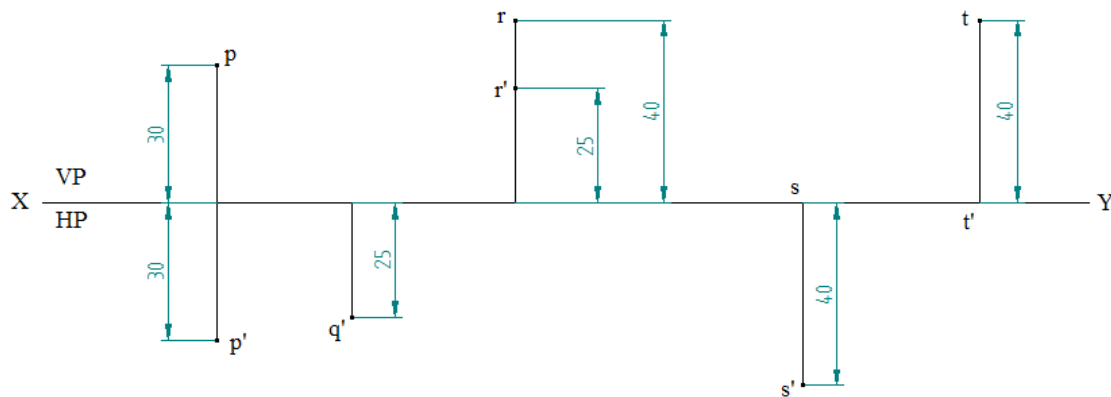


2. A point P is 25 mm above HP & 20mm in front of VP. Another point Q is on HP and 30mm behind VP. The distance between their projection measured parallel to the line of intersection of VP and HP is 50mm. Find the distance between the top view of points P and Q

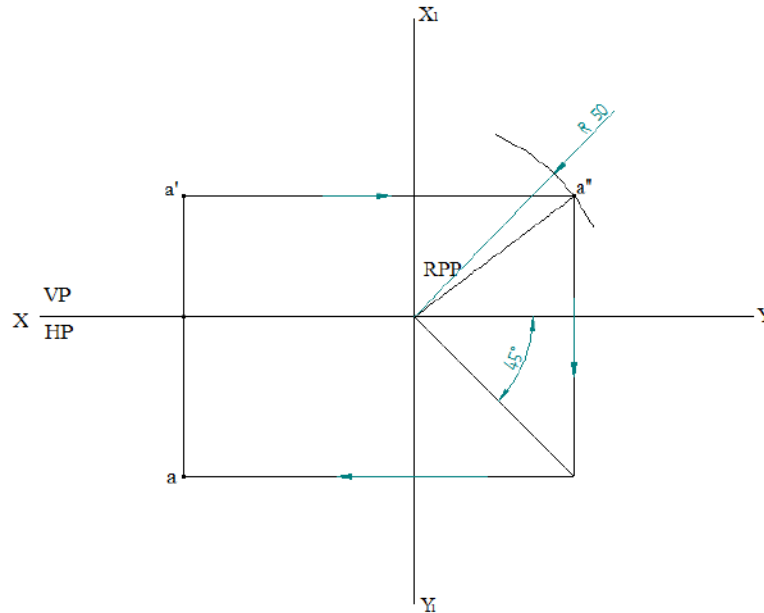


Distance Between TVs of P and Q is 70.71mm

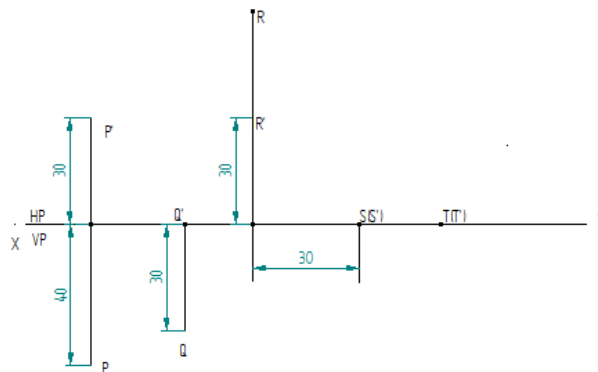
3. Draw the projections of the following points keeping the reference XY line common and distance between the projections (any two consecutive points) as 20mm.
- Point P 30mm in front of VP and 30mm above HP.
 - Point Q in HP and 25mm in front of VP
 - Point R 25mm above HP and 40mm behind VP.
 - Point S in VP and 40mm below HP.
 - Point T in HP and 40mm behind VP.



4. Draw the projections of a point A lying 30mm above HP and in first quadrant, if its shortest distance from the line of intersection of HP and VP is 50mm. Also find the distance of the Point from VP.



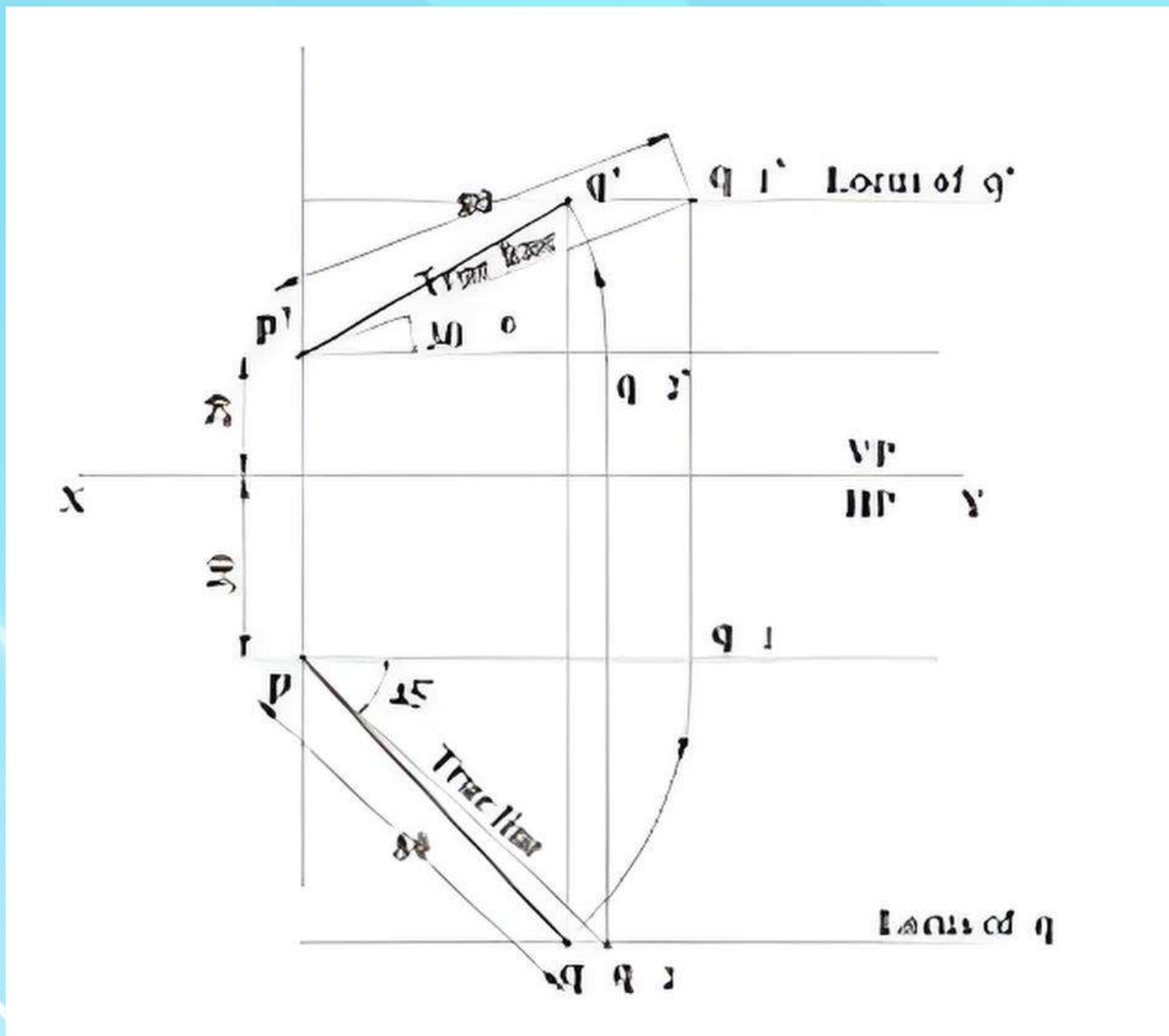
5. Draw the front and top views of the following points on common ground line XY, the Taking distance between the projections as 30mm. Draw the positions of the points in different quadrants.
- Point P 40mm in front of VP and 30mm above HP.
 - Point Q 30mm in front of VP and in HP.
 - Point R 60mm behind VP and 30mm above HP.
 - Point S in HP and VP.
 - Point T on HP VP.



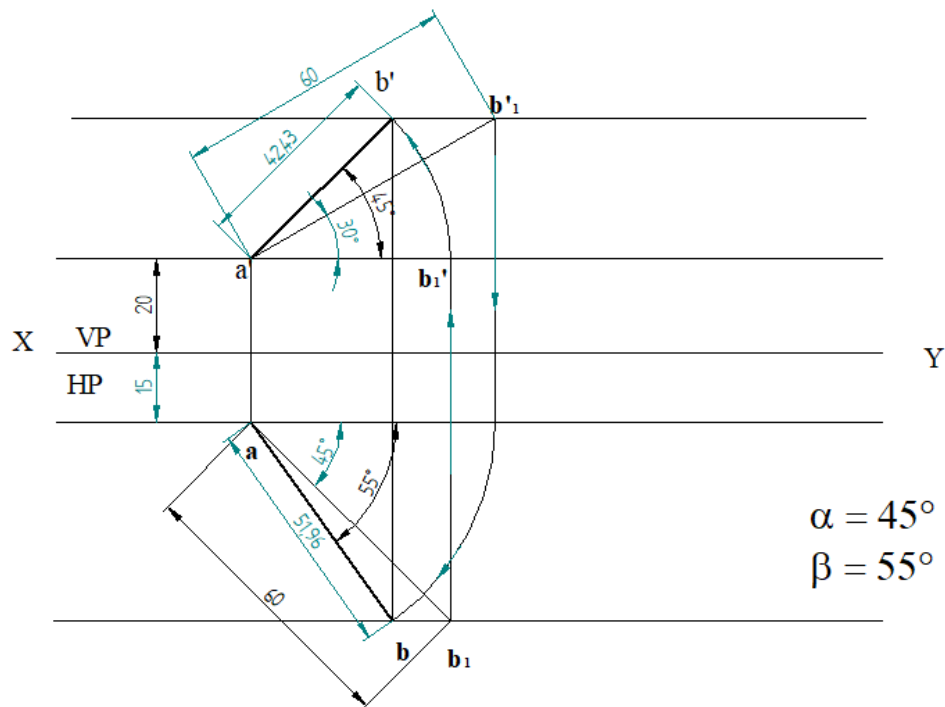
DEPARTMENT OF MECHANICAL ENGINEERING

PROJECTION OF STRAIGHT LINE

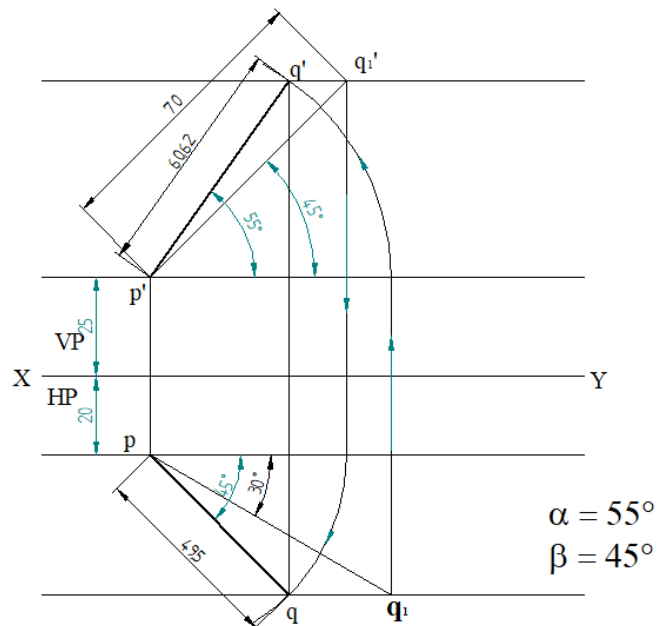
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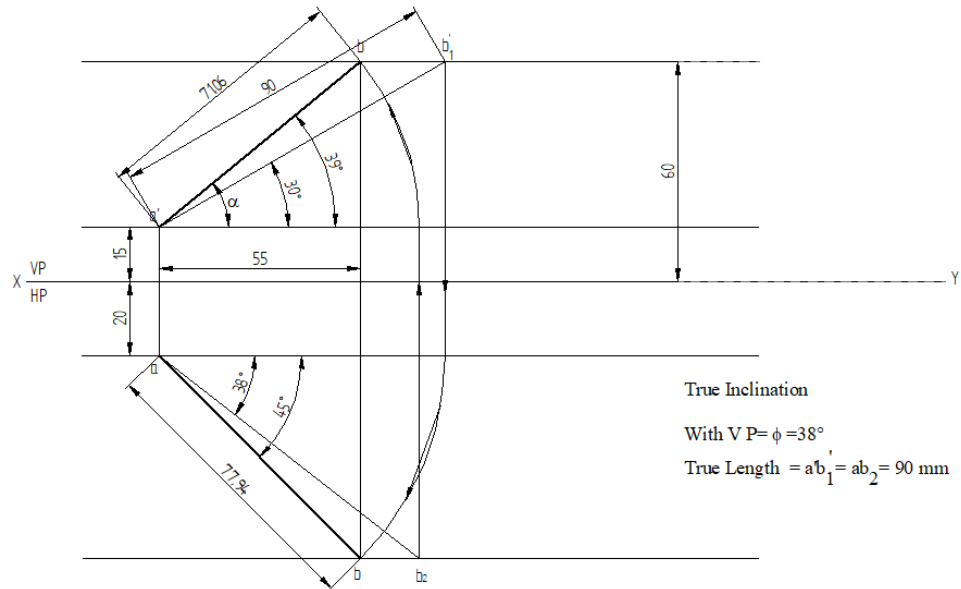
1. A straight line AB 60mm long makes an angle of 30° to HP and 45° to VP. The end A is 15mm in front of VP and 20mm above HP. Draw its projections.



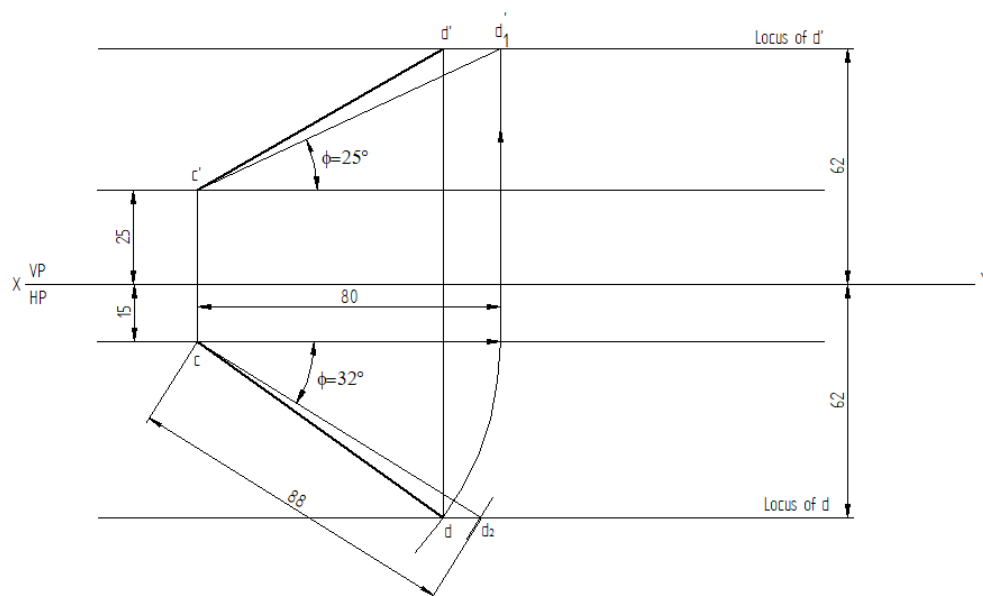
2. A straight line PQ 70mm long has its point P 25mm above HP and 20mm in front of VP. The line slopes upward right from point P to Q. Draw the projections when the line is inclined at 45° to HP and 30° to VP



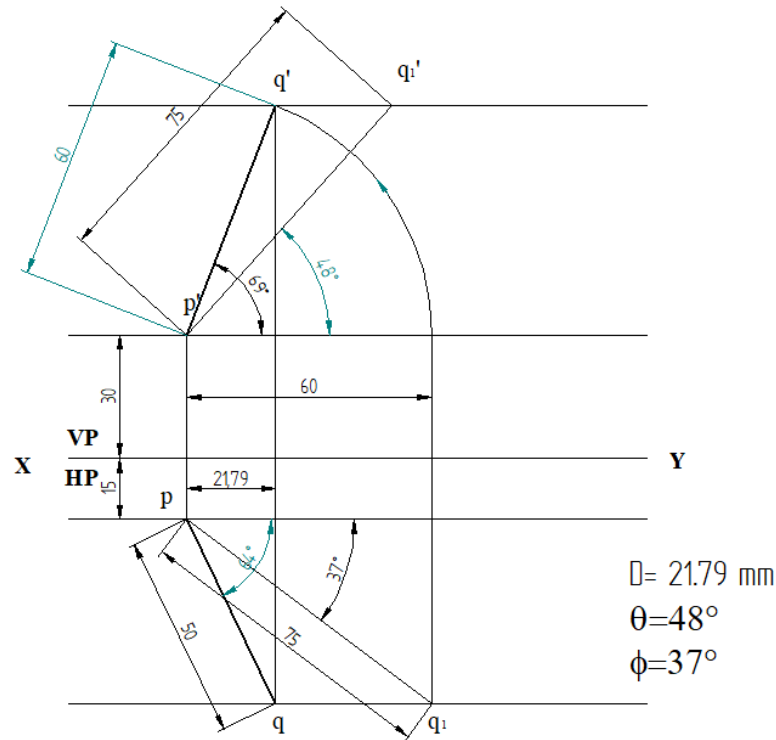
3. A line AB has its end A 15mm above HP and 20mm in front of VP. The end B is 60mm above HP and the line is inclined at 30° to HP. The distance between the end projectors of the line when measured parallel to the line of intersection is 55mm. Draw the projections and find its inclination with VP.



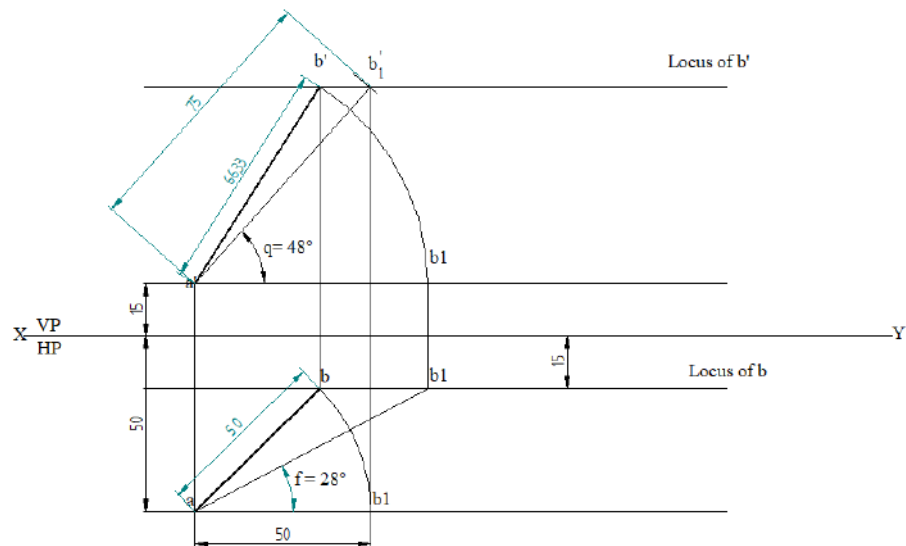
4. A line CD inclined at 25° to HP measures 80mm in top view. The end C is in the first Quadrant with 25mm and 15mm from HP and VP respectively. The end D is at equal distance from both the reference planes. Draw the projections and find true length and true inclination with the VP.



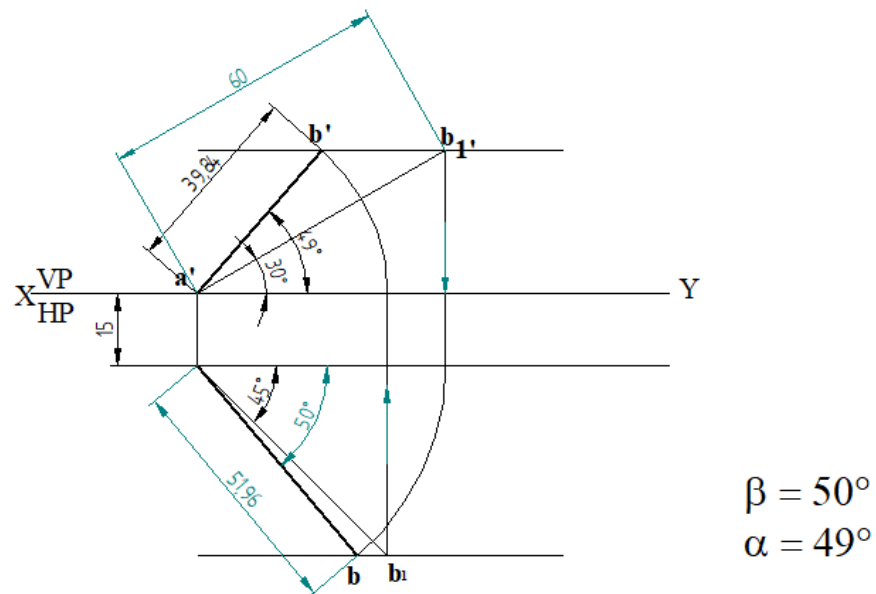
- 5 . The top view of the line PQ 75mm long measures 50mm and front view measures 60mm. The end P is 30mm above HP and 15mm in front of VP. Draw the projections Of the line and find its true inclinations with HP and VP. Find length of front view and distance between the end projectors.



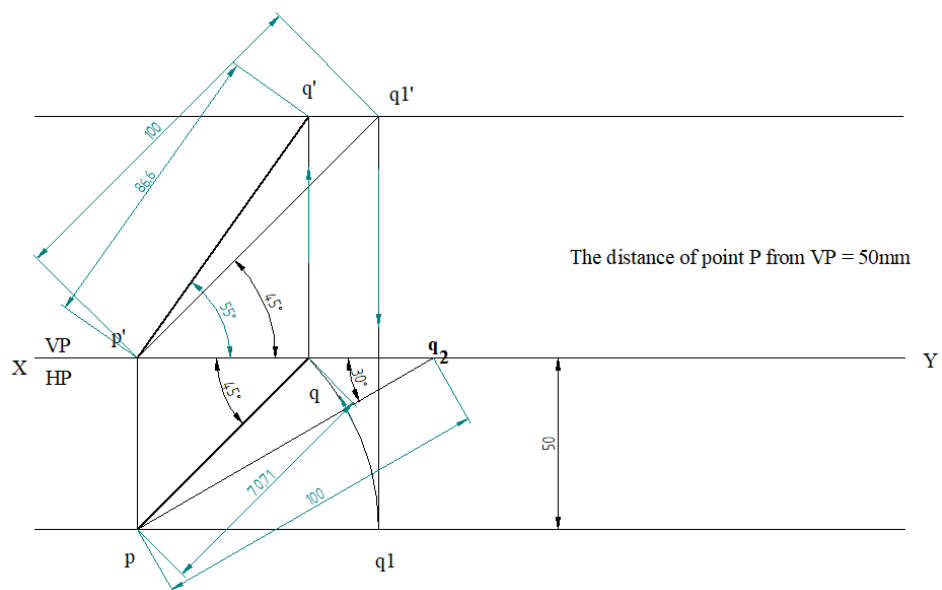
- 6 . Line measuring 75mm long has one of its ends 50mm in front of VP and 15mm above HP. The top view of the line is 50mm long. The other end is 15mm in front of VP and is above HP. Draw the projections of the line and find the true inclination



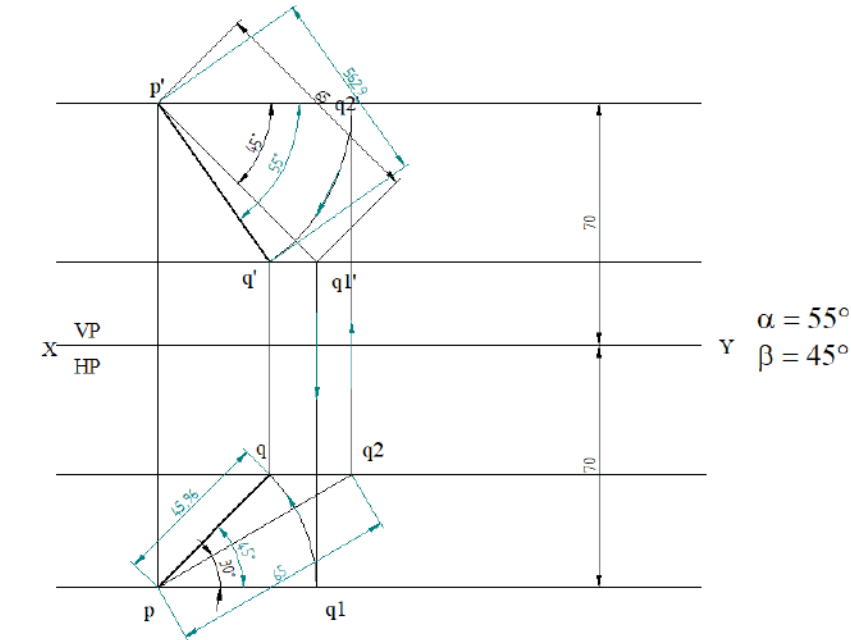
7. A straight line AB 60mm long makes an angle of 30° to HP & 45° to VP. The end A is 15mm in front of VP and in HP. Draw its projections.



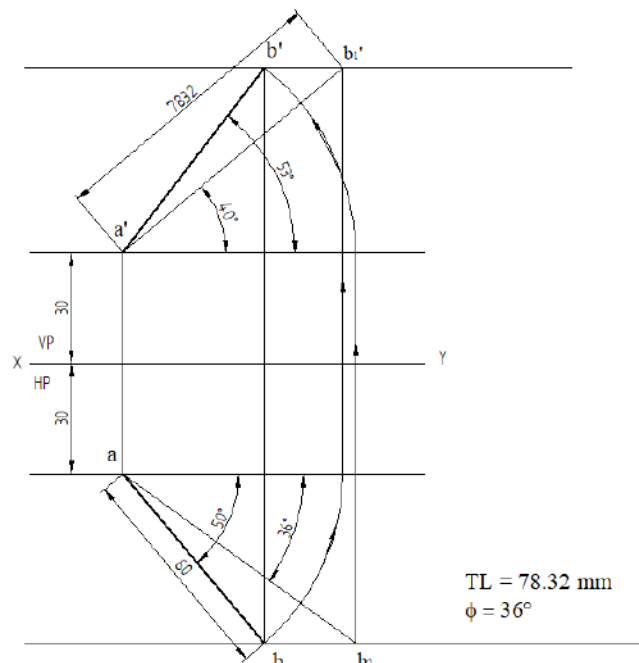
8. Draw the projections of a straight line PQ 100mm long, inclined at 45° to HP and 30° to VP. The end P is in HP and the end Q is in VP. Find the apparent length.



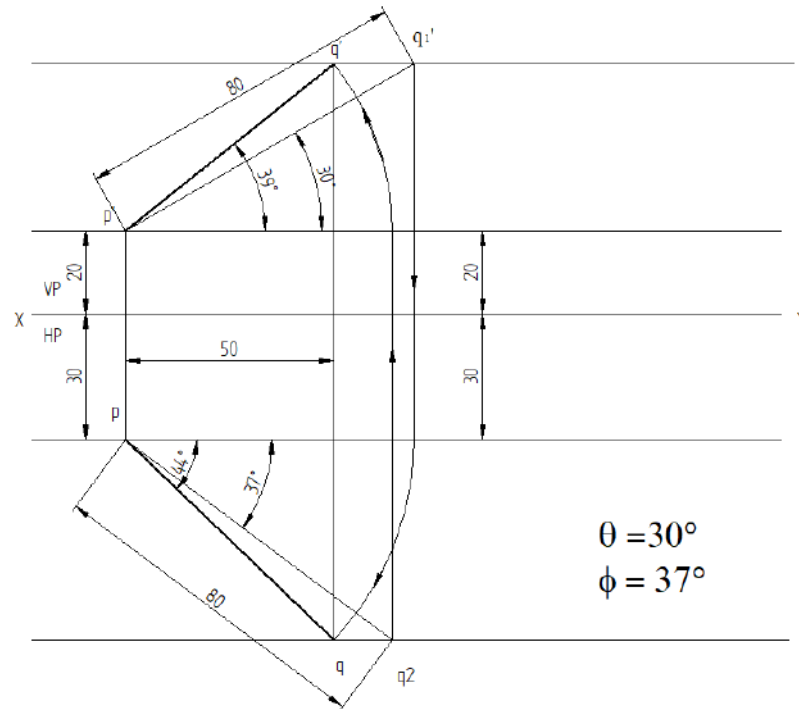
9. A straight line PQ 65mm long is inclined at 45° to HP and 30° to VP. The point P is 70mm from both the reference planes and the point Q is towards the reference planes. Draw the projections



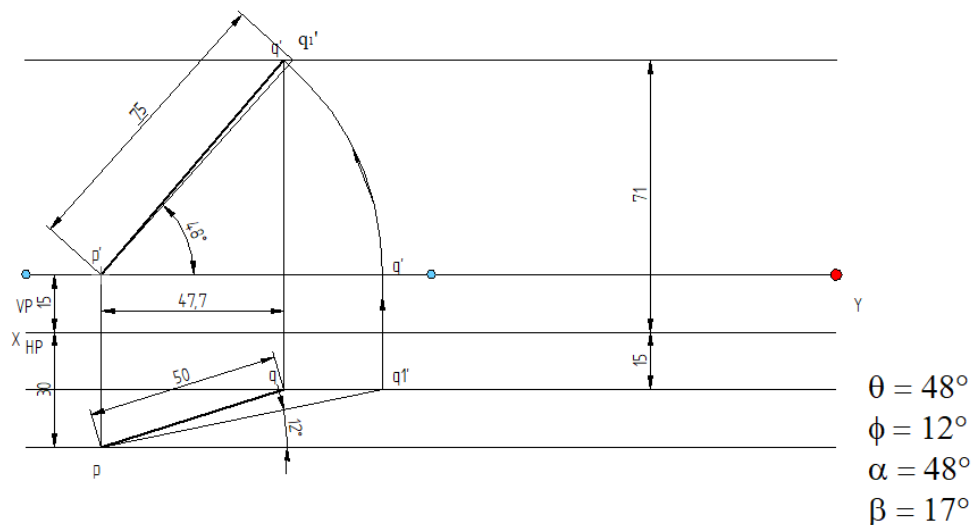
10. One end of a line AB is 30mm in front of VP and 30mm above HP. The line is inclined 40° to HP and its top view measuring 60mm inclined at 50° to XY line. Draw the projection of lines and determine the true length and inclination with VP.



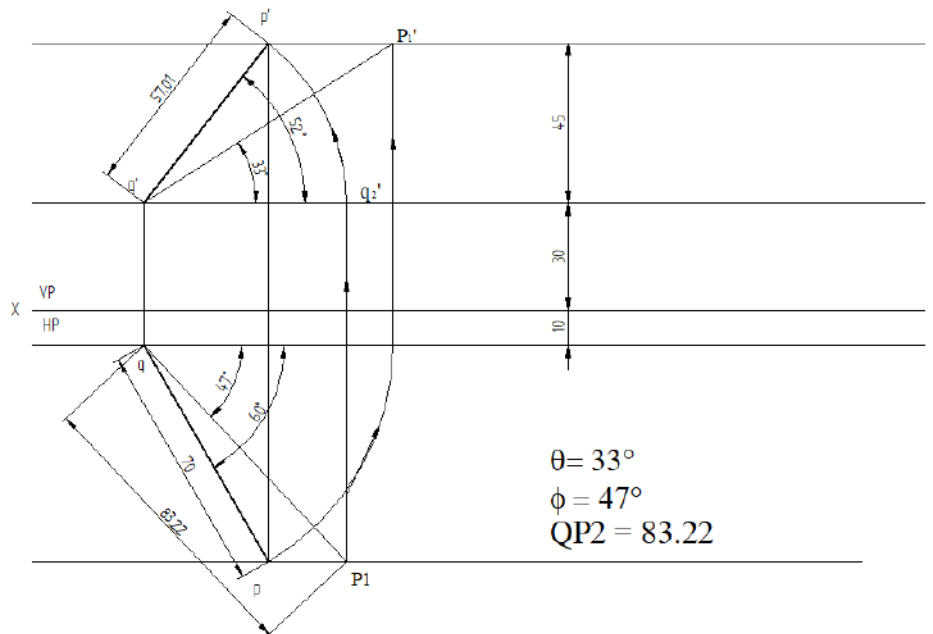
11. A line PQ measures 80mm in length. And inclined at 30° to HP the point P is above HP and in front of VP by 20 mm and 30 mm respectively. The distance between end projections of the line is 50mm specify the true inclination with VP.



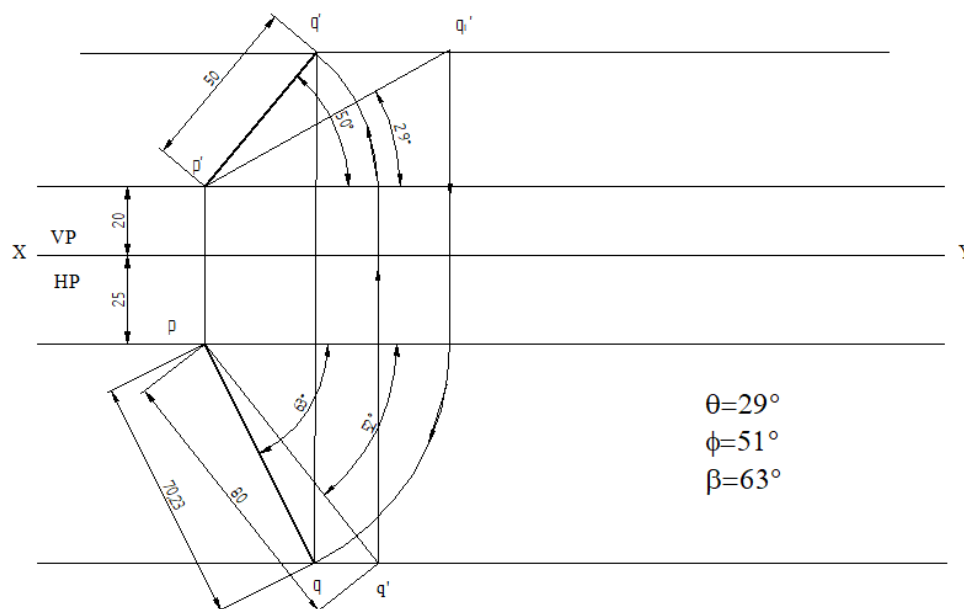
12. The top view of line PQ 75 mm long measures 50 mm. The end P is 30 mm in front of VP and 15 mm above HP. the end Q is 15 mm in front of VP and above HP. Draw the projections of the line and find its true inclinations with HP and VP. Also find the length of front view and distance between end projectors.



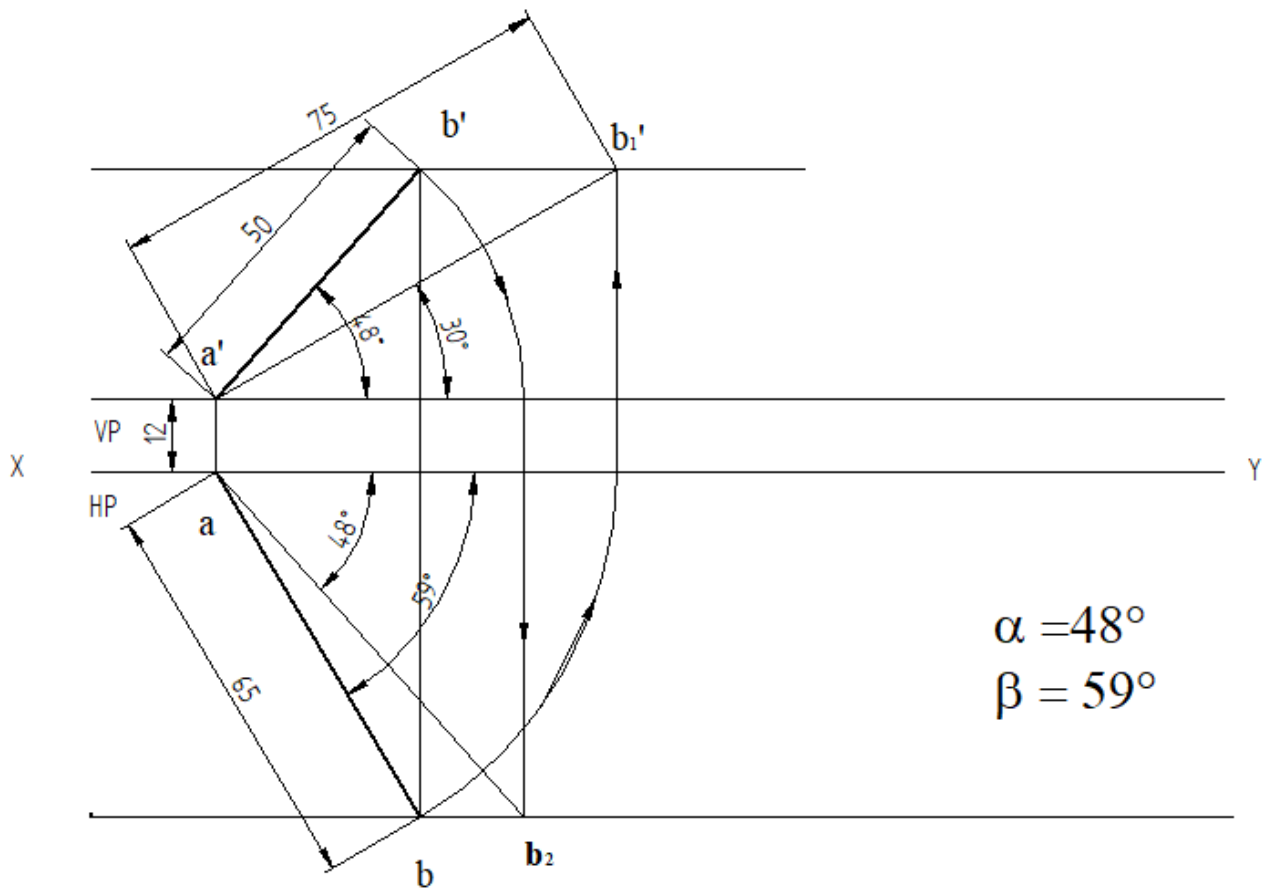
13. The top view pq of a straight line is 70 mm and makes an angle of 60° with XY line. The end Q is 10 mm in front of VP and 30mm above HP . The difference Between the distances of P and Q above HP is 45mm. Draw the projections Determine its length and true inclinations with HP and VP .



14. The front view of the line PQ 80 mm long measures 50 mm and it is inclined to XY line at 50° . One end of line P is 20mm above HP and 25mm in front of the VP Draw the front and top view of the line and find its inclinations of the line With HP and VP .



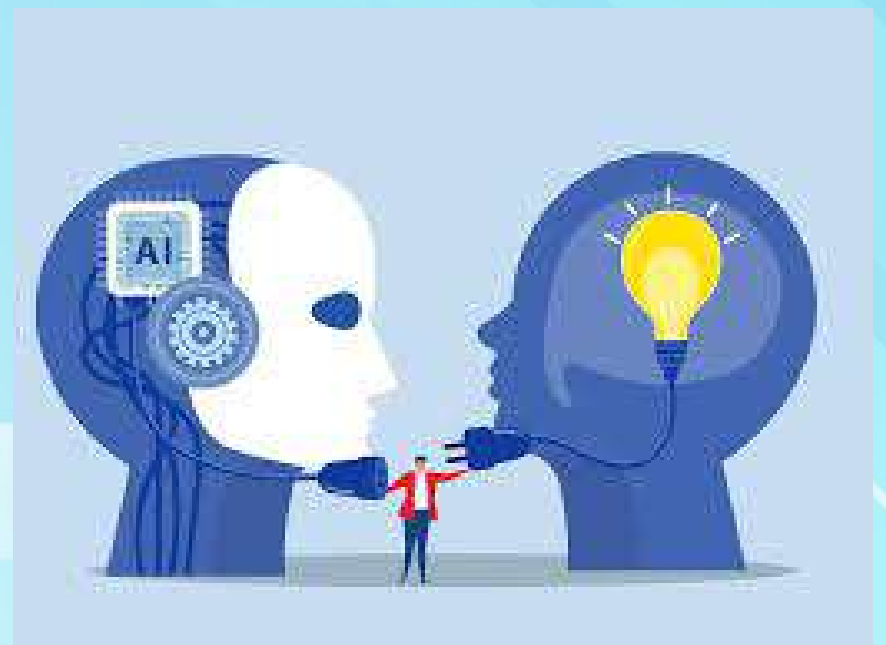
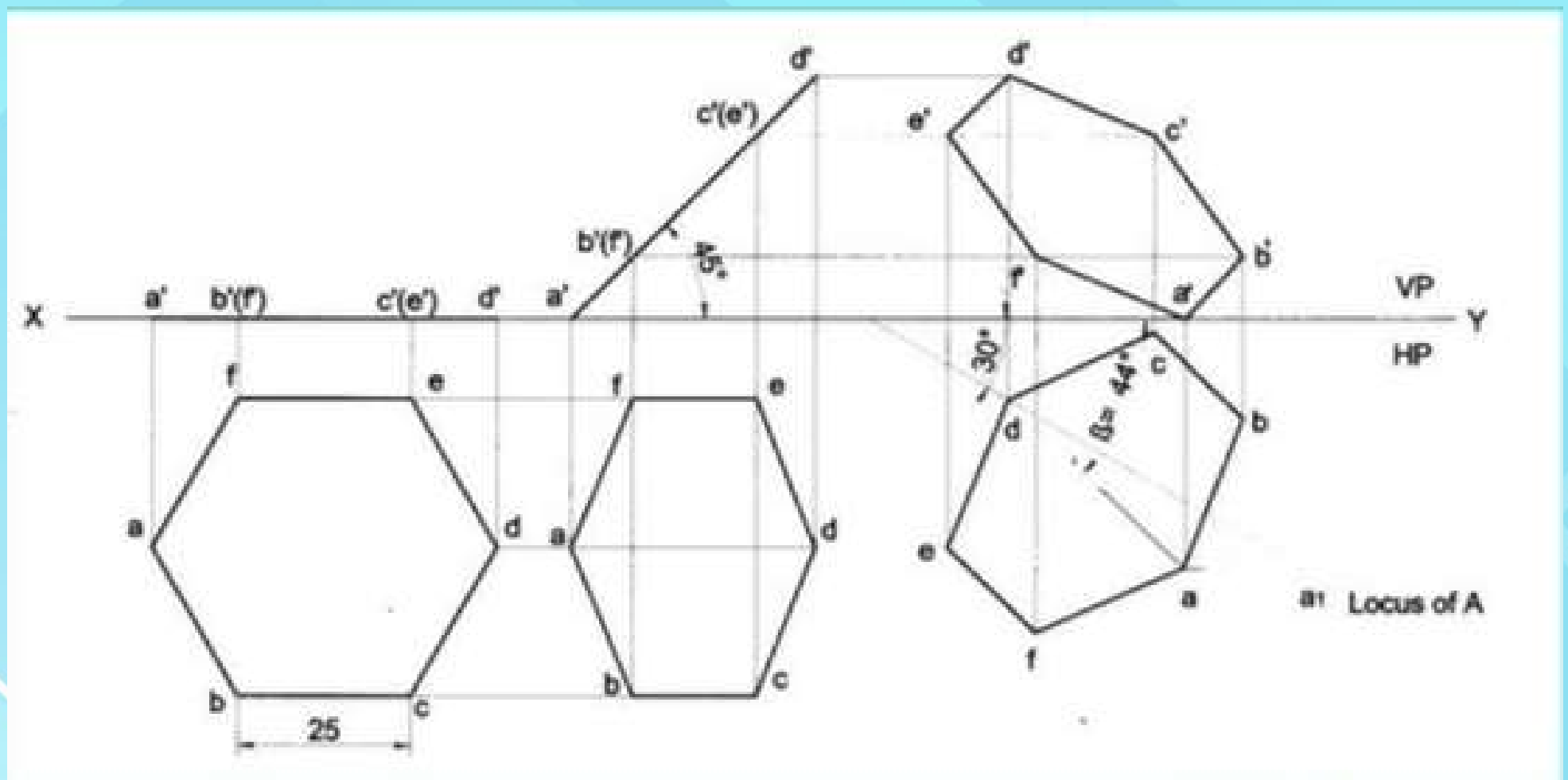
15. The top view of a 75mm long line AB measures 65mm, while the front view is 50mm. Its one end A is in the VP and 12mm above HP. Draw projections of AB and determine its inclinations with HP and VP.



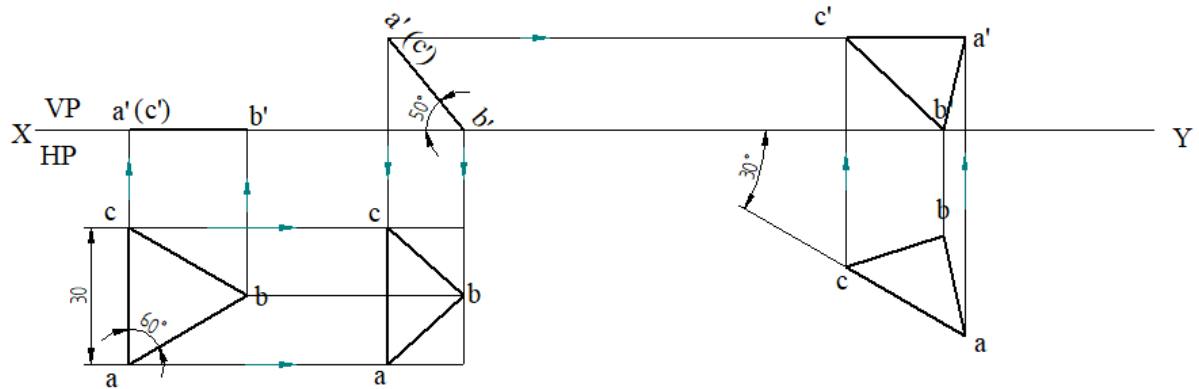
DEPARTMENT OF MECHANICAL ENGINEERING

PROJECTION OF PLANE SURFACES

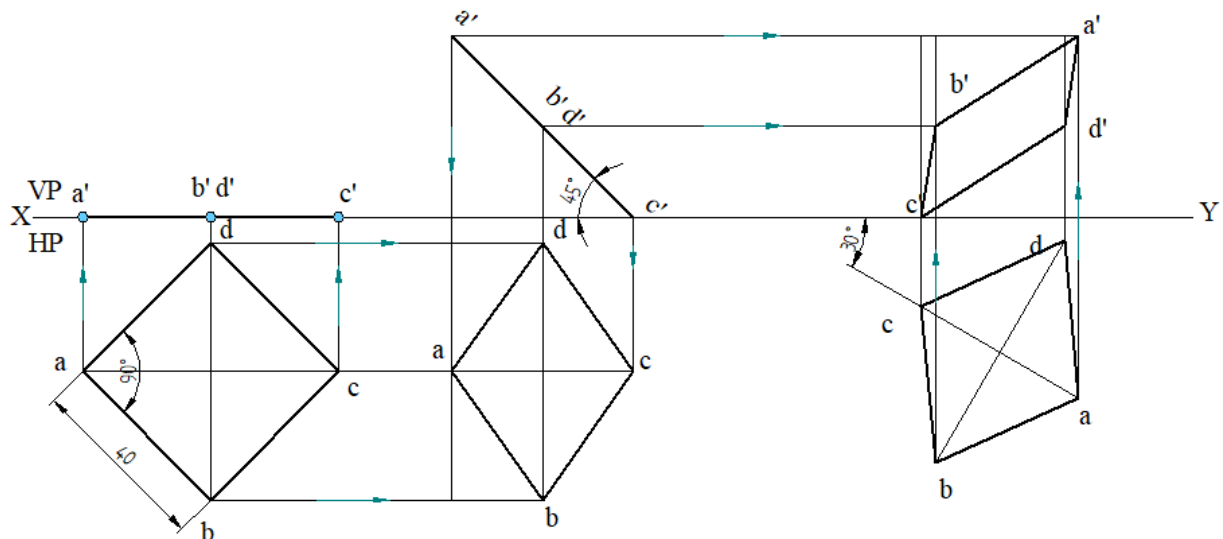
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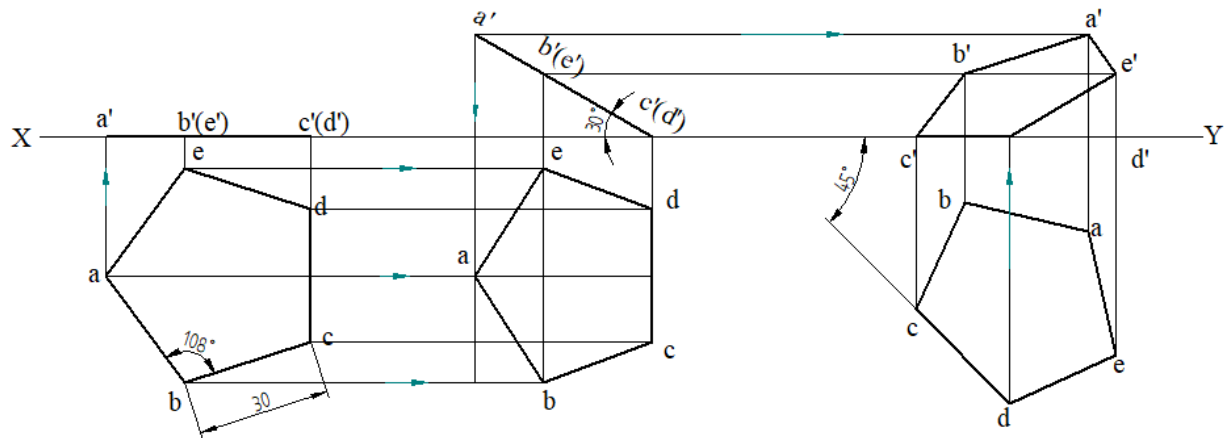
1. An equilateral triangular lamina of 30mm sides rests on HP on one of its corners such that the surface is inclined at 50° to HP. Draw the projections of lamina when the edge opposite to the corner on which lamina rests is inclined at 30° to VP.



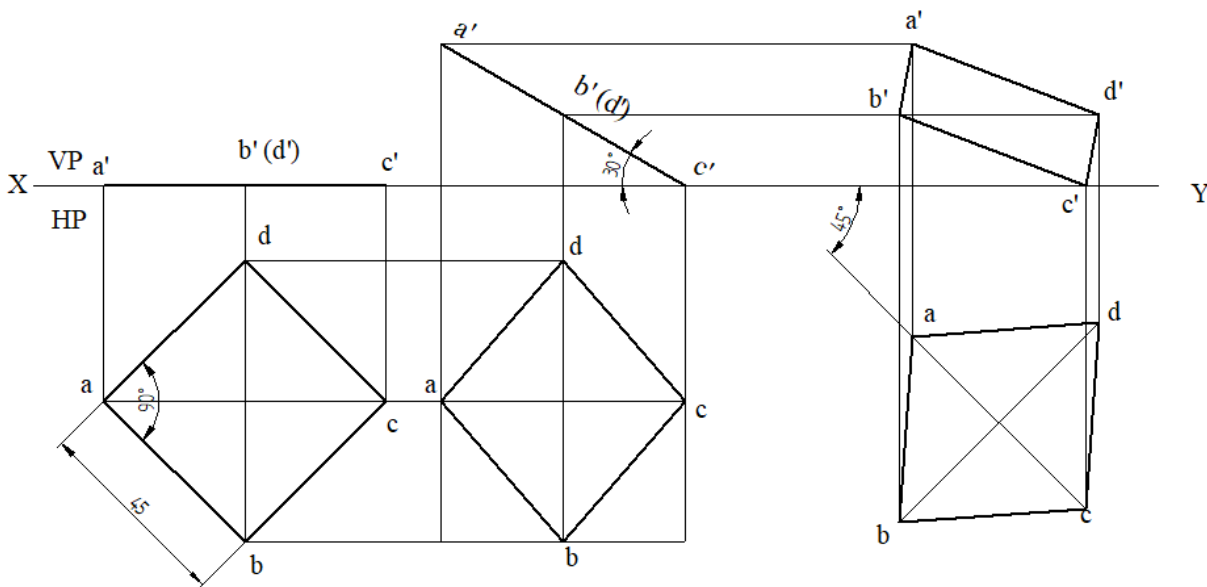
2. A square lamina ABCD of 40mm side rest on the corner C such that the diagonal AC appears to be at 30° to the VP in the top view. The two sides BC and CD containing the corner C make equal inclinations with the HP. The surface of the lamina make 45° with HP. Draw the top and front views.



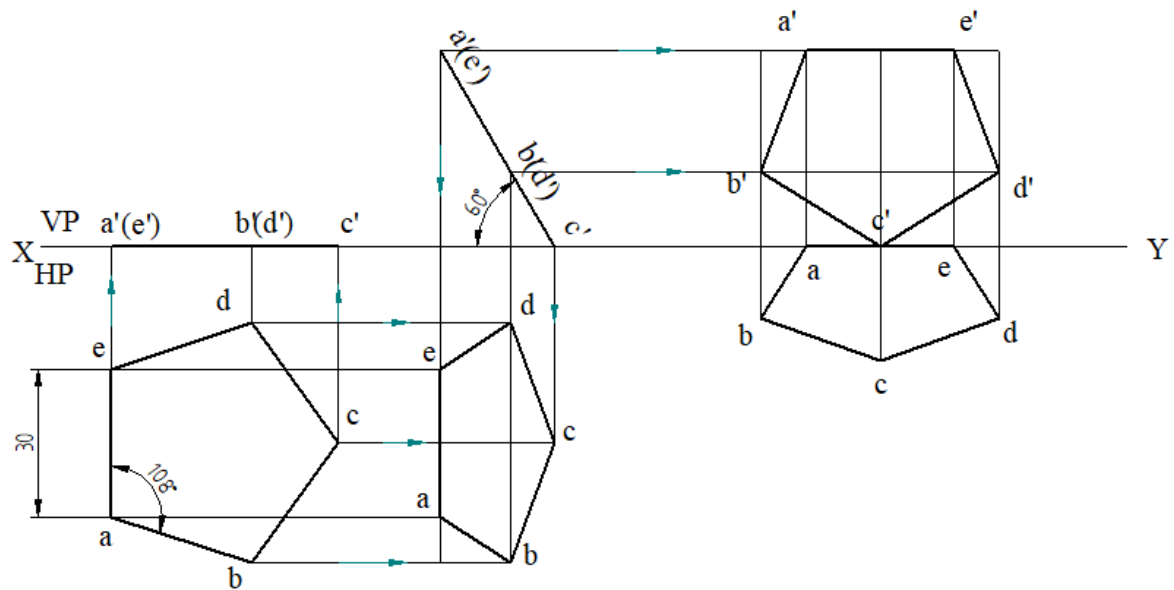
3. A regular pentagon of 30mm side, is resting on one of its edge on HP which is inclined at 45° to VP. Its surface is inclined at 30° to HP. Draw its projections.



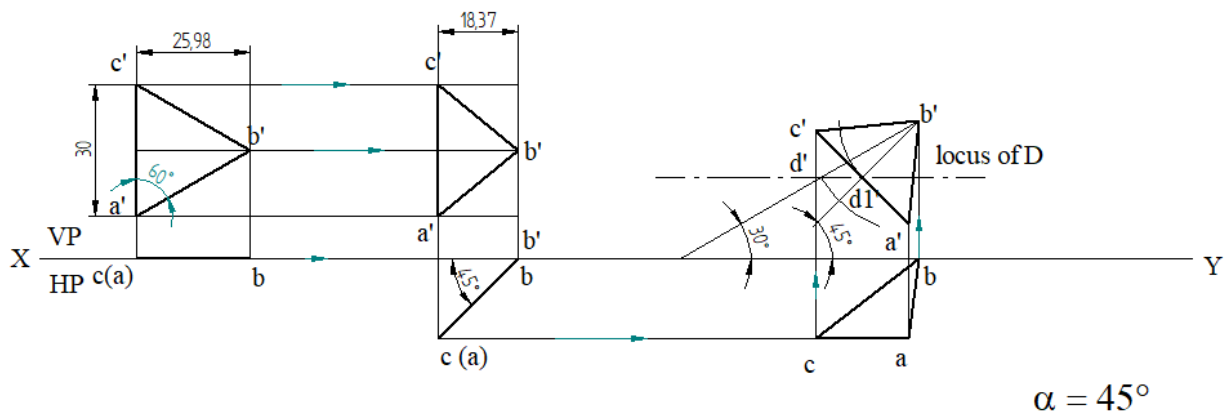
4. A square lamina of 45mm sides rest on corner on HP with one of diagonals inclined at 30° to HP and appears to be inclined at 45° to VP. Draw its top and front views.



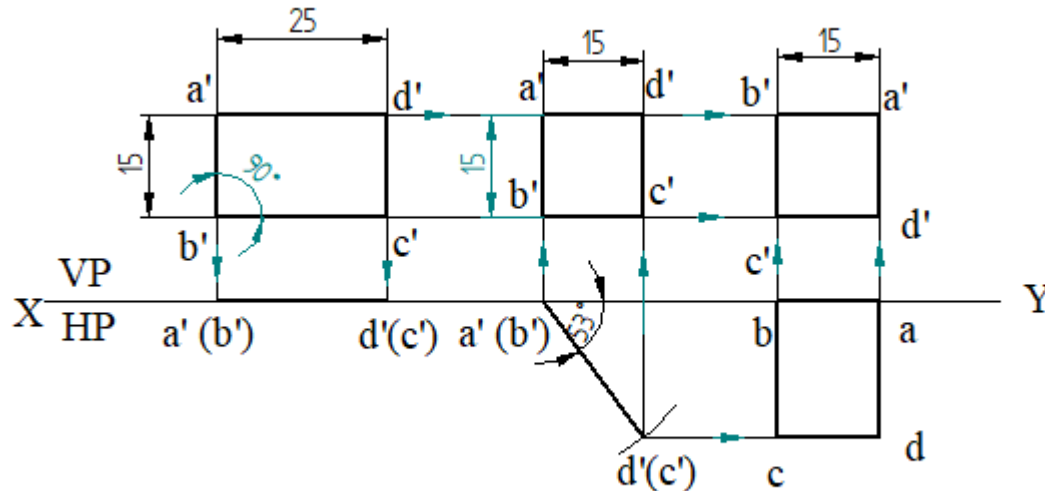
5. A regular pentagonal lamina of 30mm sides is resting on one of its corners on HP while The side opposite to this corner touches VP. If the lamina makes an angle of 60° with HP and 30° with VP, draw the projections.



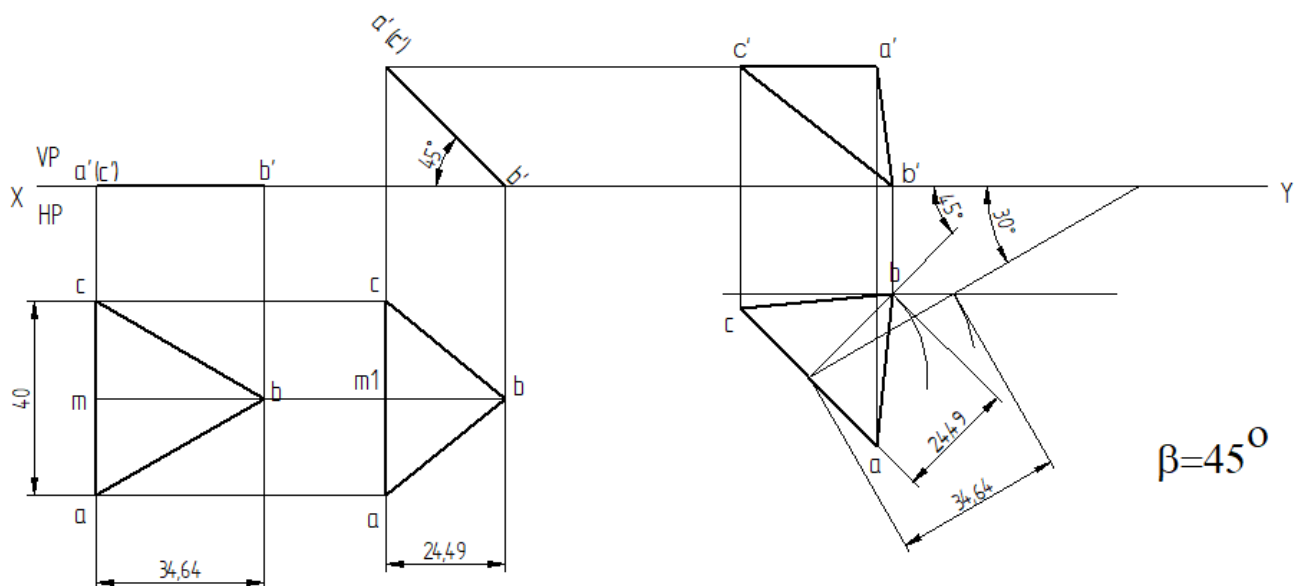
6. A triangular lamina of 30mm sides rest on one of its corner on VP such that the median passing through the corner on which it rests is inclined at 30° to HP and 45° to VP. Draw its projections



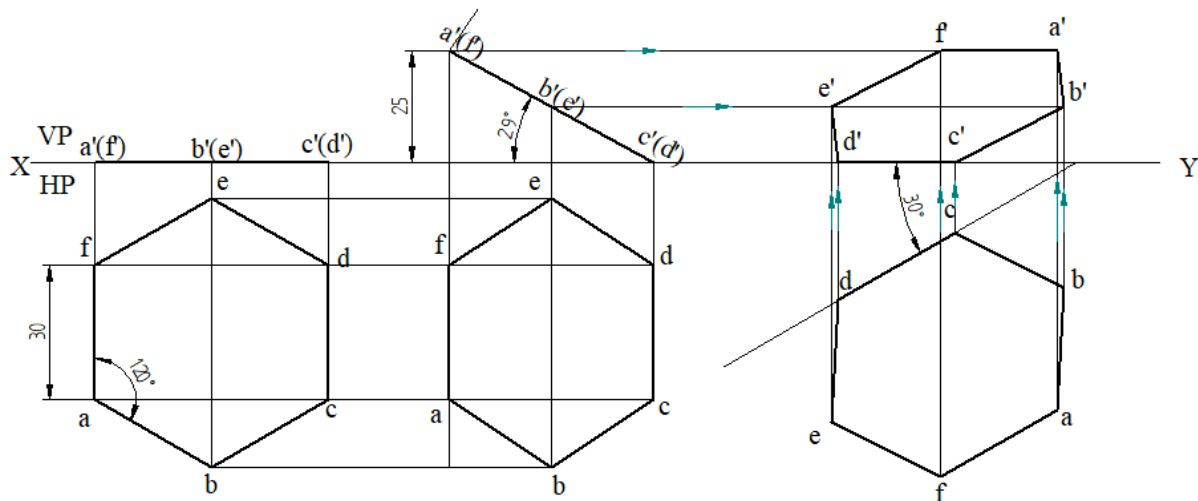
7. The front view of a rectangular lamina of sides 15mm x 25mm is a square of 15mm Sides. The longer sides of the lamina appear perpendicular to HP. Draw its projections Find the inclinations of lamina with HP and VP.



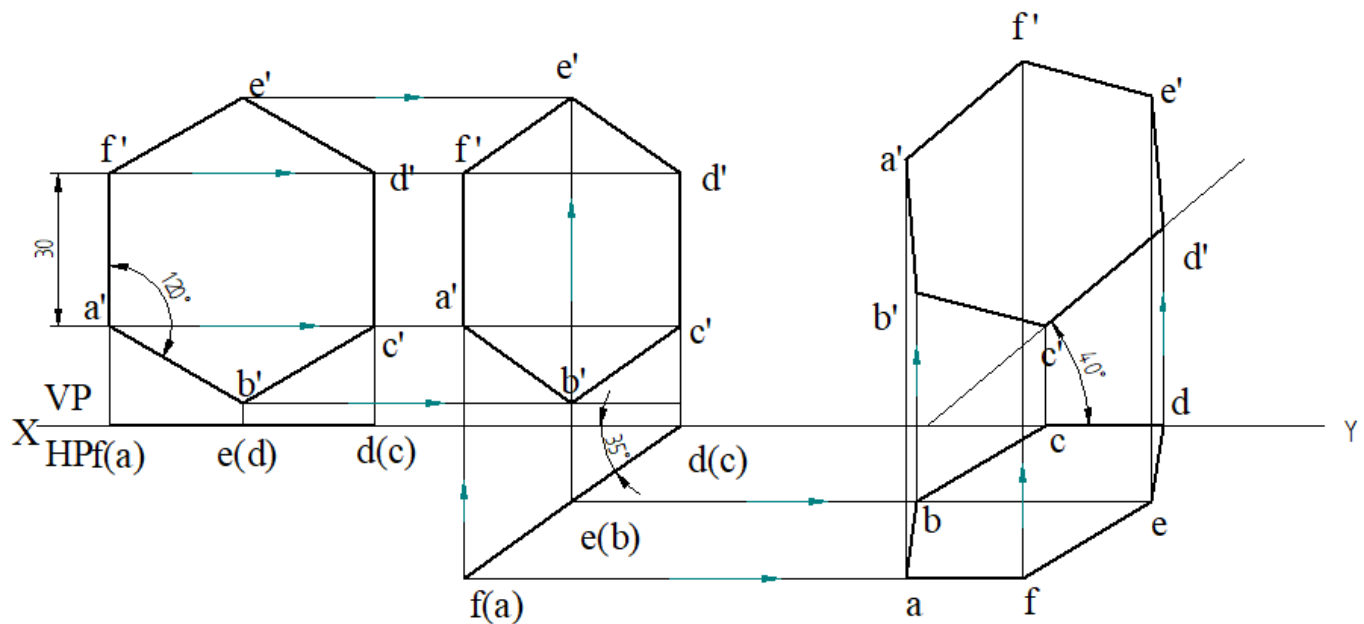
8. An equilateral triangular lamina of sides 40mm is resting on a corner on HP such that the surface makes an angle of 45° to HP and the median passing through the resting corner is inclined at 30° to VP. Draw its top and front views.



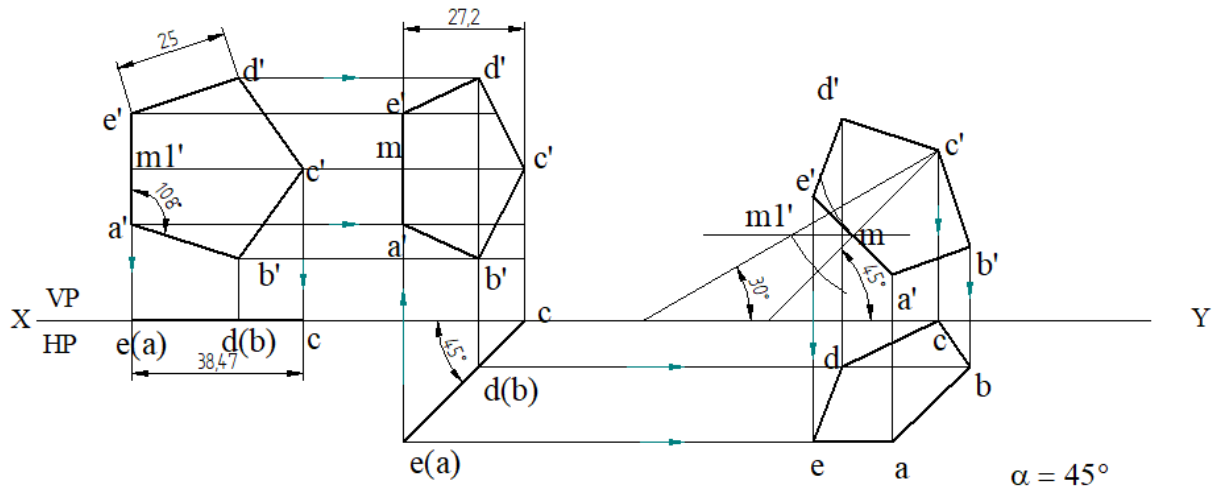
9. A hexagonal lamina of 30mm sides rest on HP on one side and is inclined at 30° to VP. If the top edge of the plate is 25mm from HP, draw the projections of the plate.



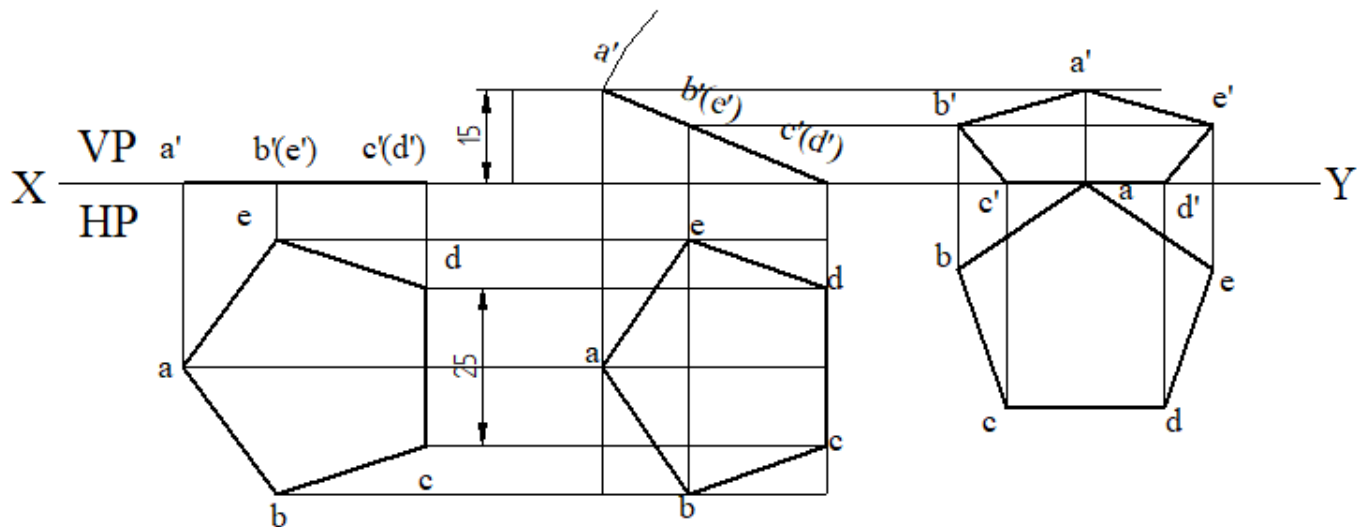
10. A hexagonal lamina of side 30mm is resting on VP on one of its side and inclined at 40° to HP. Its surface is inclined at 35° to VP. Draw the projections.



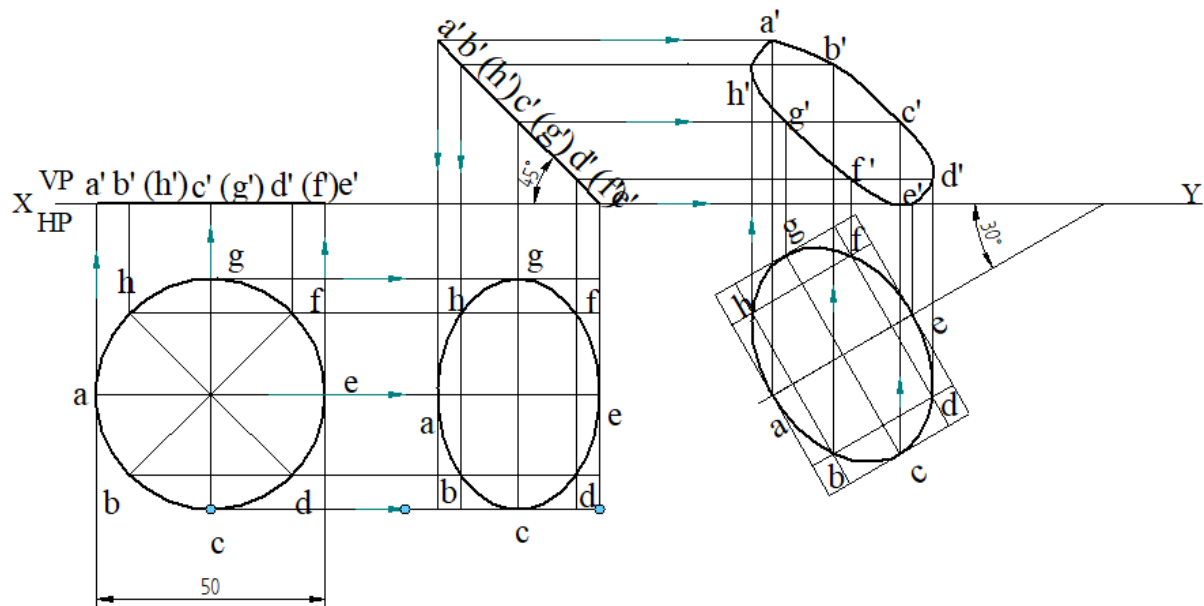
11. A pentagonal lamina of 25mm sides is placed on one of its corner on VP such that the perpendicular bisector of one of the edge is inclined at 30° to HP and 45° to VP. Draw the projections of lamina.



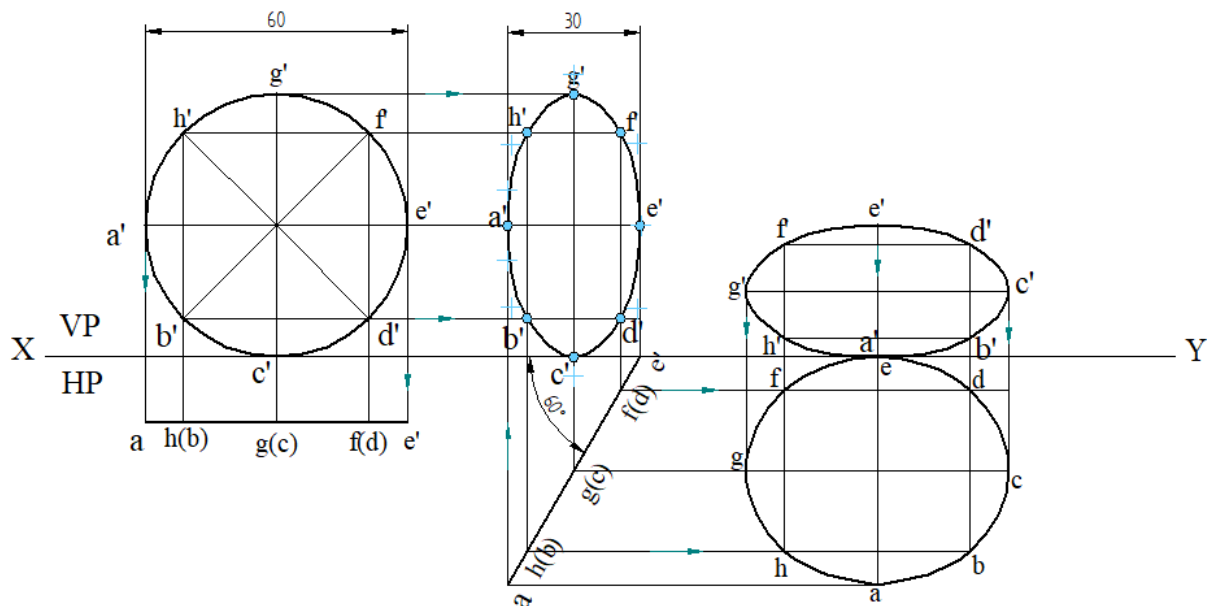
12. A pentagonal lamina of 25mm sides rest on HP on one of its edge with the corner Opposite to that edge touching VP at a height of 15mm. the edge on which the lamina rest on HP is parallel to VP. Draw the projection and determine the inclinations of surface with HP and VP.



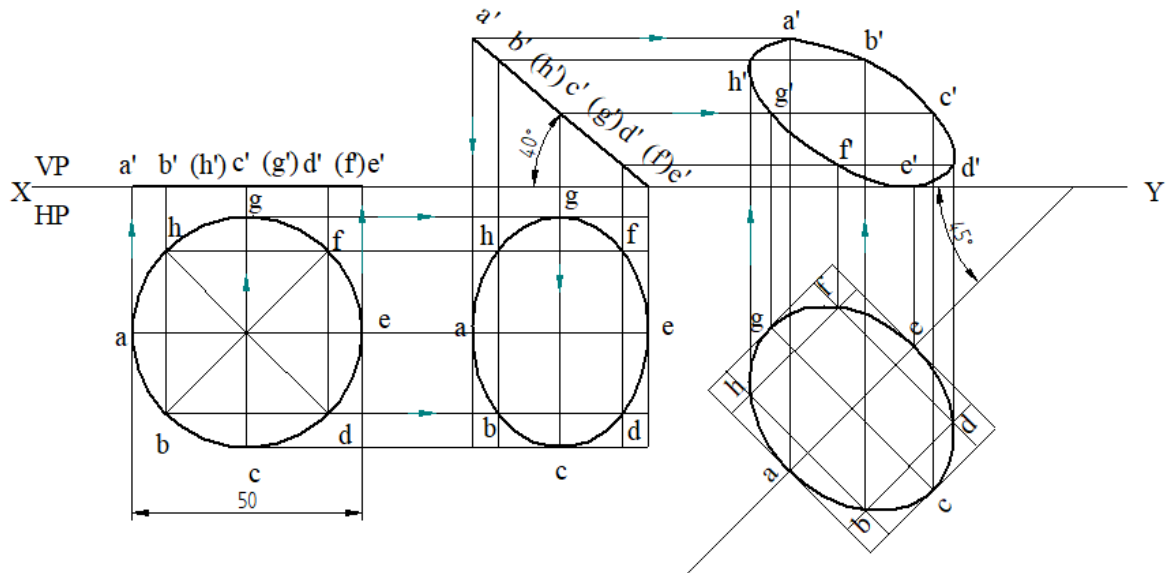
13. Draw the projections of a circular lamina of 50mm diameter is standing with one of its points on the rim on HP and its surface is inclined at 45° to the HP and the top diameter AE makes an angle of 30° with the VP.



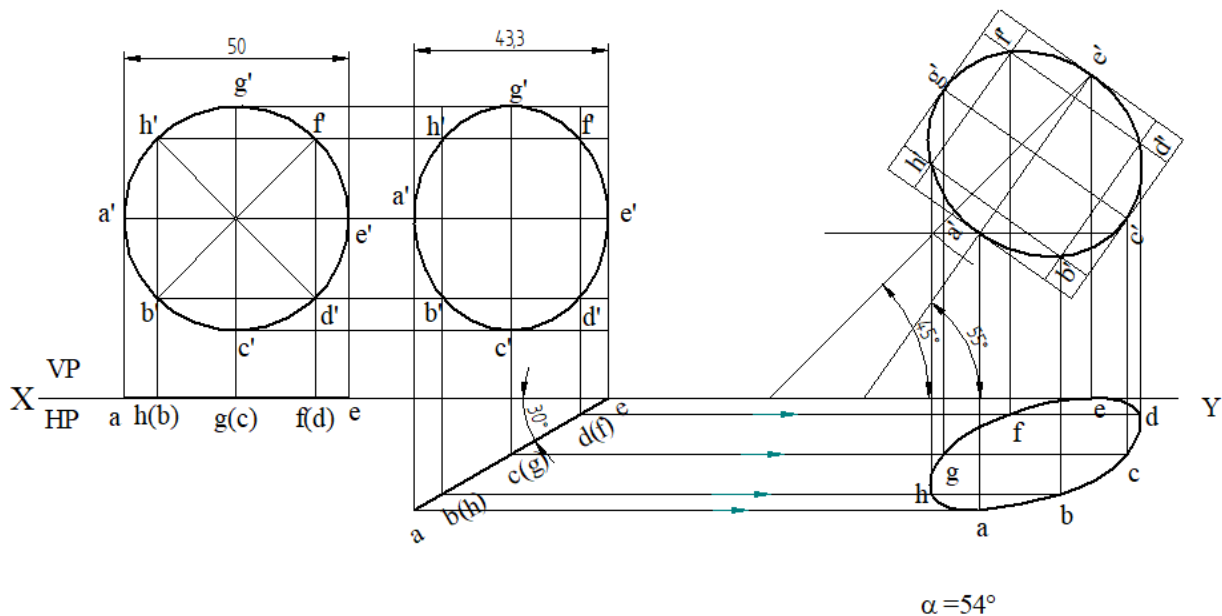
14. A circular plate inclined to VP appears in the front view as an ellipse of major axis 60mm and minor axis 30mm. The major axis parallel to both HP and VP. Determine the inclinations of the plate with HP and VP and complete the projections of the plate assuming thickness of the plate as negligible



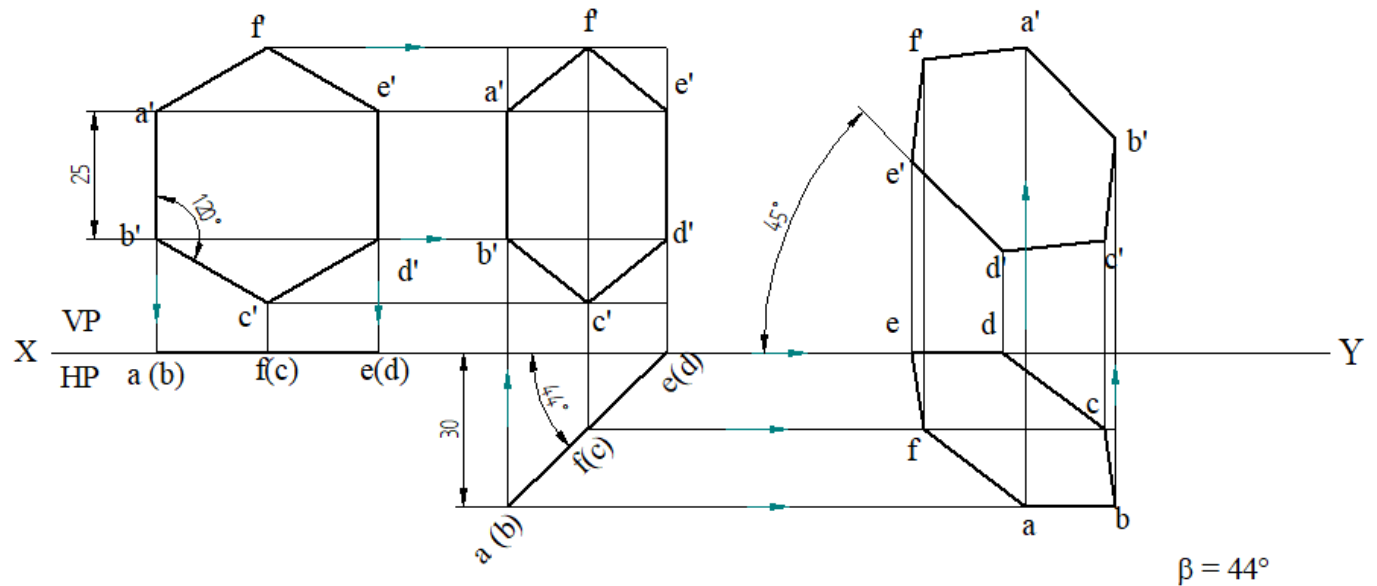
15. Draw the projection of a circular lamina of 50mm diameter standing with one of its points of the rim on HP. The plane is inclined 40° to HP. The top view of diameter, through point A is making an angle of 45° with VP. Determine the inclination of the plane with HP



16. A circular lamina of 50mm diameter rests on VP such that one of its diameters is inclined at 30° to VP and 45° to HP. Draw its top and front views in this position.



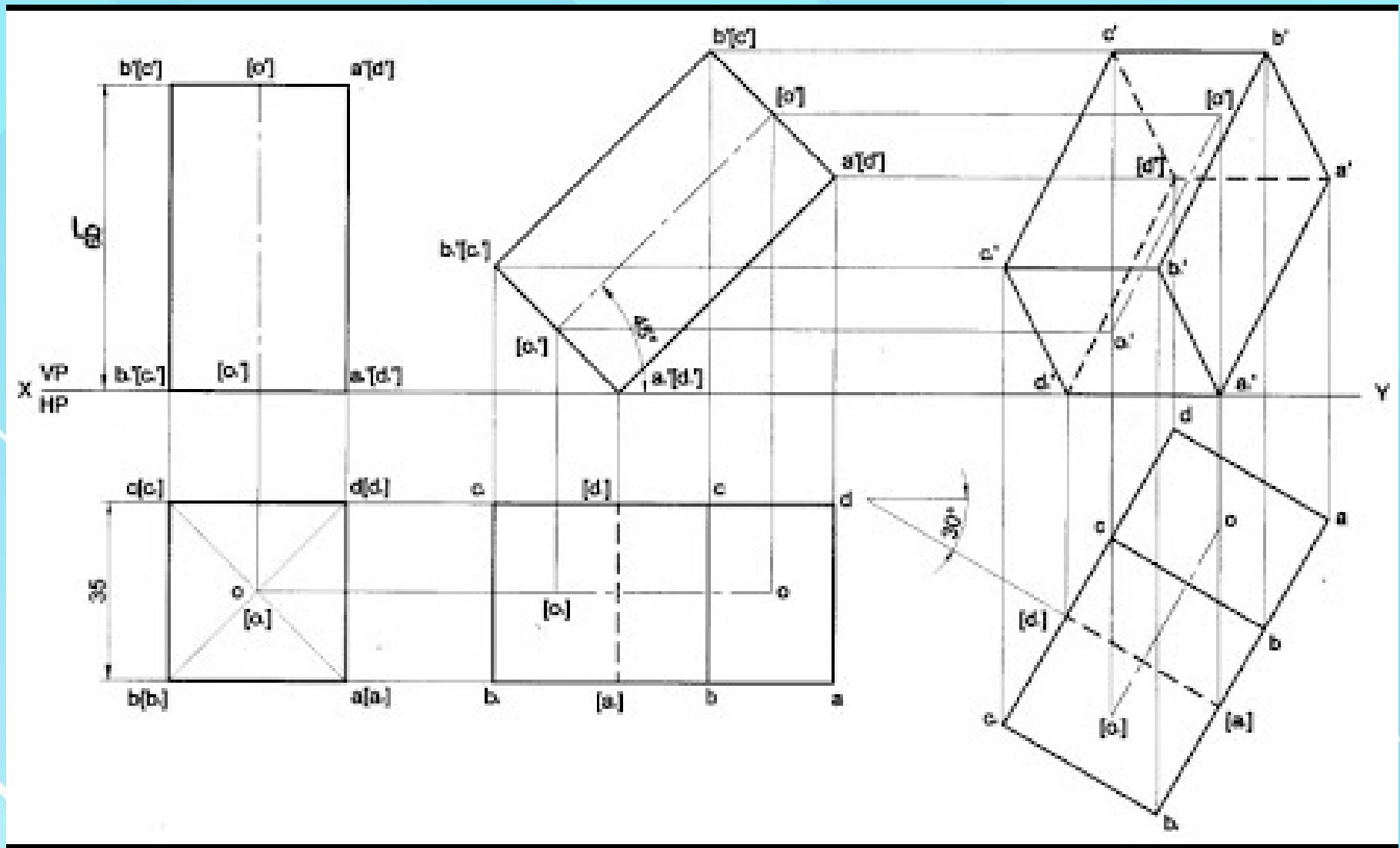
17. A hexagonal lamina of sides 25mm rests on one of its sides on VP. The side opposite to the side on which it rests is 30mm in front of VP & the side on which it rests makes 45° to HP. Draw its projections. Also determine the inclination of the lamina with the reference plane.



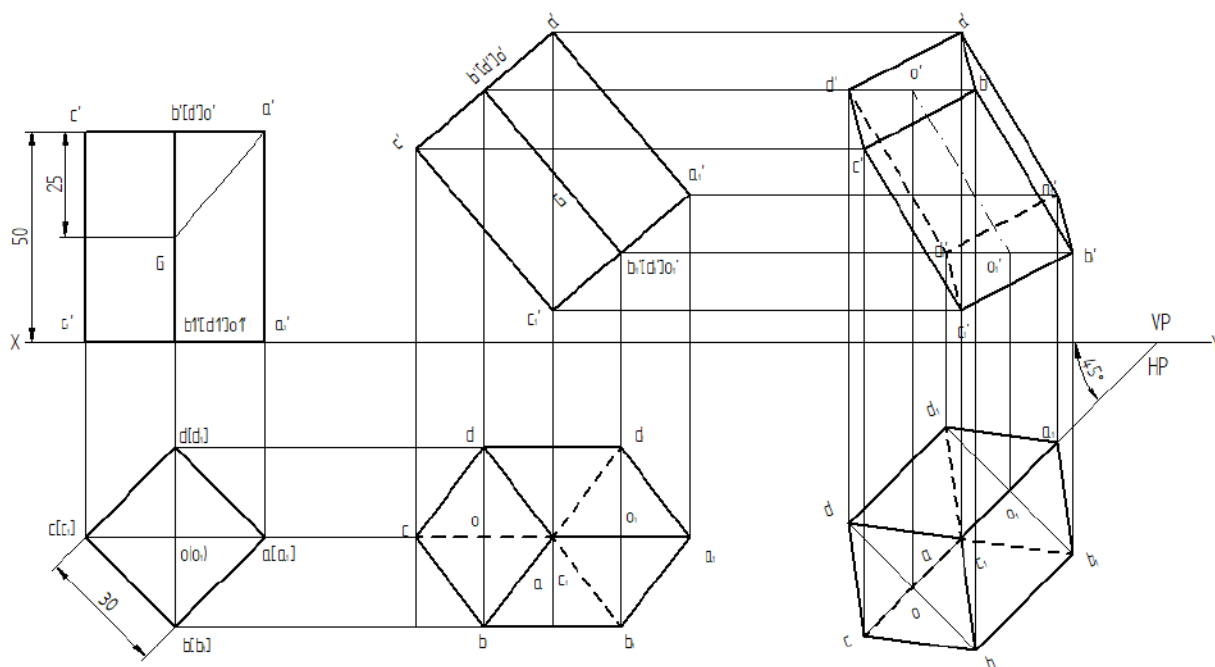
DEPARTMENT OF MECHANICAL ENGINEERING

PROJECTION OF SOLIDS

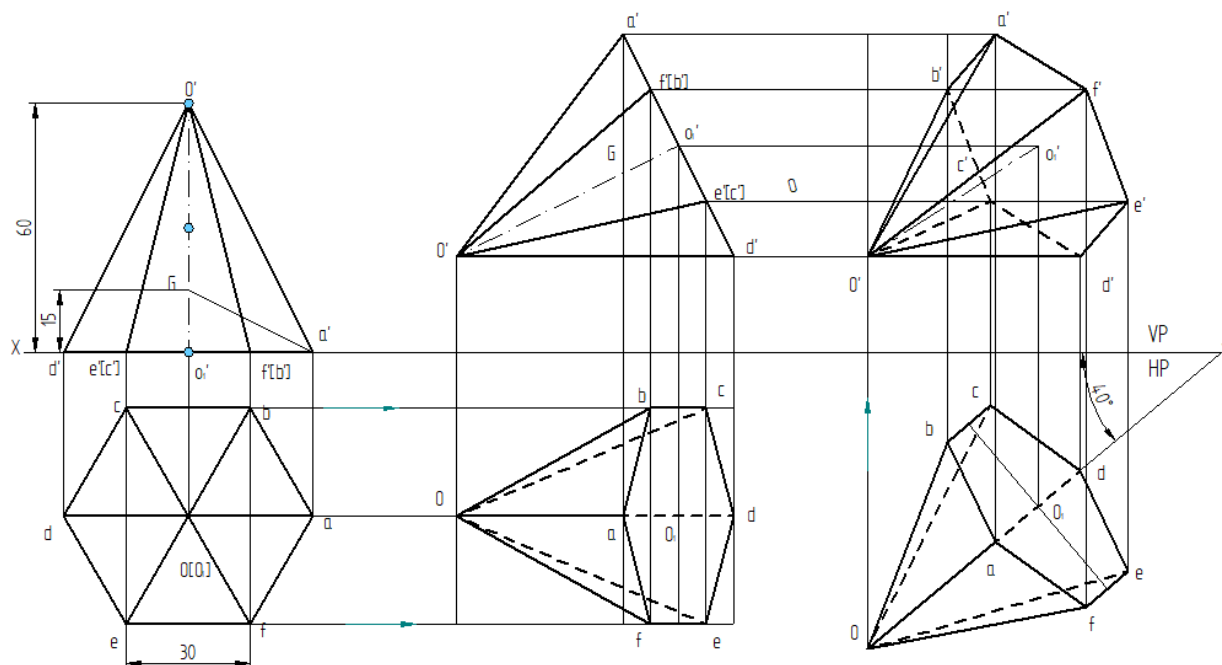
25EGD114/124



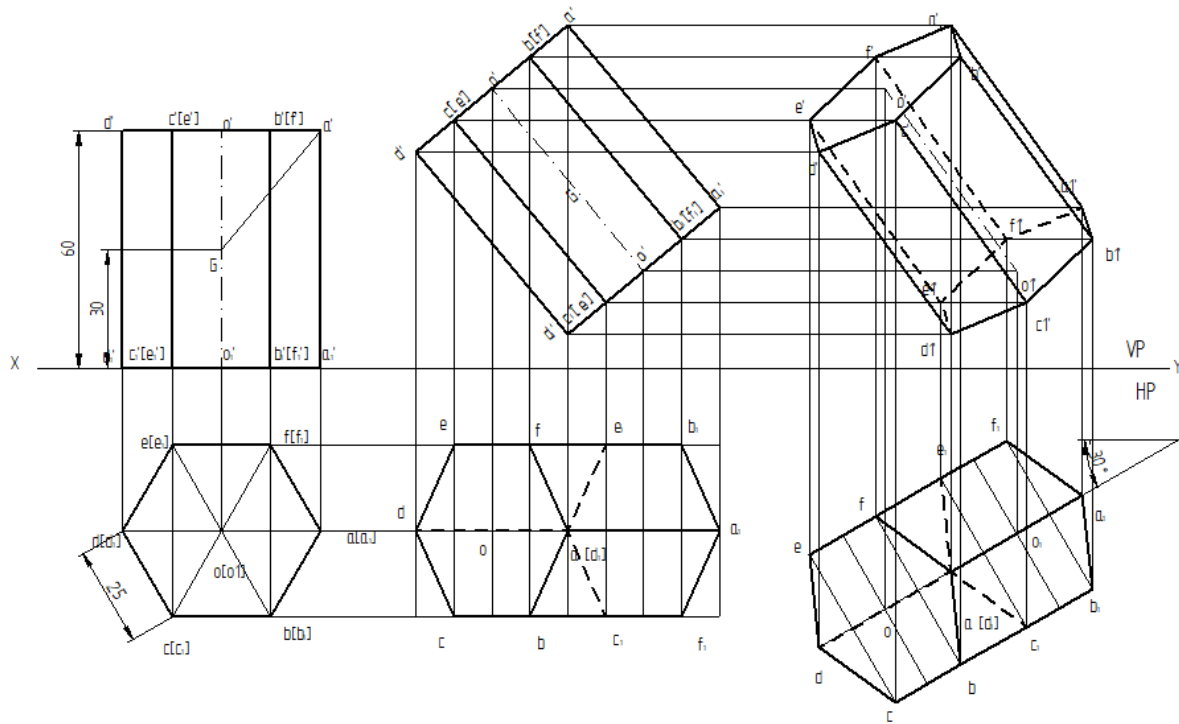
2. A square prism of base side 30mm and axis 50mm, suspended freely from one Of its corner of the top face such that the axis appears to be inclined at 45° to VP. Draw the projections of the prism.



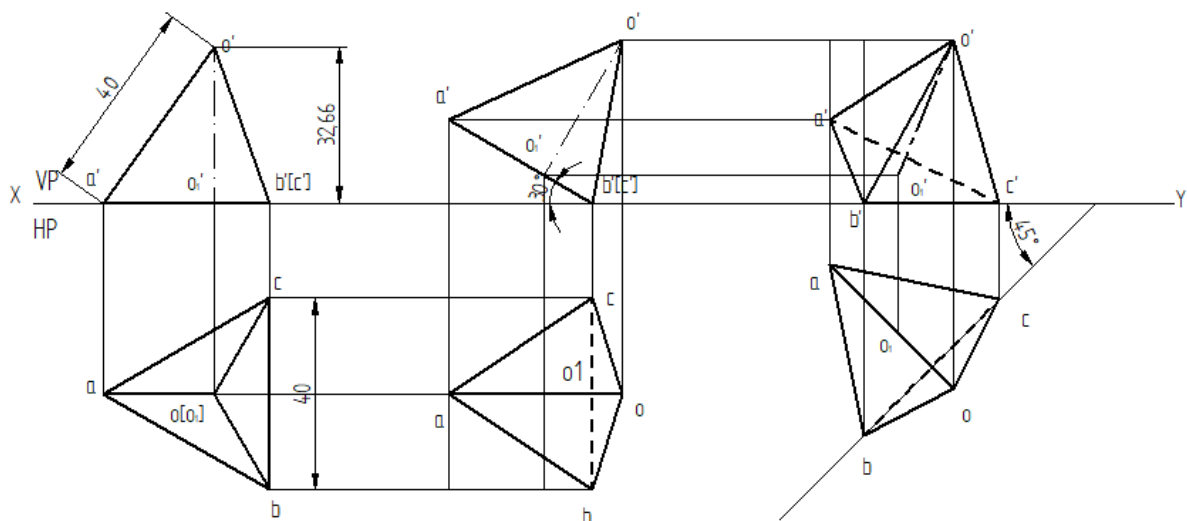
4. A hexagonal pyramid of base side 30mm and axis 60mm is freely suspended from a base corner such that the axis appears to be inclined at 40° to VP. Draw its projections.



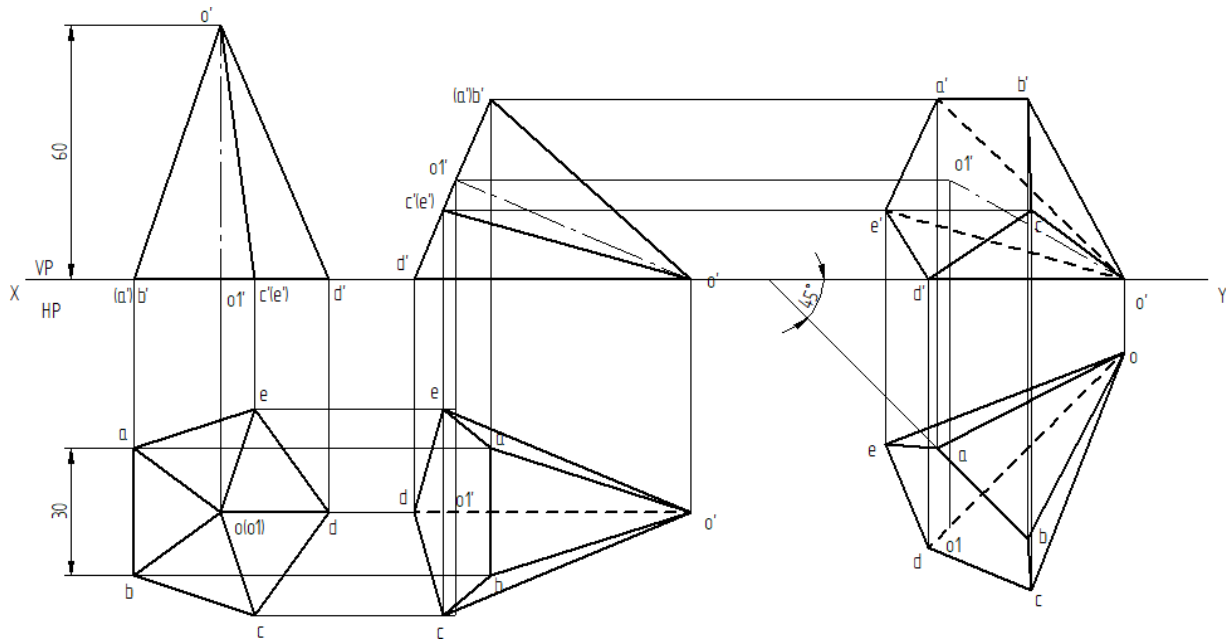
5. A hexagonal prism of base side 25mm and axis length 60mm is suspended freely from a corner such that axis appears to be inclined at 30° to VP. Draw its projection.



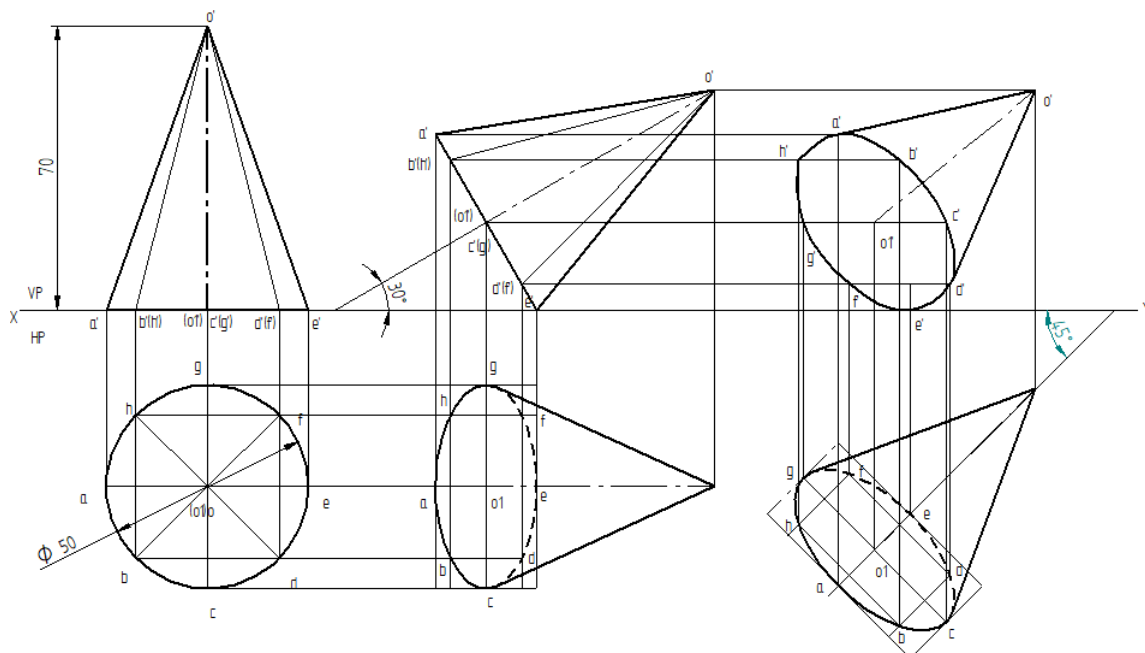
6. A tetrahedron of 40mm side rests on HP on one of its sides such that the base is inclined to HP at 30° and the side on which it rests on HP is inclined at 45° to VP. Draw its projections.



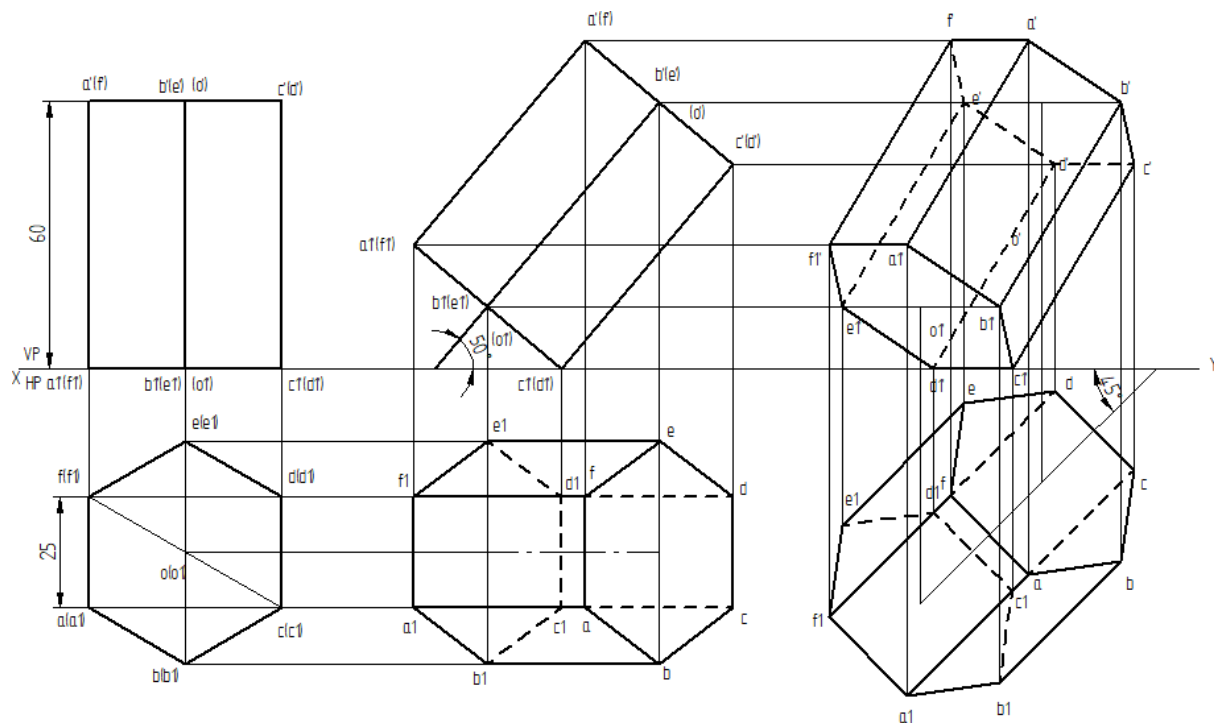
7. A right regular pentagonal pyramid, side of the base 30mm and height 60mm, has one of its slant edges in HP. One of the base edges parallel to HP & inclined at 45° to VP. Draw its projections.



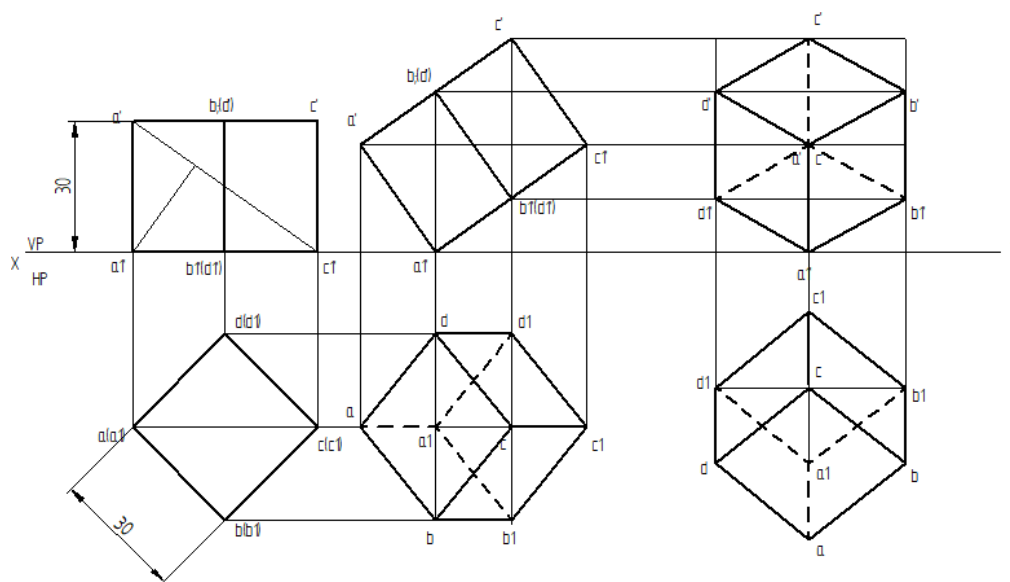
8. A right circular cone, base 50mm diameters, axis 70mm long, when it is resting on HP on a point of its base circle, with axis making an angle 30° with HP and plan of axis 45° with VP.



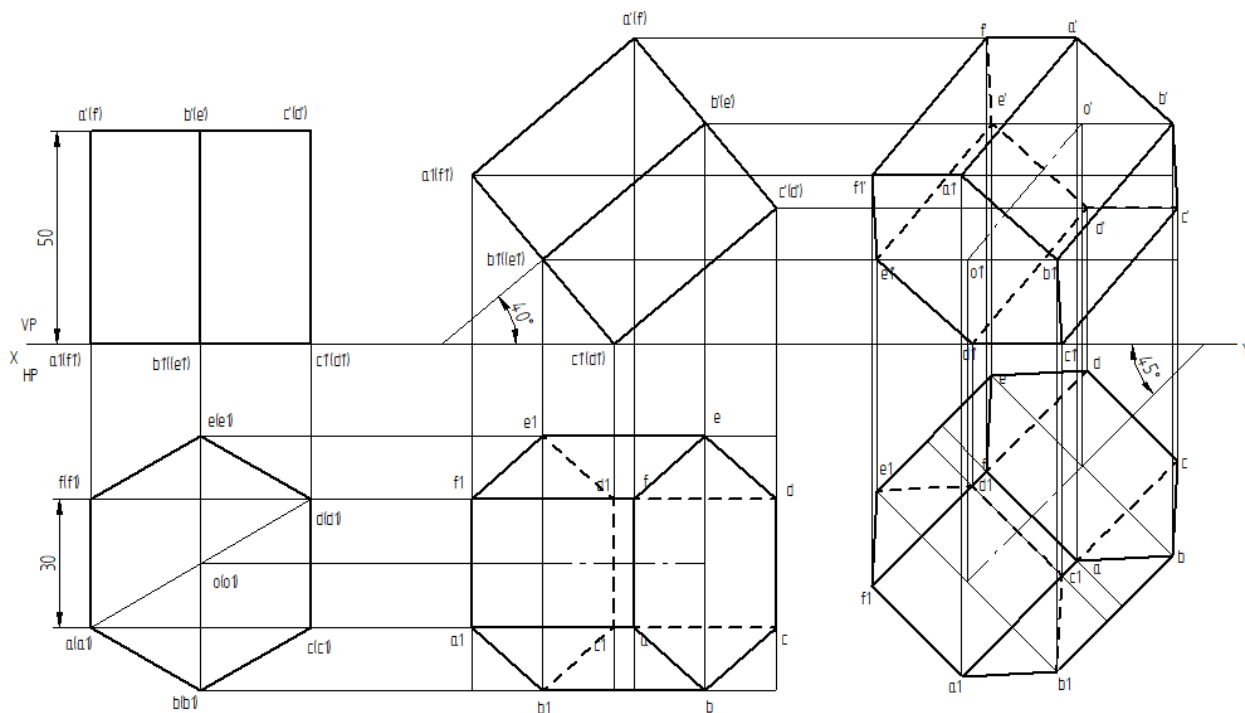
9. A hexagonal prism with side of base 25mm lies on HP on one of its base edges. The length of axis is 60mm and is inclined at 50° to HP and the top view of the axis is inclined at 45° to XY line. Draw the projections.



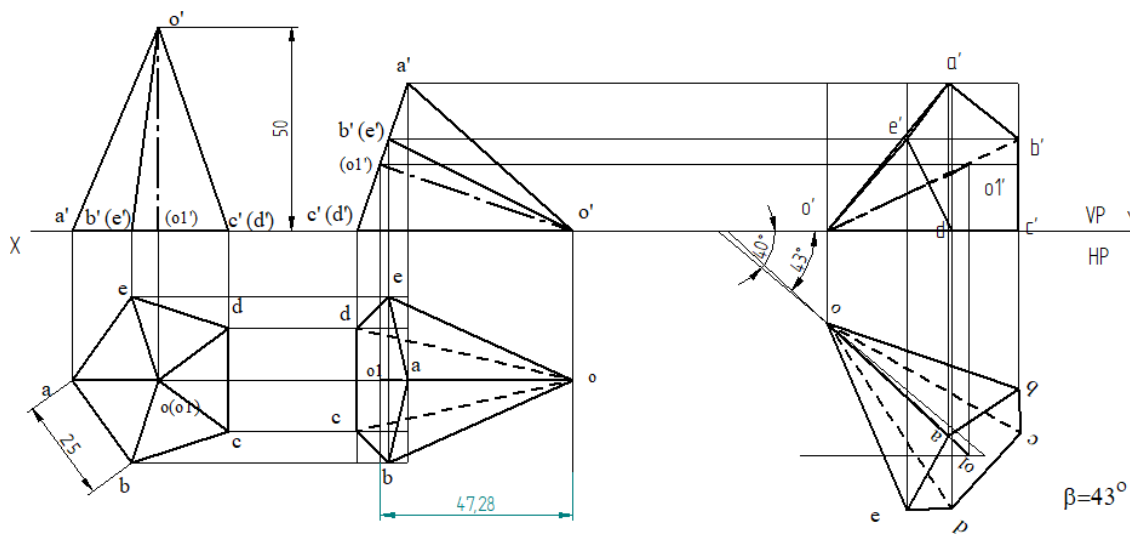
10. A cube with the edge length 30 mm rests on HP on one of its corners with solid diagonal perpendicular to the VP. Draw the projections



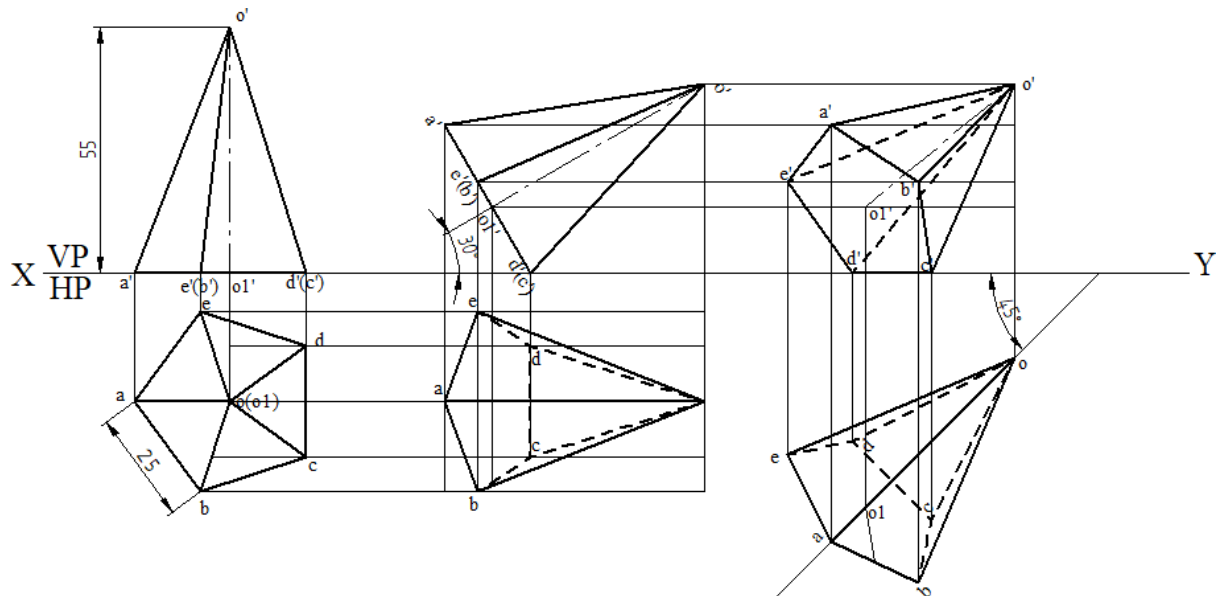
11. Draw the projections of a Hexagonal prism 30mm side of base and axis 50mm long resting on one of its base edges on HP such that its axis is inclined at 40° to HP and appears to be inclined at 45° to VP.



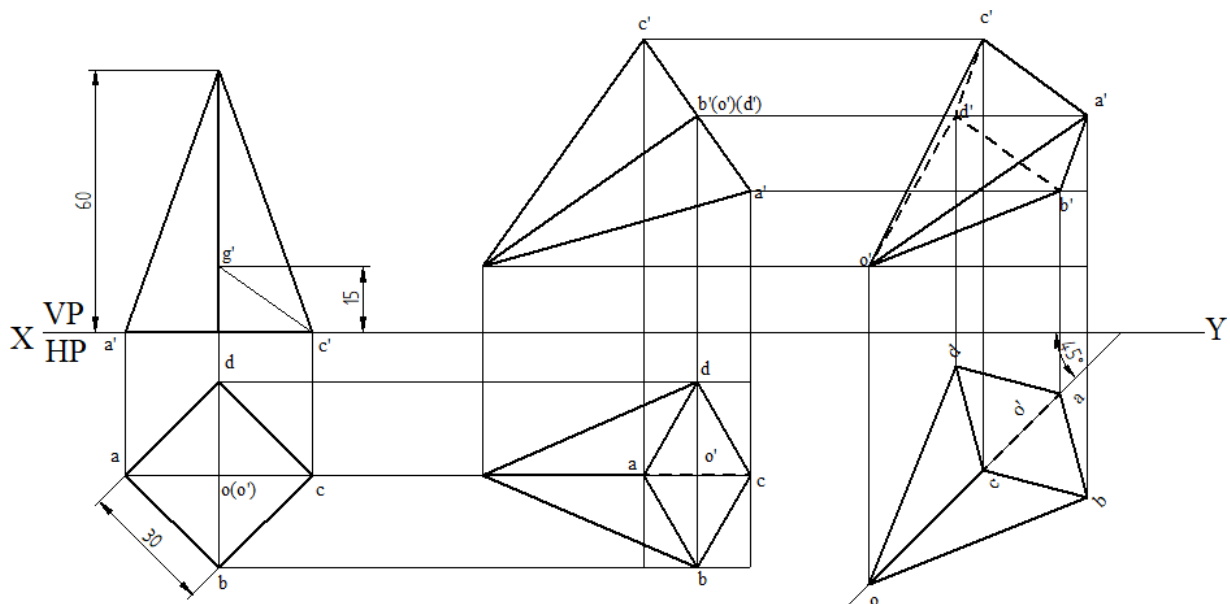
12. A Pentagonal pyramid of 25mm base side and 50mm axis length placed one of its slant triangular surfaces on HP, with the axis inclined to VP at 40° . Draw the projections of the pyramid.



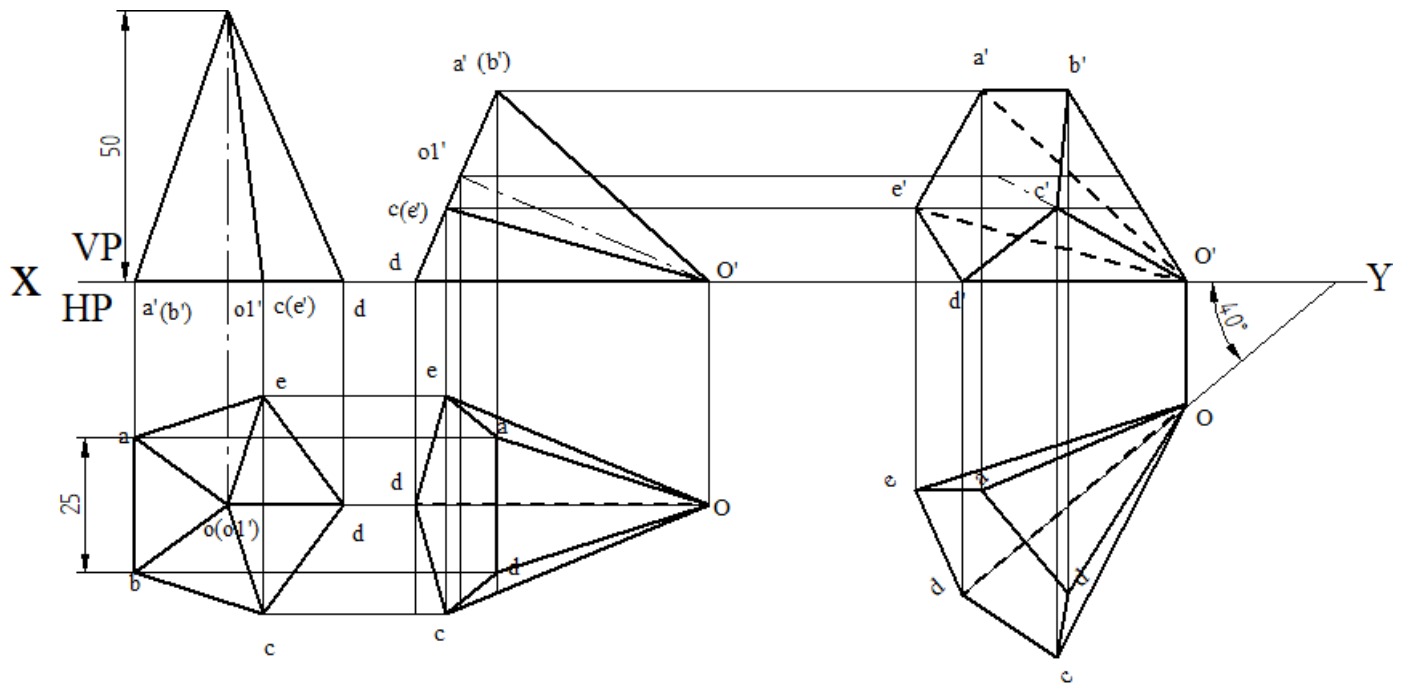
13. A right regular pentagonal pyramid, edges of base 25mm and height 55mm is held on ground plane On one of its base edges, such that its axis is inclined at 30° to HP and appears to be inclined at 45° To VP. Draw its projections.



14. A square pyramid with base side 30mm and axis 60mm is freely suspended from one of the corner Of its base. Draw the projection when the top view of the axis makes an angle of 45° with line XY.



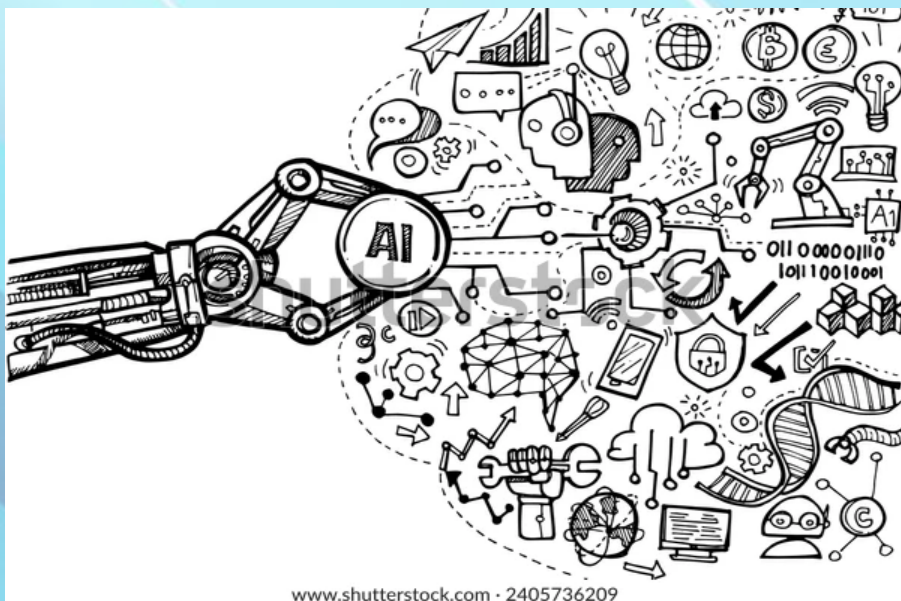
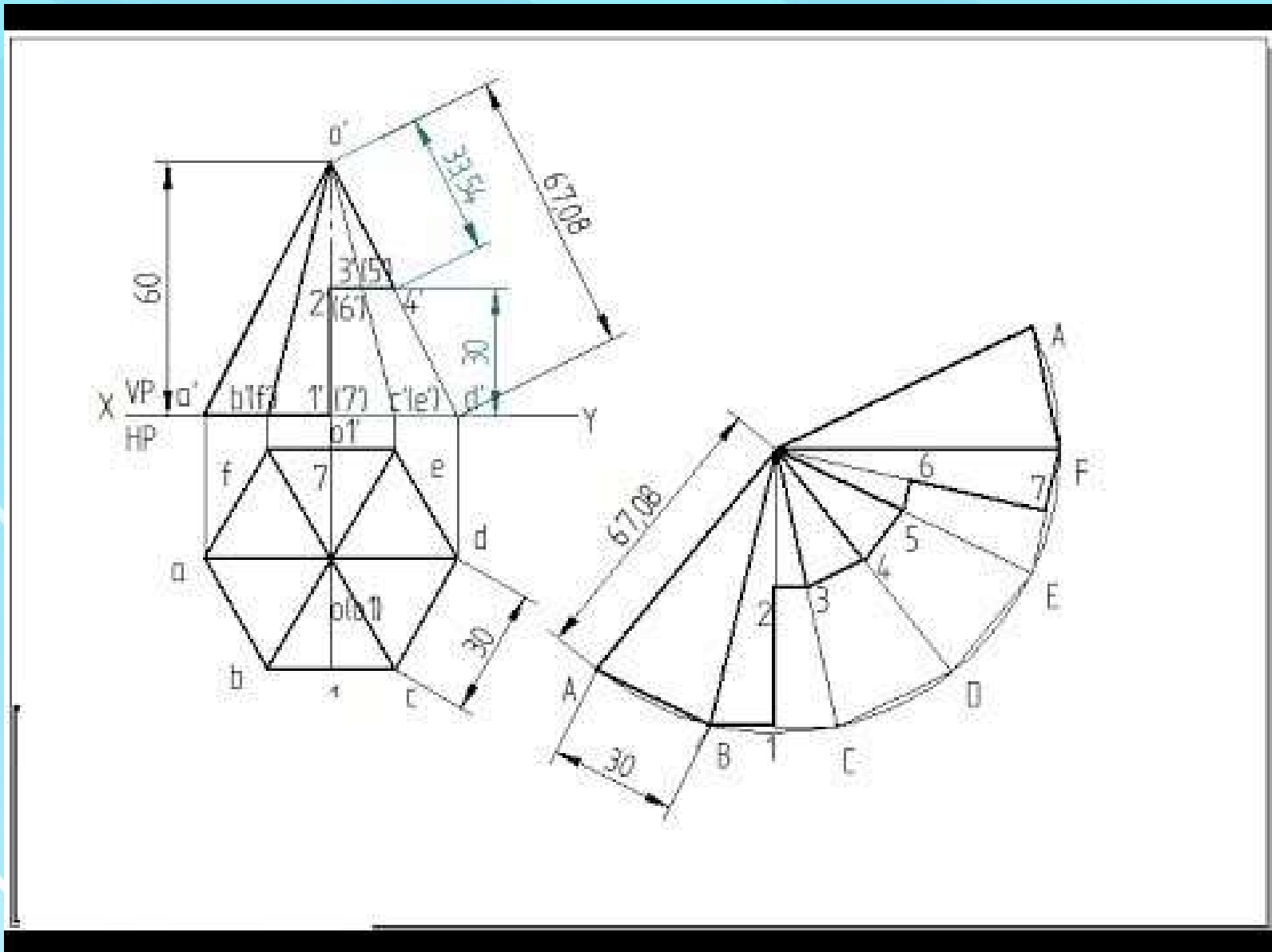
15. A pentagonal pyramid of 25mm base side and 50mm axis length is placed with one of its slant edge on HP such that the two triangular faces containing that edges are equally inclined to HP and the axis appears to be inclined to VP at 40° . Draw the projection of the pyramid.



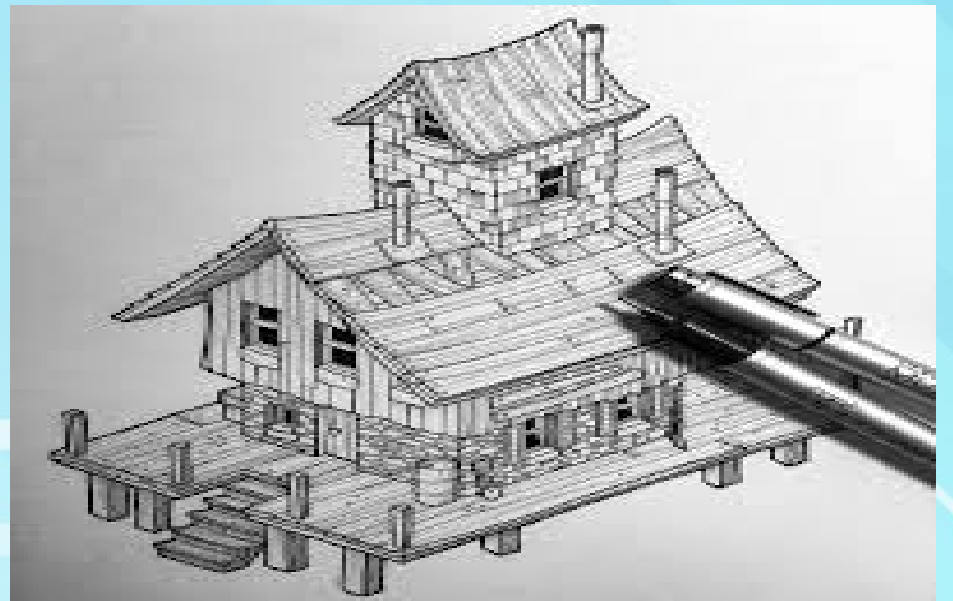
DEPARTMENT OF MECHANICAL ENGINEERING

DEVELOPMENT OF LATERAL SURFACES

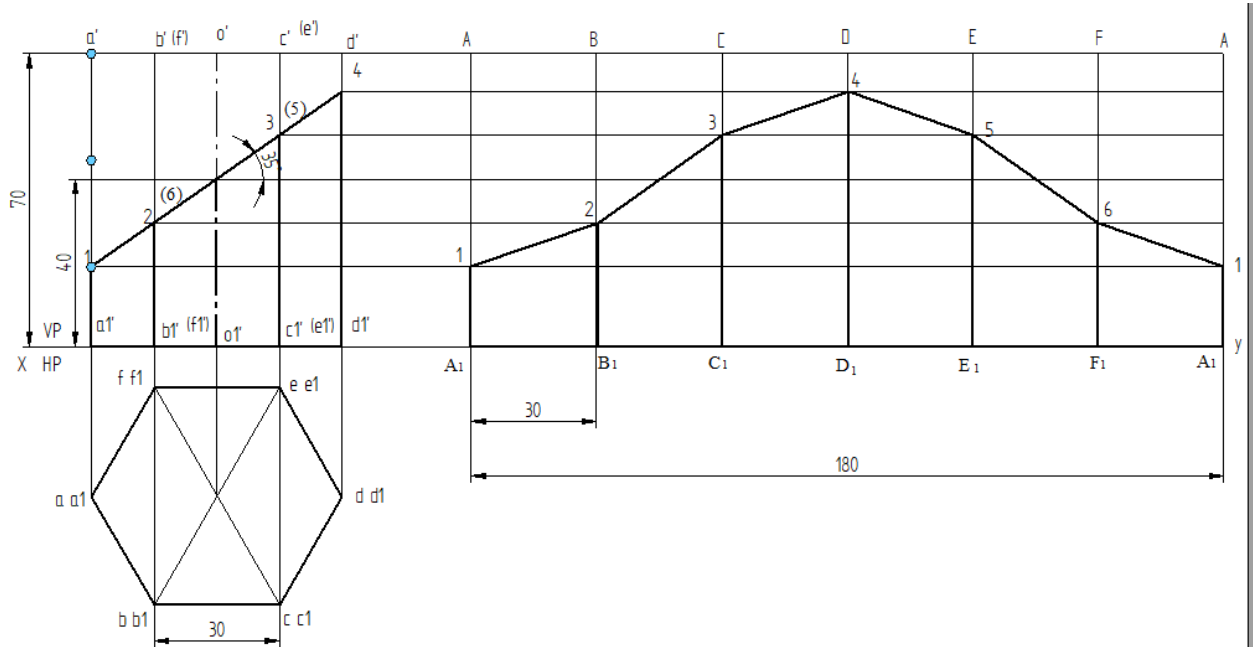
25EGD114/124



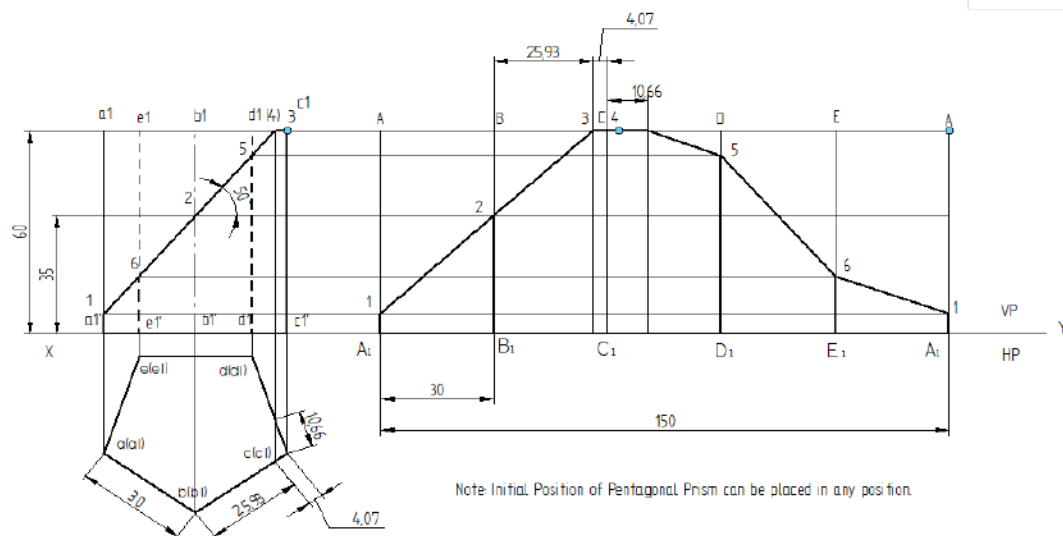
www.shutterstock.com · 2405736209



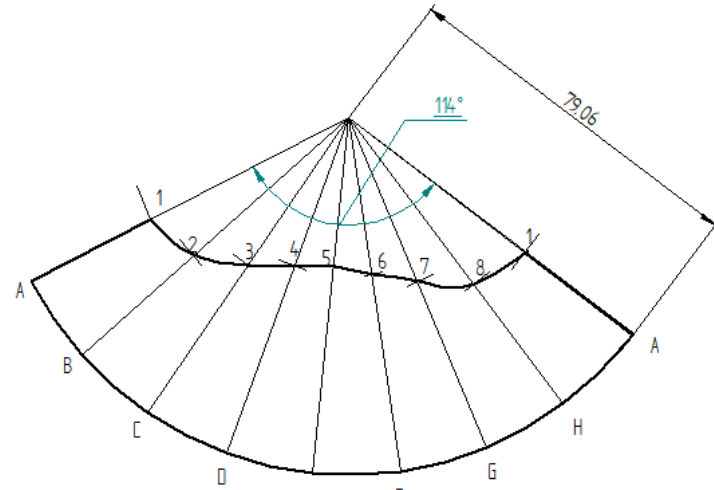
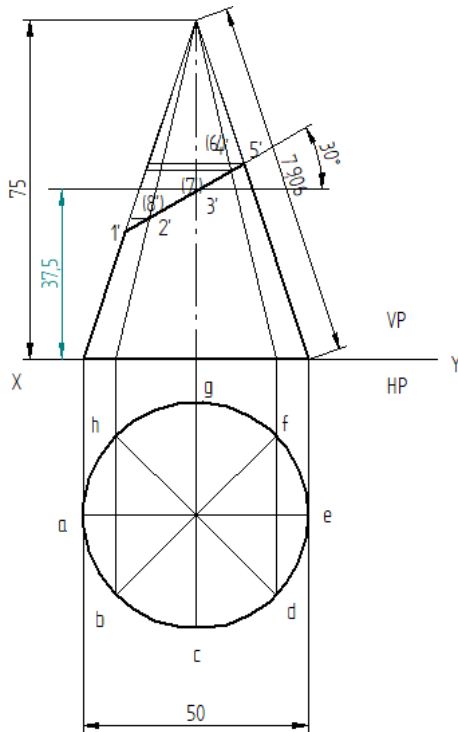
1. A hexagonal prism of base side 30mm and axis height 70mm is resting on its base on HP with one of its base edges parallel to VP and it is cut by plane perpendicular to VP and inclined at 35° to HP, meeting the axis at a distance of 40mm from the base. Draw the development of lateral surfaces of the lower portion of the prism by parallel line development method.



2. A pentagonal prism of base side 30mm and height 60mm is cut by a section plane perpendicular to VP and 50° to HP and passing through the axis at a height of 35mm above the base the prism is resting on VP in such a way that, one of its rectangular faces is parallel to VP and nearer to it. Draw the development of the lower portion of the Solid.



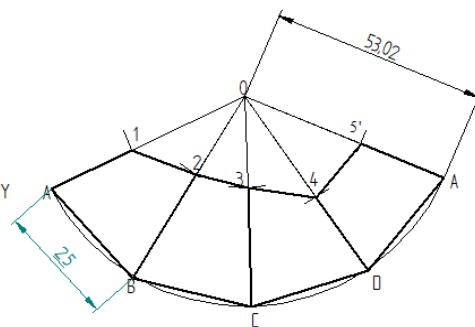
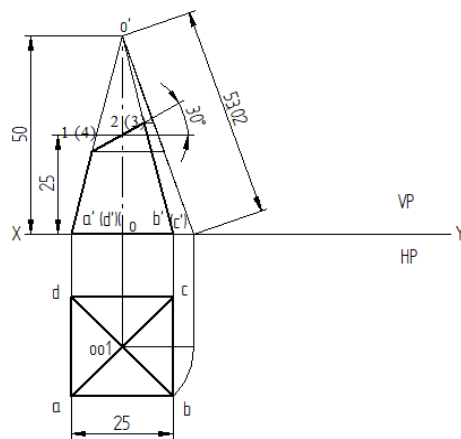
3. A right circular cone of base diameter 50mm and height 75mm is resting on its base on the ground. It is cut by a section plane perpendicular to VP and inclined at 30° to HP. The cutting plane bisects the axis of the cone. Draw the development of the lateral surfaces of the truncated cone.



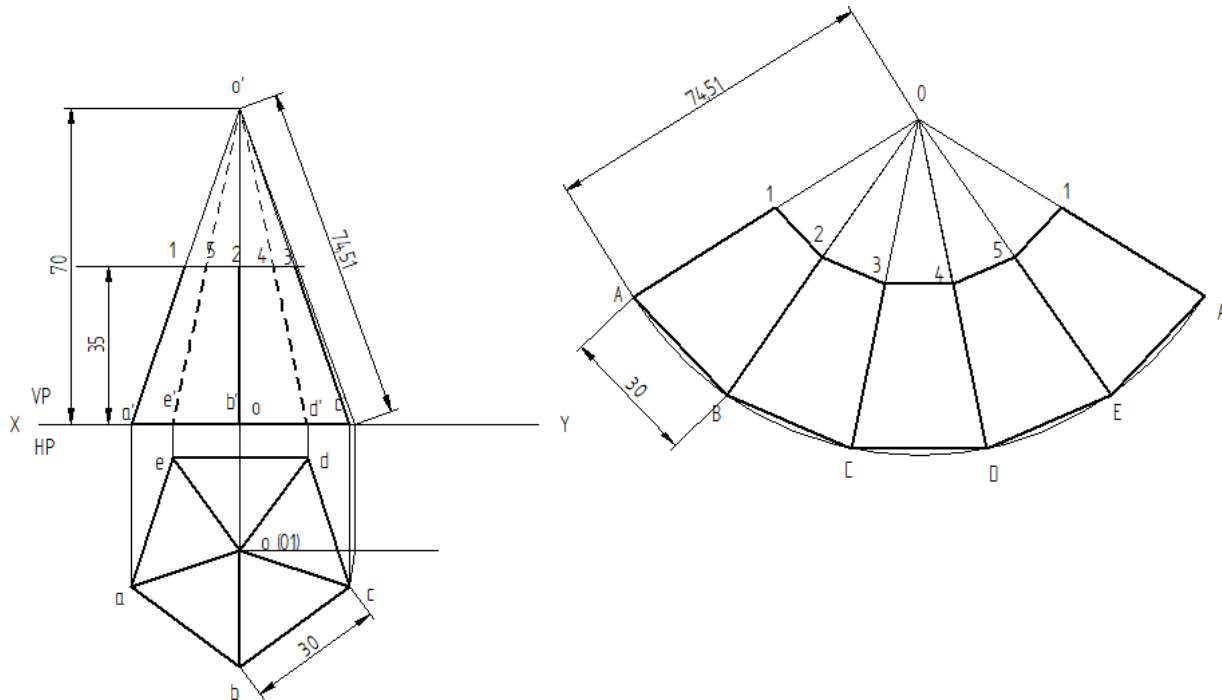
$$\theta = \frac{\text{Radius of the cone}}{\text{True length of generator}} \times 360^\circ$$

$$\theta = \frac{25}{79.06} \times 360 = 113.837$$

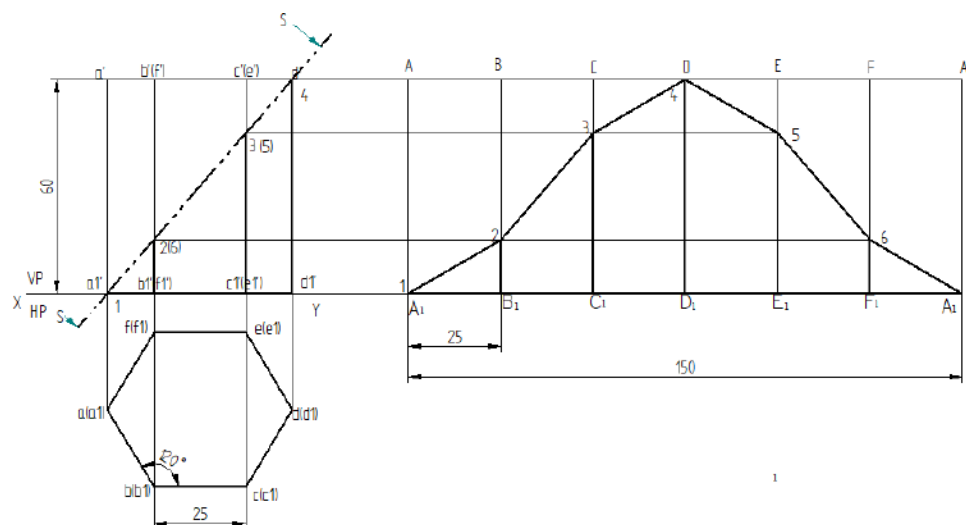
4. A square pyramid of base side 25mm and altitude 50mm rests on its base on HP with two sides of the base parallel to VP. It is cut by a plane bisecting the axis and inclined at 30° to the base. Draw the development of the lower part of the pyramid.



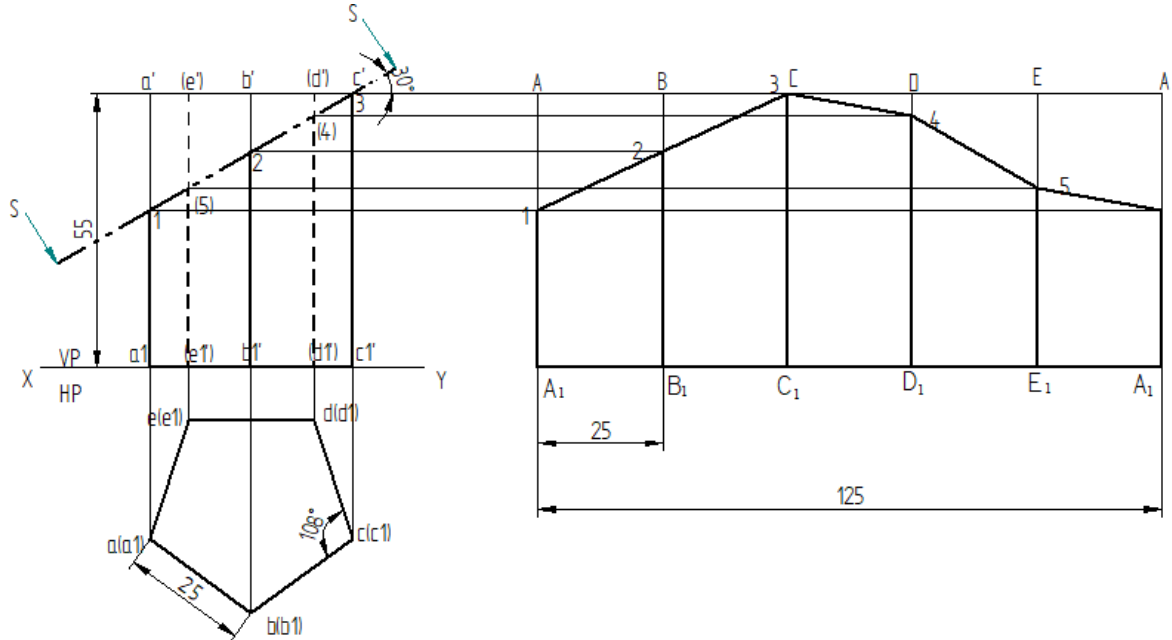
5. A pentagonal pyramid of base side 30mm and height 70mm is resting vertically with its base on the ground with one of its base edge being parallel to VP. and nearer to it. It is cut by a plane perpendicular to VP and parallel to HP at a distance of 35mm above the base. Draw the development of the lateral surface of the frustum of pyramid.



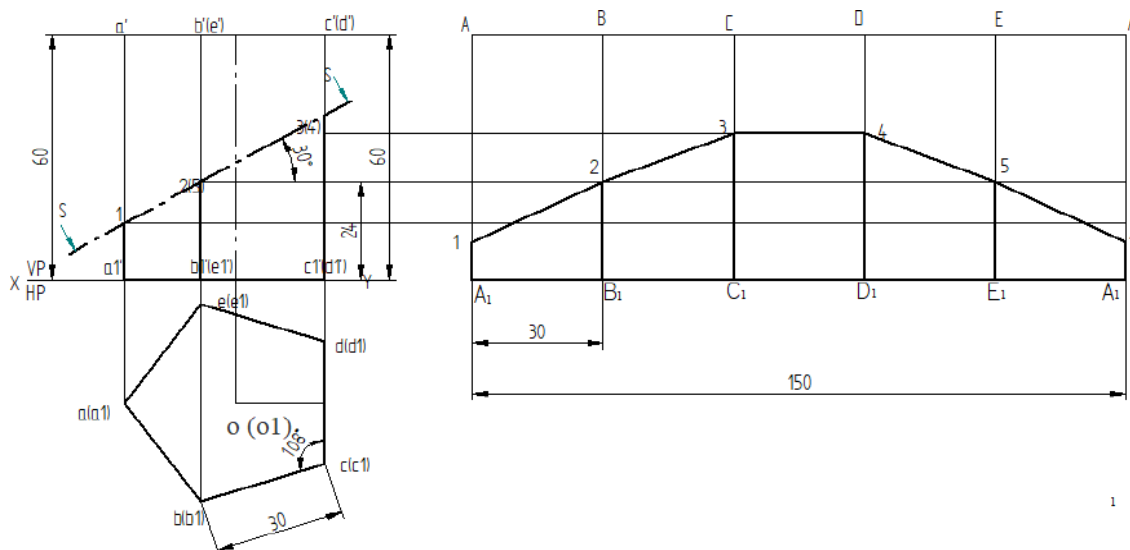
6. A hexagonal prism, having a 25mm base side and a 60mm axis height, has its two sides of base parallel to VP. A section plane cuts the prism in such a way that the section plane passes through extreme left bottom corner and extreme right top corner of the prism. Show the DLS of the half-cut prism



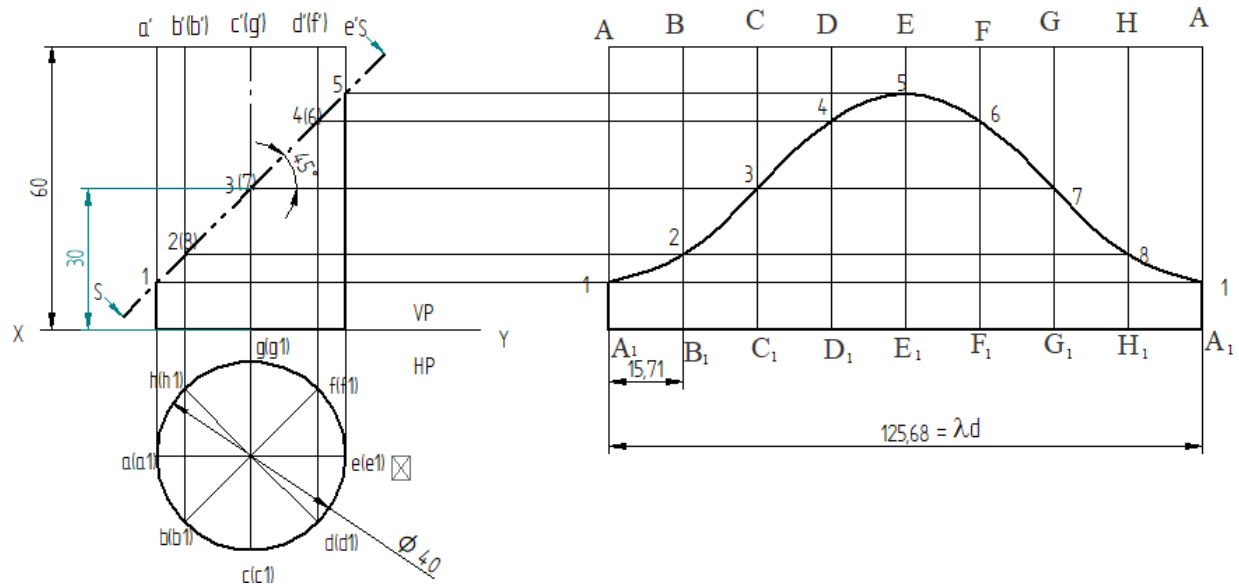
7. A pentagonal prism, edges base 25mm and axis 55mm long, rest with its base on HP such that one of its rectangular faces is parallel to VP. It is cut by a plane perpendicular to VP inclined at 30° to HP and passing through the right corner of the top face of the prism. Draw the sectional top view and develop the lateral surfaces of the truncated prism.



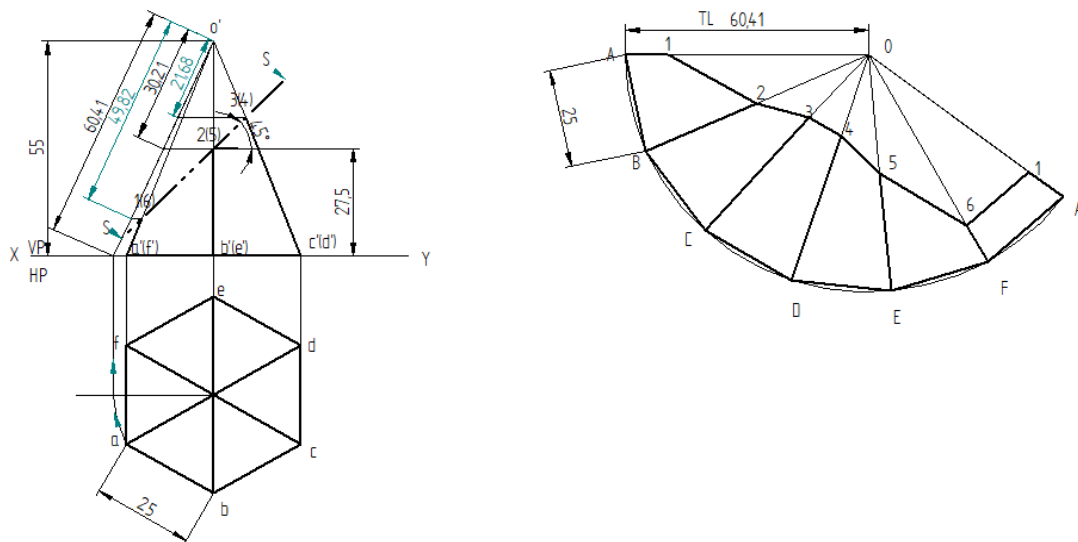
8. A right regular pentagon prism of side 30mm and 60mm height is resting on its base on HP having One of its base edge perpendicular to the VP. An auxiliary inclined plane inclined to HP at 30° and perpendicular to the VP cuts its axis at a distance of 24mm from the base. Develop the lateral surfaces of the truncated prism.



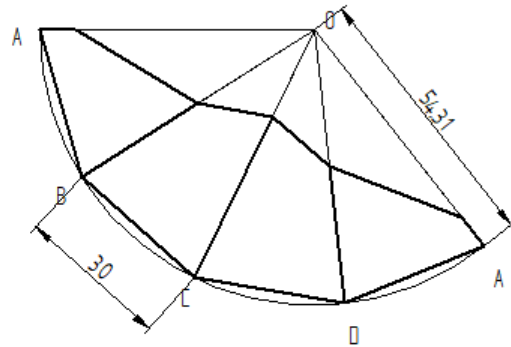
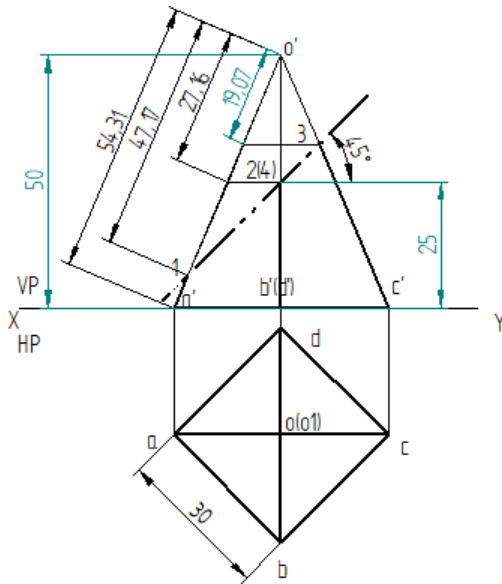
9. Draw the development of lateral surfaces of lower portions of a cylinder of diameter 40mm and axis 60mm. It is cut by a plane inclined at 45° to HP and perpendicular to VP and bisects the axis.



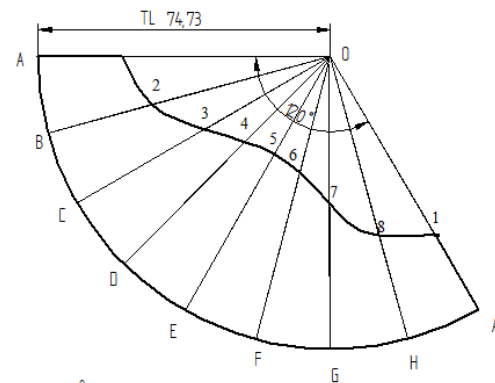
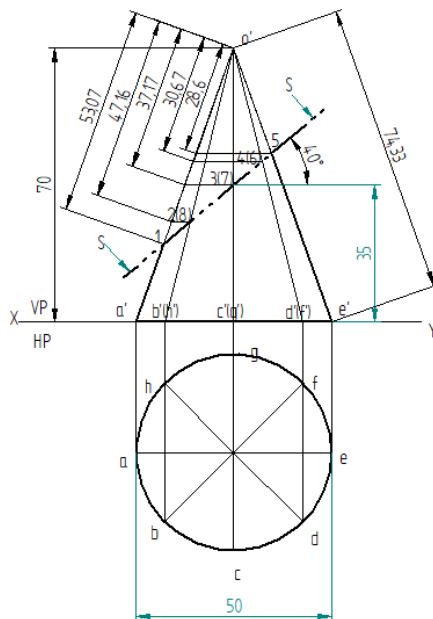
10. A hexagonal pyramid of base side 25mm and height 55mm is resting on its base on HP such that two sides of the triangle base are perpendicular to VP. A cutting plane is cut at an angle of 45° to HP and perpendicular to VP. The cutting plane passes through the middle of the axis of the pyramid. Obtain the development of the lateral surfaces of the truncated pyramid.



11. A square pyramid, having 30mm edge of base and 50 mm axis length, rests vertically on its base with adjacent edges of base are equally inclined to VP. It is cut by a cutting plane perpendicular to VP and inclined at 45° to HP, such that it bisects the axis. Draw DLS of the major part of the pyramid.



12. A cone of base diameter 50mm and height 70mm is resting on its base on HP. It is cut by a plane perpendicular to VP and inclined at 40° to HP. The plane bisects the axis of the cone. Draw the development of its lateral surfaces.



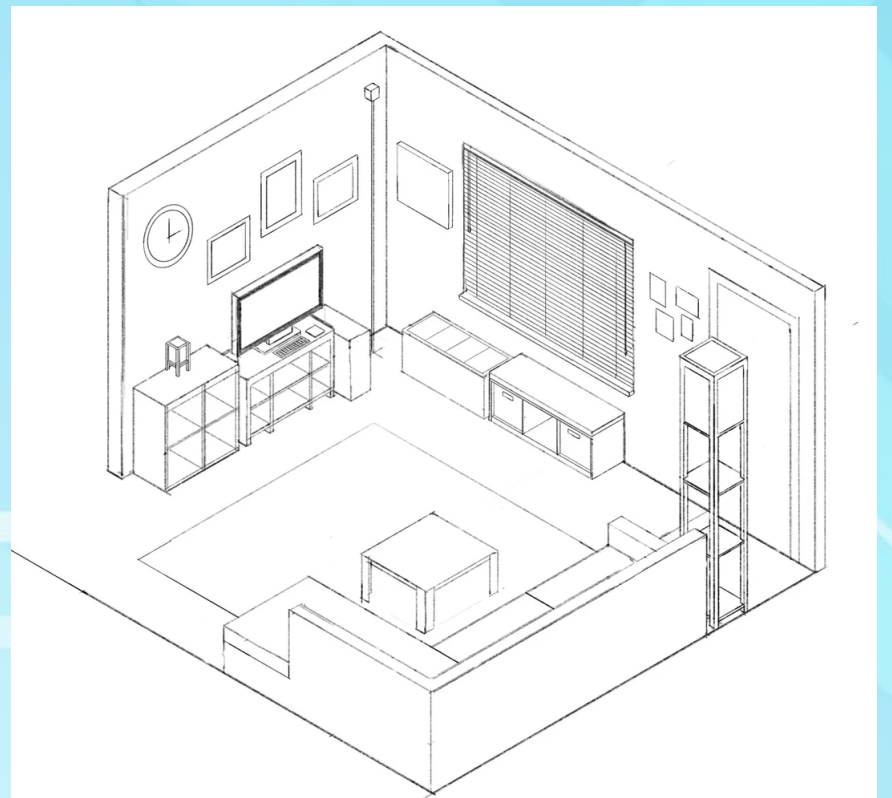
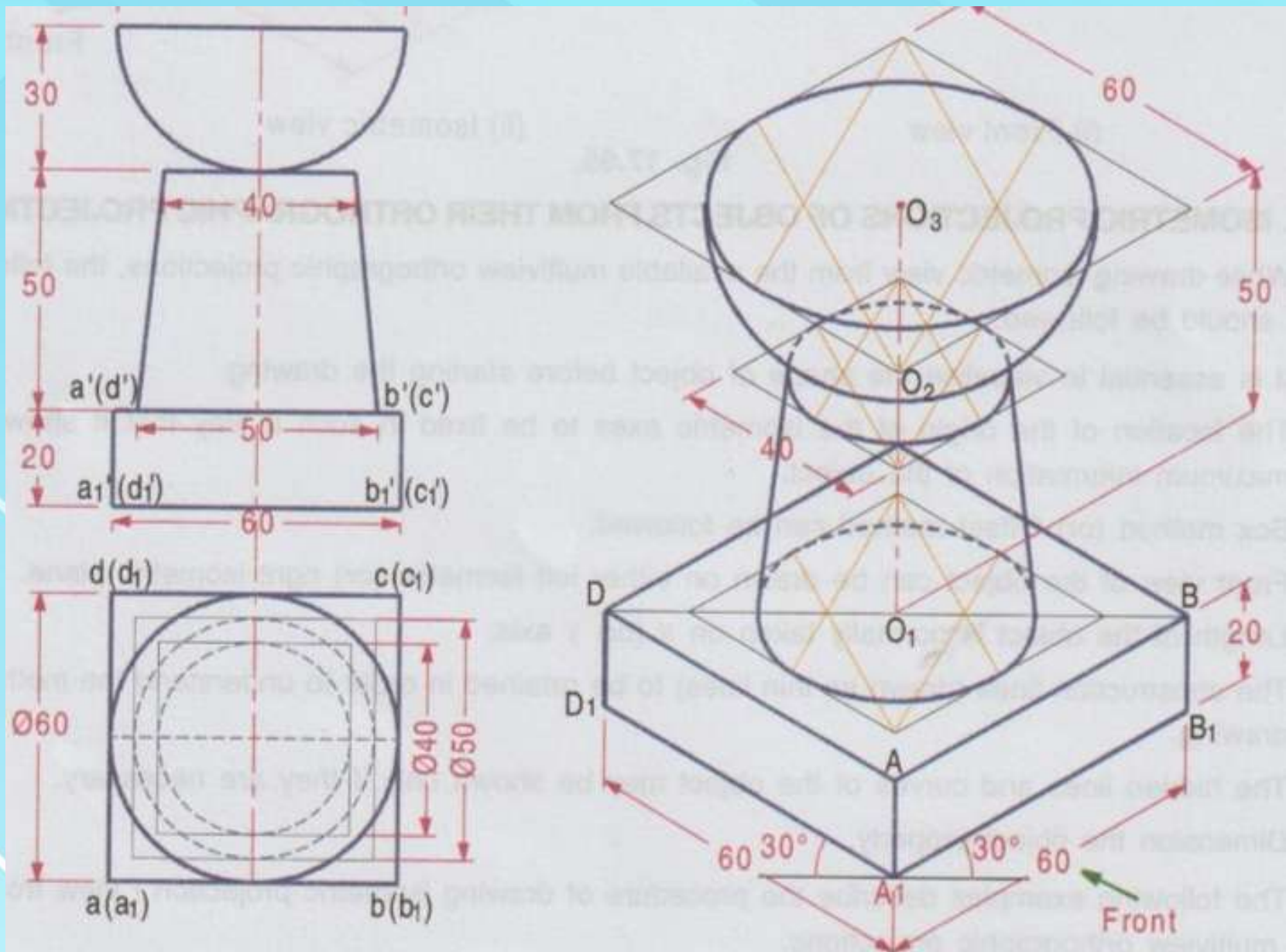
$$\theta = \frac{r}{R} \times 360^\circ$$

$$\theta = \frac{25}{74.33} \times 360^\circ = 121.081$$

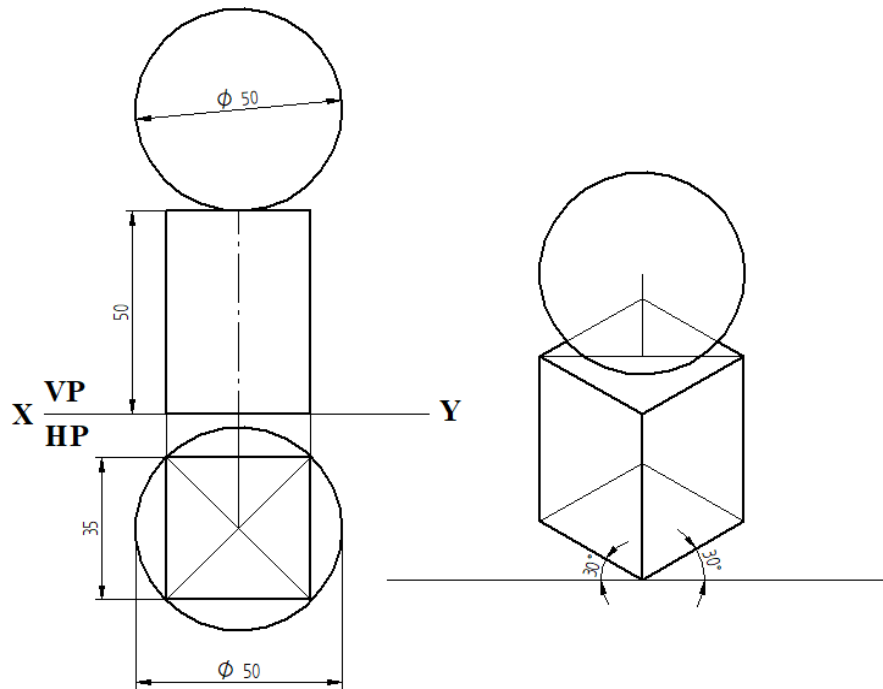
DEPARTMENT OF MECHANICAL ENGINEERING

ISOMETRIC PROJECTION

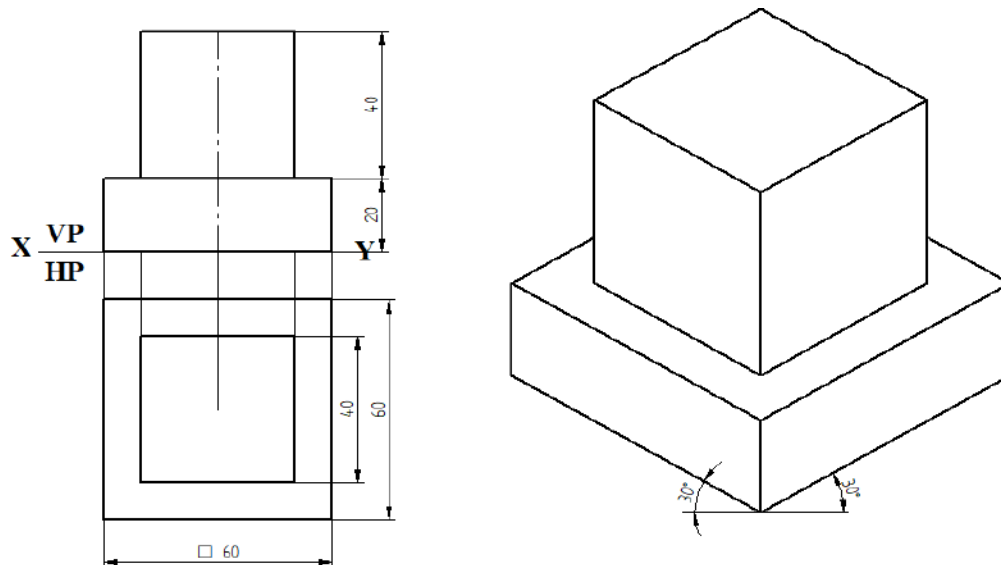
25EGD114/124



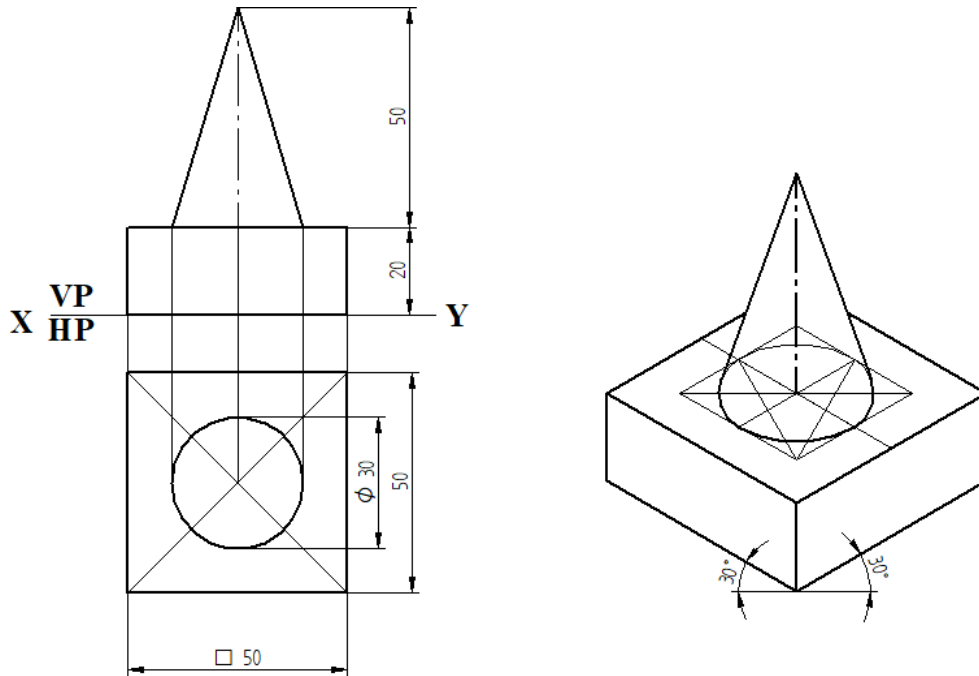
1. A sphere of diameter 50mm rests centrally on top of a square prism of base side 35mm and height 50mm. Draw the isometric projection of combination solids.



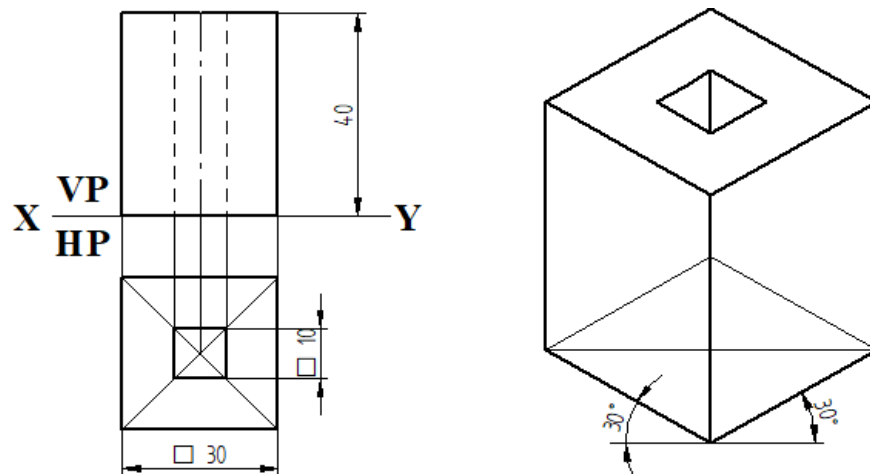
2. A cube of side 40 mm side rests centrally on a square block of 60mm edges and 20mm thick. Draw the isometric projections of the two objects with edges of the two blocks mutually parallel to each other.



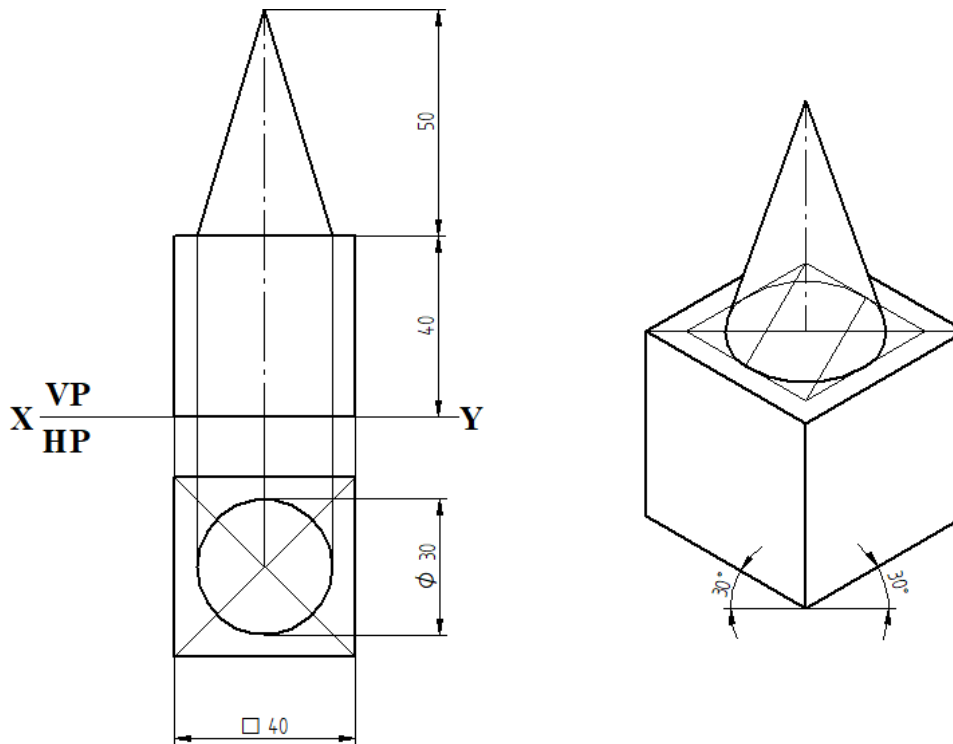
3. A right circular cone of base diameter 30mm and height 50mm rests centrally on top of a square block of 50mm side and 20mm thickness. Draw the isometric projection of the two solids.



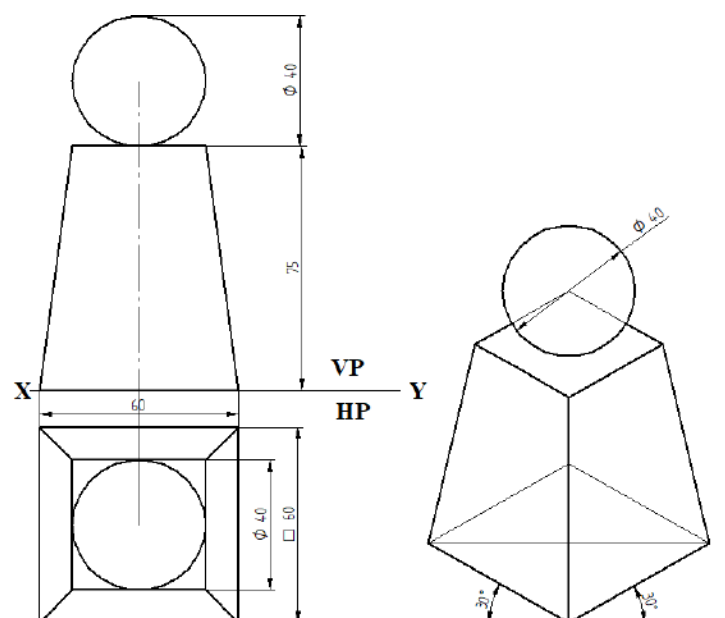
4. A square prism of side 30 mm 40 mm and height is resting on HP A vertical square bore of 10 mm side is cut through its face reaching other square face of the prism. Draw the isometric projections of the prism



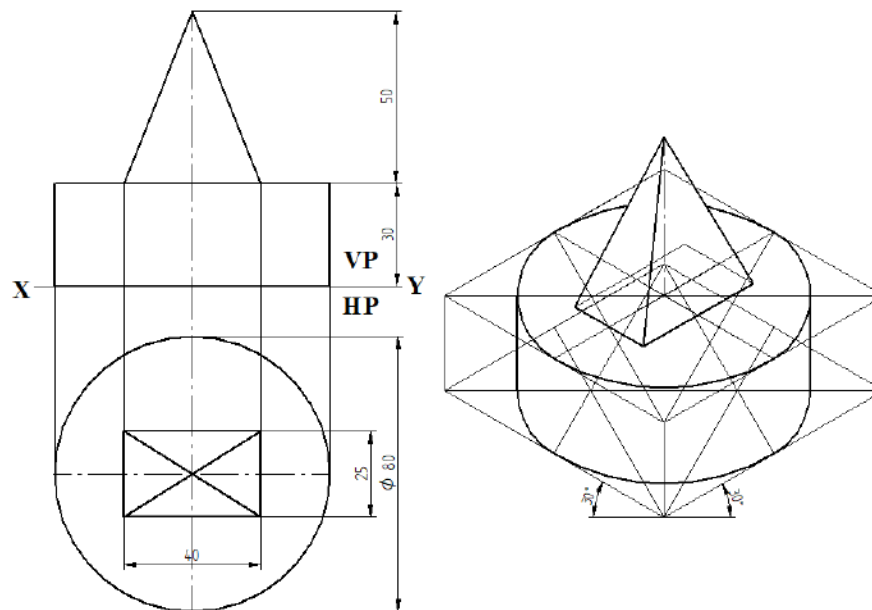
5. A right circular cone of base diameter 30mm and height 50mm rests on top of a cube of 40mm side such that the axes of the two solids are along the same vertical line. Draw the isometric projection of the combination of solids.



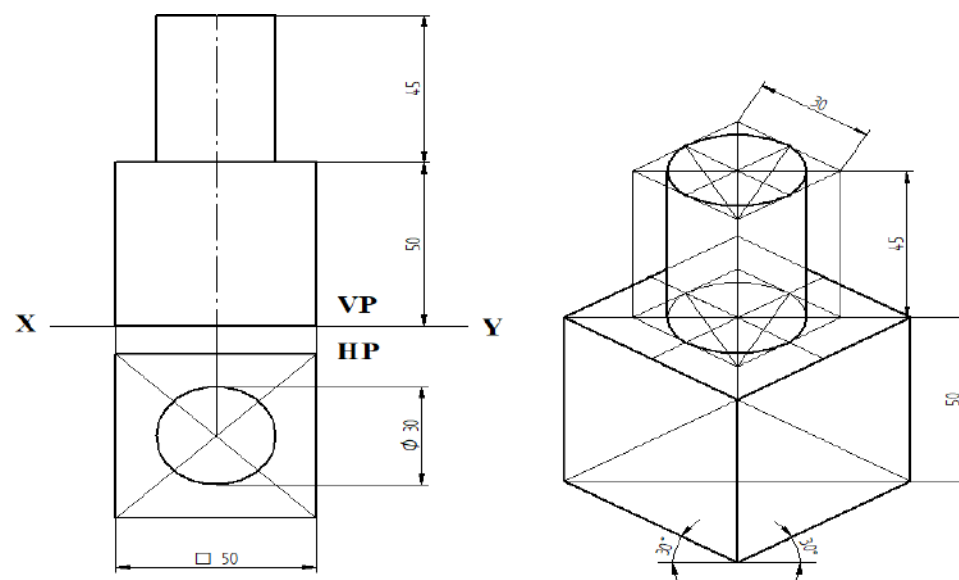
6. A sphere of diameter 40mm rests on the centre of a square frustum base side 60 mm, top side 40 mm and height 75 mm. Draw the isometric projection of the combination of solids.



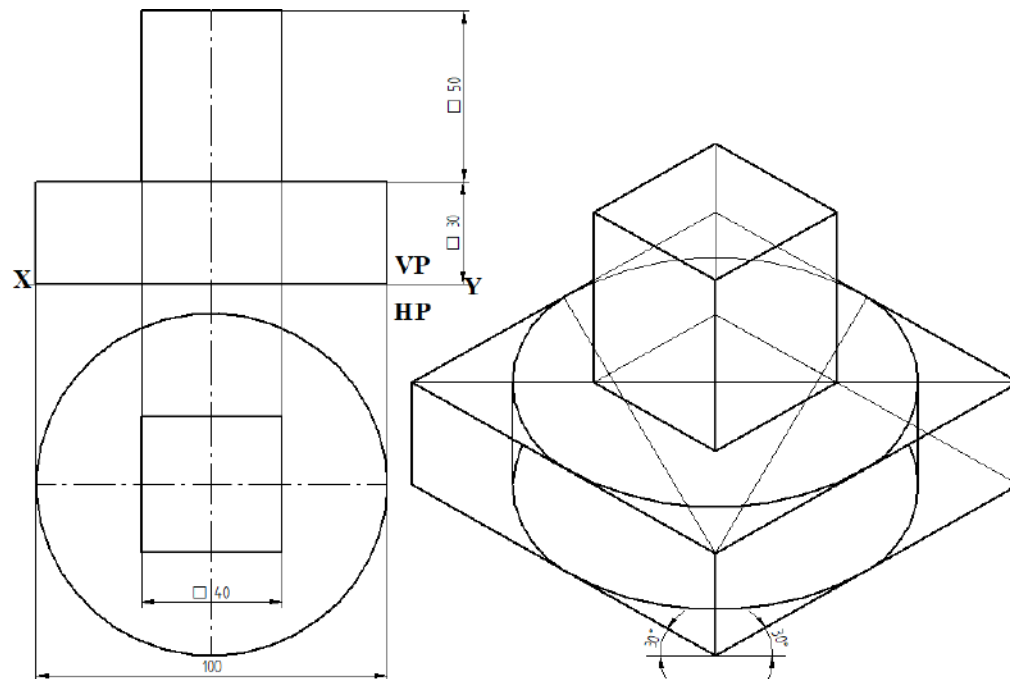
7. A rectangular pyramid of base 40x25mm and height 50mm is placed centrally on a cylindrical slab of diameter 80mm and thickness 30mm. Draw the isometric projection of combination of solids.



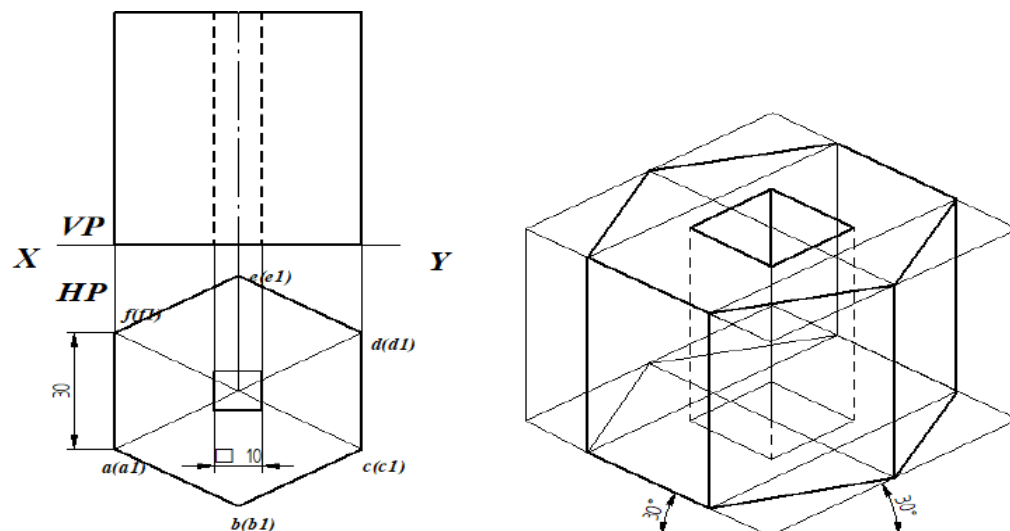
8. Draw the isometric projection of the two solids when one of them a cylinder of 30mm dia and 45mm long resting centrally on a cube of 50mm edge.



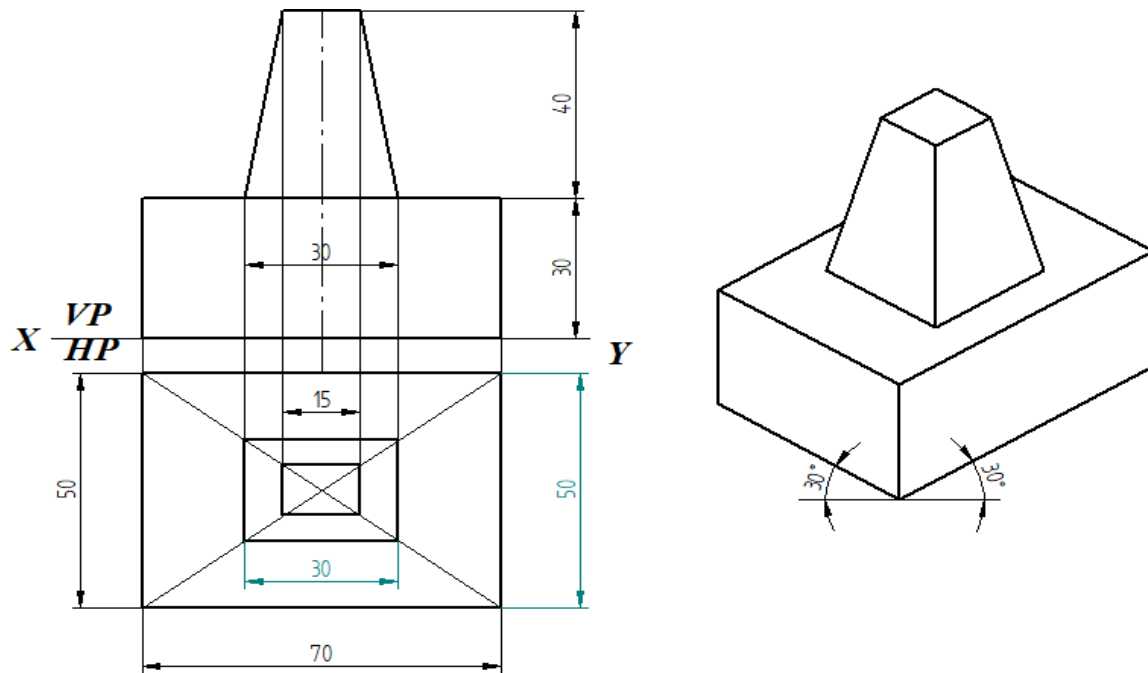
9. A square prism of base side 40 mm and height 50 mm is placed centrally on a cylindrical slab of diameter 100 mm and thickness 30 mm. Draw the isometric projection of the combination of solids.



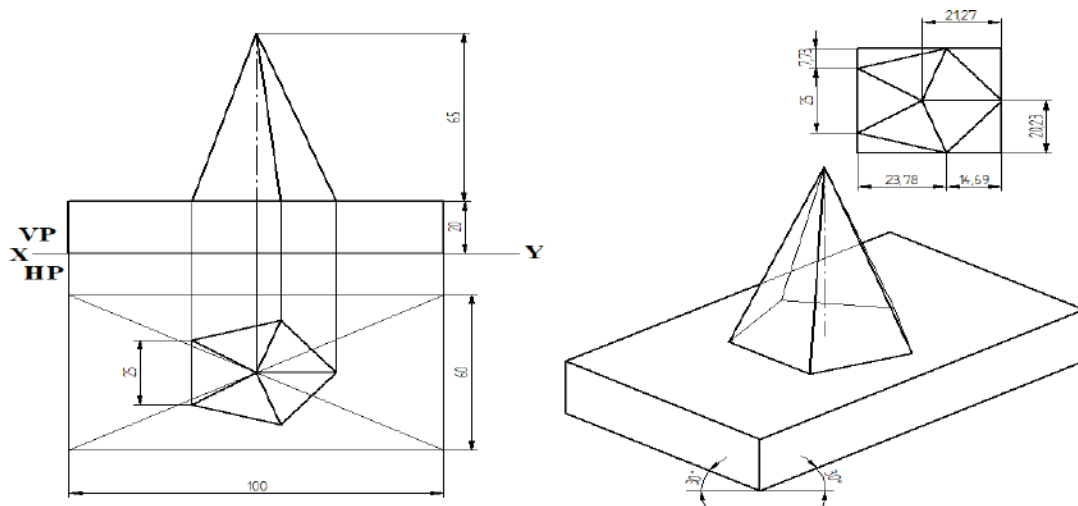
10. Draw the isometric projection of a hexagonal prism having side of base 30 mm and axis height 60 mm with a square hole of 20 mm at the centre. The side of square face is parallel to the rectangular face of the hexagonal prism.



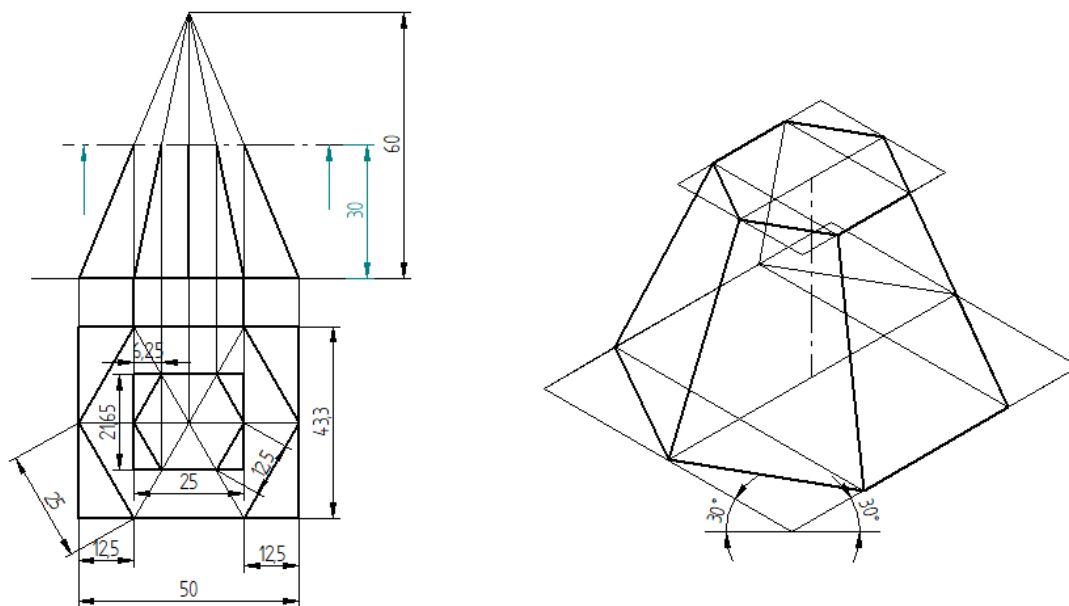
11. Draw the isometric projection of a frustum of square pyramid of base side 30 mm, top side 15 mm and height 40 mm, placed centrally on a rectangular prism of sides 70 mm x 50 mm and height 30mm, the shorter base side of the prism being perpendicular to VP.



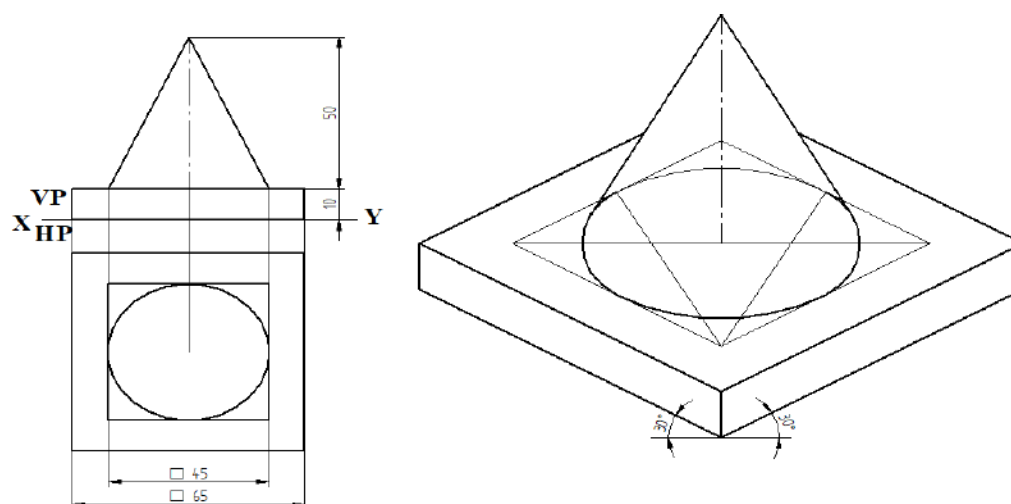
12. A pentagonal pyramid of base side 25 mm and height 65mm is placed centrally on a rectangular slab of 100mmx60mm and 20mm thick. Draw the isometric projection of the combination of solids.



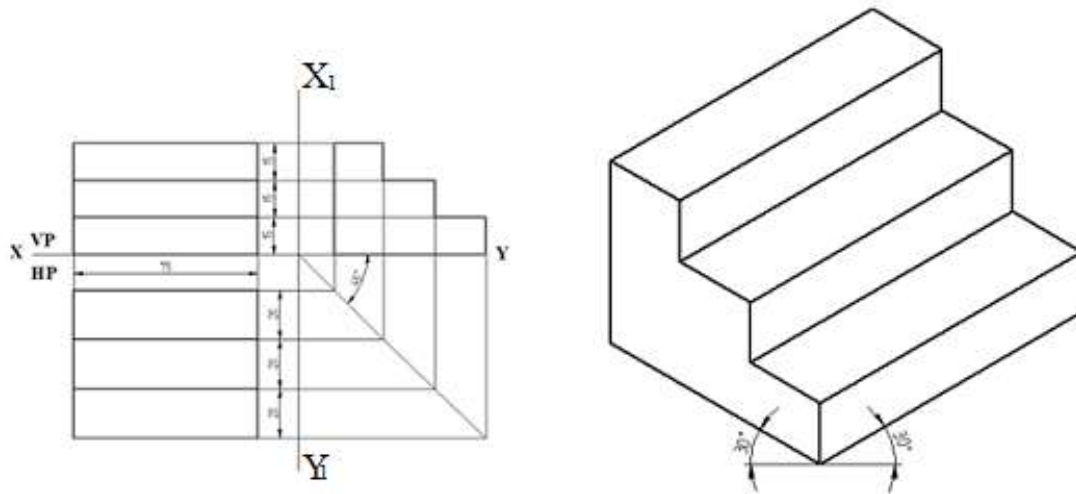
13. A hexagonal pyramid of base 25 mm and height 60 mm stands with its base on HP with an edge of base parallel to VP. A horizontal plane cuts the pyramid and passes through a point on the axis at a distance of 30 mm from the apex. Draw the isometric projections of the frustum of pyramid.



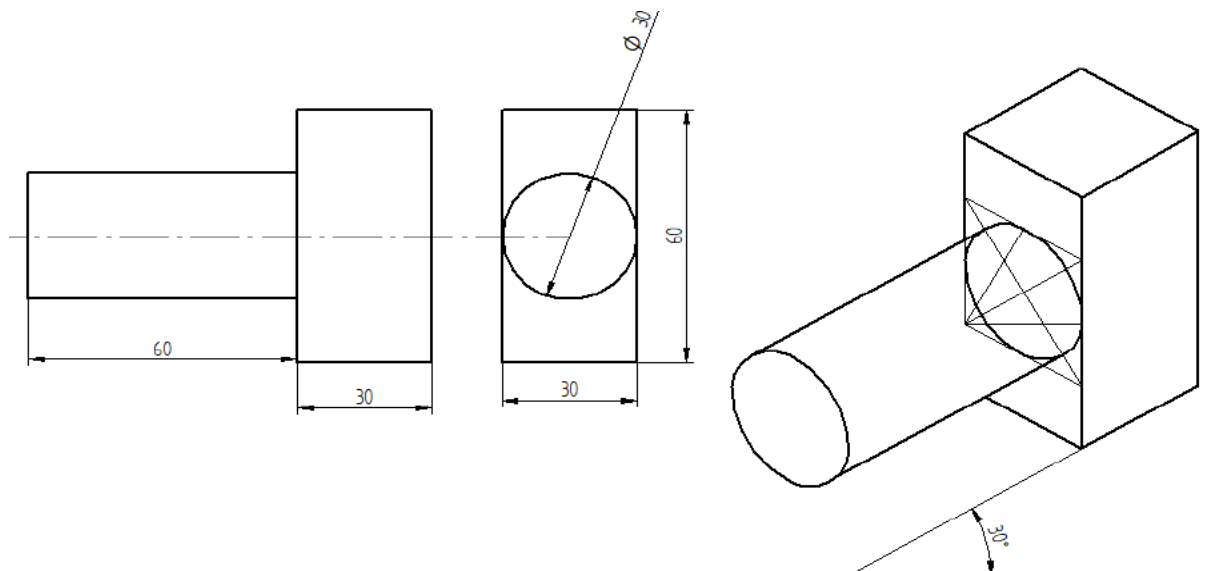
14. A cone diameter of base 45mm and height 50mm is mounted centrally on the top of a square slab of thickness 10mm & 65mm side. Draw the isometric projections of the combination of solids.



15. Draw the isometric view of door – steps having three steps of 20mm tread and 15mm rise. The steps measure 75mm width wise.



16. Following figure shows the orthographic views of an object. Draw the isometric Projection of the object.





DEPARTMENT OF MECHANICAL ENGINEERING

OVERVIEW

The Mechanical Engineering branch was started in the academic year 2011-2012. All the laboratories have been established with full-fledged and state-of-art equipment facilities. The Department has a team of well-qualified members of staff, with a blend of industrial and academic experience. The Faculty Members with Master's and Doctorate degrees, having specialization in machine design, thermal engineering, product design, and maintenance engineering are involved in academic and research activities.

VISION

To be a premier department for education in Mechanical engineering in the state of Karnataka, that moulds students into professional engineers.

MISSION

To provide a teaching-learning process that prepares engineers to meet the needs of industry and higher learning.

To Provide environment for self learning to meet the challenges of changing technology and inculcate team spirit and leadership qualities to succeed in professional career.

To instill professional ethics and concern for environment for the benefit of society.



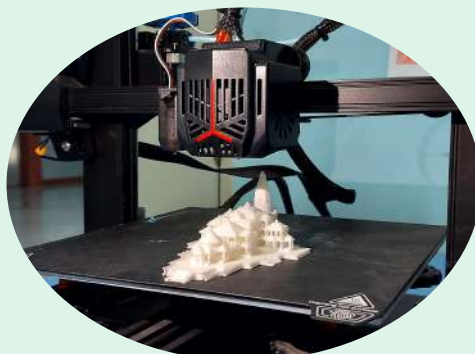
FLUID MECHANICS LAB



MACHINE SHOP LAB



PROJECT LAB



3D PRINTING COE



BNMIT-TOYOTA COE